

# Constant- $\ell$ rotating-dust galaxy models, Part I: local realizability, sharp constraints, and global-completion obstructions

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Stationary, axisymmetric rotating-dust spacetimes have been proposed as a general-relativistic account of flat galactic rotation curves. We give a theorem-level analysis of the constant- $\ell$  subclass, separating local realizability from regularity, positivity, asymptotic completion, junction, and observational requirements. The proposed nonzero constant ZAMO-speed field solves neither the exact nor the retained-order equation on an open region. Nevertheless, analytic flat equatorial traces exist locally away from the axis. Their continuation is controlled by a sharp density discriminant, a fold of the point-linearising map, axis regularity, and quantitative span bounds. We prove automatic analyticity on every regular sheet, exact Hadamard instability of elliptic Cauchy continuation, global period obstructions for toroidal bodies, and Liouville rigidity of the global regular-axis  $\ell = 0$  branch. The auxiliary equation possesses a weighted variational principle, a De Donder–Weyl multisymplectic formulation, Noether currents, and a Lagrangian boundary-data space; these structures degenerate precisely at the density fold and must not be identified with the physical Einstein–dust phase space. A zero-density surface is only a candidate boundary: shell-free completion requires the complete Darboux data and compatibility with a gauge-fixed exterior Ernst problem. Finally, the local ZAMO speed does not determine a spectroscopic rotation curve. The resulting hierarchy distinguishes proved identities, global obstructions, conditional matching criteria, and questions reserved for the numerical Part II.

## I. INTRODUCTION

The persistence of nearly flat orbital-speed profiles in the outer parts of many disk galaxies is one of the clearest manifestations of the galactic mass discrepancy [1]. The standard interpretation invokes a dominant non-luminous matter component [2], but it is nevertheless legitimate to ask whether the global, nonlinear structure of general relativity can modify the mass inferred from galactic kinematics. That question is more demanding than a comparison of a single velocity profile with data. A viable relativistic alternative must simultaneously solve the field equations, possess a physically acceptable matter source, remain regular on the rotation axis, admit a controlled outer completion, and predict the frequency shifts actually measured by an observer. We address this global viability problem for a specific subclass of stationary, axisymmetric rotating-dust models.

The relevant solution space is the stationary, axisymmetric dust family described by Winicour and in the standard exact-solution literature [3, 4]. Balasin and Grumiller used its rigidly rotating sector as a galactic model [5]. A later proposal by Astesiano, Cacciatori, Gorini, and Re enlarged the construction to differential rotation by parametrizing the geometry with a field  $\eta(r, z)$  and a negative function  $H(\eta)$  [6]. The speed measured by ZAMOs is  $v = \eta/r$ , whereas the departure from rigid rotation is encoded in  $\ell(\eta) = H_{,\eta}/H$ . In Sect. 4 of Ref. [6], the choice  $\ell = \ell_0 = av_c/R_G$  is inserted into a low-velocity truncation of the velocity-field equation (VFE). After a further near-edge replacement  $r \simeq R_G$ , the resulting frozen-coefficient equation is reported to admit the constant value  $v = av_c$ , which is then associated with a flat rotation curve. The same approximation is used to argue for a density reduction and a finite density cutoff. These claims raise a precise question: can they be realized by a solution of the parent constant- $\ell$  equations on a finite region, and can that branch be completed to a regular isolated spacetime?

The proposal belongs to a broader strong-gravitomagnetic programme that connects exact  $(\eta, H)$  dust solutions, ZAMO kinematics, and weak-field limits [7–9]. Its claims must therefore be compared both with the exact branch equations and with earlier global results for rotating dust. Most decisively, Zingg, Aste, and Trautmann derived a finite-radius density restriction for the general non-rigid Winicour family, argued that an asymptotically flat dust region cannot extend to arbitrarily large cylindrical radius under their hypotheses, and identified the tension caused by prescribing both metric and normal data at a smooth vacuum interface [10]. Caporali’s global theorem, Pfister’s conditional analysis, and the distributional singularity results for rigid van Stockum–Bonnor flow supply further

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precedents [11–13]. Accordingly, finite radial extent, asymptotic confinement, and matching tension are not claimed here as new general mechanisms.

Several independent analyses have also identified serious obstacles for large gravitomagnetic explanations of galactic dynamics. In physically sourced weak-field disks, corrections of order  $10^{-6}$  and the dangers of incorrect scaling or homogeneous solutions were established in Refs. [14–16]; Rowland gives earlier context for the low-velocity correspondence issue [17]. Costa, Natário, Frutos-Alfaro, and Soffel related the large dragging of the rigid Balasin–Grumiller spacetime to its asymptotic frame and axial singular structure [18]. Broader stationary-spacetime arguments use both massive-particle dynamics and lensing [19, 20]. These analyses do not directly test whether the specific constant- $\ell$  edge ansatz solves its parent equations.

Complementary work has developed the nonlinear rotating-dust programme itself. Ruggiero emphasized its lack of a direct Newtonian analogue and its likely global singular structure [21]. Direct redshift formulae and inverse-reconstruction difficulties precede the present observability argument [22, 23]; a recent locally inertial and radially locked construction gives a complementary local-observer programme [24]. Further studies have treated observer-dependent velocities, lensing, explicit non-rigid subclasses, topological regularity, and quasilocal pressure [25–29]. Related approaches include nonstandard outer boundary behaviour, dark-Coriolis profiles, and CMB multipoles [30–32]. Conically Minkowskian ends in the broader dust family show that our asymptotic results concern the standard isolated asymptotically flat class, not every possible outer geometry [33].

The present contribution is narrower and complementary. We give the direct exact and retained-order residual test of the proposed edge field; the explicit sharp constant- $\ell$  density interval and quantitative annular span bounds; the fold identification and its variational degeneration; local analytic and semi-global Runge realization together with exact Hadamard instability; a metric-only axis-jet obstruction; exact asymptotic locking; toroidal period obstructions; and a gauge-fixed linearized Ernst boundary test. These are branch-specific sharpenings and structural results, not priority claims for the broad finite-radius or matching mechanisms. The central question is whether the constant- $\ell$  subclass can support a nonzero flat, or quantitatively approximately flat, ZAMO-speed profile while maintaining the field equations, nonnegative density, regularity, and an admissible exterior.

This question separates into eight parts. Does the proposed constant field solve either VFE on an open region? Can transverse curvature support a flat equatorial trace, and can that trace extend to a regular axis? What sharp pointwise and radial-span restrictions follow from density positivity? What regularity and stability properties govern continuation? Can the branch possess a standard isolated asymptotically flat end? Does zero density yield a shell-free boundary? Which topology and period conditions are required for global Weyl and twist potentials? Finally, which auxiliary variational structures survive the fold, and which local quantities determine an observed spectroscopic shift?

The principal conclusions are collected in Theorem 1. The proposed constant field fails both equations on every open radial region, although flat traces exist locally off the axis. Their continuation is constrained by the density fold, axis regularity, and the invariant vertical-shape condition. Regular sheets are automatically analytic but elliptic Cauchy continuation is Hadamard unstable. The exact asymptotic end is locked to a single limiting value of  $rv$ , while a finite density cutoff is only a candidate boundary. Toroidal matter requires global period tests. The auxiliary action has exact multisymplectic and Noether structure, but its Legendre map collapses at the density fold and it is not the Einstein–dust phase space. The local ZAMO speed does not by itself determine a spectroscopic rotation curve.

Section II reviews the exact family, and Sect. III states the synthesis theorem. Sections IV–X give the field-equation, density, fold, analytic-continuation, exact-flatness, and approximate-flatness arguments. Section VII develops the auxiliary variational geometry, while Sect. VIII separates it from the Brown dust and Einstein–Hilbert–GHY variational problem and records conditional EMDA extensions. Section XI treats the outer end, and Sect. XII adds the helical, toroidal-period, and linearized Ernst analyses. Section XIII derives the local matching alternatives, while Sect. XIV formulates the observer-aware spectroscopic forward problem. Section XV closes with the proved scope and the remaining physical-completion programme.

## II. ROTATING-DUST FRAMEWORK AND THE CONSTANT- $\ell$ REDUCTION

This section identifies the precise field-equation framework underlying the subsequent viability analysis. We first recall the exact velocity-field equation for the general  $(\eta, H)$  family and then, specialize it to the constant- $\ell$  branch. We also distinguish the exact reduced equation from the retained-order equation used in Ref. [6], thereby separating consequences of the constant- $\ell$  restriction from those of the additional low-velocity approximation. We begin by fixing the notation and conventions used throughout. We work with the exact stationary, axisymmetric rotating-dust family described in Refs. [3, 4, 6]. We use coordinates  $(t, \phi, r, z)$  adapted to the stationary and axial Killing fields  $k = \partial_t$  and  $m = \partial_\phi$ . In these coordinates, the circular orbits of  $m$  have period  $2\pi$ . We use signature  $(-, +, +, +)$  and units  $c = 1$ , while retaining  $G_N$  in gravitational actions and junction formulae. The coordinates  $t, r, z$  and the potential  $\eta$  have dimensions of length;  $\phi, v, H$ , and  $\mu$  are dimensionless; and  $\Omega$  and  $\ell$  have dimensions of inverse length. In particular,

$x = r\ell$  is dimensionless. The circular line element takes the form

$$ds^2 = g_{tt}(r, z) dt^2 + 2g_{t\phi}(r, z) dt d\phi + g_{\phi\phi}(r, z) d\phi^2 + e^{\mu(r, z)}(dr^2 + dz^2). \quad (1)$$

The metric functions in Eq. (1) are not independent. The matter source is pressureless dust,  $T_{\mu\nu} = \rho u_\mu u_\nu$ , where  $\rho$  is the dust density and  $u$  is the dust four-velocity. Since the dust worldlines are circular, their four-velocity is tangent to a linear combination of the two Killing directions. The exact solution family can be parametrized by a scalar field  $\eta(r, z)$  and a function  $H(\eta)$ , with

$$H = g(k + \Omega m, k + \Omega m) < 0, \quad \eta = g(k + \Omega m, m), \quad (2)$$

where  $\Omega(\eta)$  is the coordinate angular velocity of the dust. Thus,  $H$  is the squared norm of the unnormalized dust-flow direction  $k + \Omega m$ , while  $\eta$  is its scalar product with the axial Killing field. The inequality  $H < 0$  expresses the timelike character of the dust flow. The normalized four-velocity and the relation between  $\Omega$  and  $H$  are therefore

$$u = \frac{k + \Omega m}{\sqrt{-H}}, \quad \frac{d\Omega}{d\eta} = \frac{H, \eta}{2\eta}. \quad (3)$$

Once  $\eta(r, z)$ ,  $H(\eta)$ , and the corresponding angular velocity  $\Omega(\eta)$  are specified, the metric coefficients in the  $(t\phi)$ -sector are given by

$$g_{\phi\phi} = \frac{r^2 - \eta^2}{-H}, \quad g_{t\phi} = \eta - \Omega g_{\phi\phi}, \quad g_{tt} = H - 2\Omega\eta + \Omega^2 g_{\phi\phi}. \quad (4)$$

The remaining metric function  $\mu$  is determined, up to an additive constant, by the remaining Einstein equations. For later use, let  $G = (g_{AB})_{A, B \in \{t, \phi\}}$  denote the metric block associated with the two Killing directions. The parametrization above implies the exact identity

$$\det G = g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -r^2. \quad (5)$$

Ref. [6] introduces the ZAMO-measured speed  $v$  and the logarithmic derivative  $\ell$  through

$$v(r, z) = \frac{\eta(r, z)}{r}, \quad \ell(\eta) = \frac{H, \eta}{H}. \quad (6)$$

In this notation, the exact velocity-field equation (VFE) (see Eq. (3.1) of Ref. [6]) can be written as

$$\begin{aligned} \mathcal{V}_\ell[v] := & \ell'(\eta) \left( \frac{1}{v} - v \right) [r^2 v_z^2 + (rv_r + v)^2] + \ell(\eta) \left[ \left( \frac{1}{v} - v \right) (rv_{zz} + rv_{rr} + 3v_r) - \left( \frac{1}{v} + v \right) \frac{r}{v} (v_z^2 + v_r^2) + \frac{2}{r} \right] \\ & + 2 \left( v_{zz} + v_{rr} + \frac{v_r}{r} - \frac{v}{r^2} \right) \end{aligned} \quad (7)$$

with the field equation  $\mathcal{V}_\ell[v] = 0$ . Throughout the local analysis, we assume  $r > 0$ ,  $v \in C^2$ , and  $0 < v < 1$ . The strict inequality avoids the singular value  $v = 0$  in expressions containing  $v^{-1}$  and implements the subluminal interpretation. We now impose  $\ell(\eta) \equiv \ell_0$  with  $\ell_0 \in \mathbb{R}$ . Using Eq. (6), the definition of  $\ell$  integrates to

$$H(\eta) = -C e^{\ell_0 \eta}, \quad C > 0. \quad (8)$$

For  $\ell_0 \neq 0$ , Eq. (3) further gives, on every interval not crossing  $\eta = 0$

$$\Omega(\eta) = \Omega_* - \frac{C\ell_0}{2} \text{Ei}(\ell_0 \eta), \quad (9)$$

whereas  $\Omega$  is constant when  $\ell_0 = 0$ . Here, Ei denotes the exponential integral. Since  $\ell' = 0$ , the exact constant- $\ell$  VFE is  $\mathcal{E}_{\ell_0}[v] = 0$ , where

$$\mathcal{E}_{\ell_0}[v] := \left[ 2 + \ell_0 r \left( \frac{1}{v} - v \right) \right] \Delta_{rz} v + \left[ \frac{2}{r} + 3\ell_0 \left( \frac{1}{v} - v \right) \right] v_r - \ell_0 r \left( \frac{1}{v^2} + 1 \right) |\nabla_{rz} v|^2 + \frac{2\ell_0}{r} - \frac{2v}{r^2}, \quad (10)$$

with

$$\Delta_{rz} v := v_{rr} + v_{zz}, \quad |\nabla_{rz} v|^2 := v_r^2 + v_z^2. \quad (11)$$

If we discard the terms that are higher order in the low-velocity expansion used in Sect. 4 of Ref. [6], we obtain the retained-order equation  $\mathcal{T}_{\ell_0}[v] = 0$ , where

$$\mathcal{T}_{\ell_0}[v] := \left(2 + \frac{\ell_0 r}{v}\right) \Delta_{rz} v + \left(\frac{2}{r} + \frac{3\ell_0}{v}\right) v_r - \frac{\ell_0 r}{v^2} |\nabla_{rz} v|^2 + \frac{2\ell_0}{r} - \frac{2v}{r^2}. \quad (12)$$

In Sect. 4 of Ref. [6], the constant value

$$\ell = \ell_0 = \frac{av_c}{R_G} \quad (13)$$

is adopted, where  $v_c$  is a characteristic stellar orbital speed,  $R_G$  is a radial scale comparable to the galactic radius, and  $a$  is a dimensionless parameter. With this parameter choice, the retained-order equation  $\mathcal{T}_{\ell_0}[v] = 0$  coincides with Eq. (4.3) of Ref. [6]. The difference between the exact and retained-order operators is the algebraic identity

$$\mathcal{E}_{\ell_0}[v] - \mathcal{T}_{\ell_0}[v] = -\ell_0 [rv\Delta_{rz}v + 3vv_r + r|\nabla_{rz}v|^2]. \quad (14)$$

In particular, the discarded part vanishes for every spatially constant field. All terms in Eqs. (10) and (12) have dimension  $L^{-2}$ , since  $[\ell_0] = L^{-1}$  and  $v$  is dimensionless.

### III. PRINCIPAL CONSTRAINTS ON THE CONSTANT- $\ell$ BRANCH

The branch is governed by several logically independent tests: local field equations, inversion of the auxiliary map, density, axis jets, continuation, outer geometry, boundary data, and observability. The following synthesis keeps those levels separate. Every item inherits the hypotheses of the cited result; in particular, no local construction is being promoted to a global galaxy and no numerical evidence is used as a proof.

**Theorem 1** (Synthesis theorem and structural dichotomies). *Consider the exact stationary, axisymmetric rotating-dust family of Sect. II, with positive subluminal ZAMO speed and constant  $\ell = \ell_0$ . The following statements hold.*

1. *A nonzero spatially constant field cannot satisfy either the exact or retained-order constant- $\ell$  VFE on an open set with nontrivial radial projection. The proposed edge calibration makes its residual vanish only on  $r = R_G$ . On the regular jet stratum  $A \neq 0$ , by contrast, the exact VFE is involutive and point-equivalent to a linear elliptic equation; the fold  $A = 0$  is a genuine symbol-rank jump.*
2. *At every regular noncritical point,  $\rho \geq 0$  is equivalent to the sharp interval in Eq. (33). The lower endpoint is simultaneously a nondegenerate Whitney fold of the linearising map, whereas strict positive density selects its orientation-preserving inverse sheet.*
3. *Every analytic exactly flat equatorial trace satisfying the strict off-axis inversion conditions has a local analytic realization. Conversely, every  $C^2$  solution on a connected invertible sheet is automatically analytic, and an open constant- $\eta$  plateau propagates throughout that sheet. Continuation from unrestricted smooth Cauchy data is nevertheless Hadamard unstable. Elliptic Runge approximation gives semi-global realizations only on a protected compact core with quantitative physical margins; it gives no control on the surrounding region.*
4. *A  $C^2$  metric with a regular effective axial action forces  $H'(0) = 0$  on every branch reaching the axis. Hence a nontrivial constant logarithmic derivative can reach a regular axis only when  $\ell_0 = 0$ . No axis-connected branch can carry a nonzero exactly flat trace. Under the strong complete-half-plane, regular-axis, and decay assumptions of Theorem 5, even the global  $\ell_0 = 0$  branch is Minkowski vacuum.*
5. *Under reflection symmetry, positive density, subluminality, and the stated vertical-maximum and logarithmic-slope hypotheses, approximately flat traces have finite radial span. The nonpositive- $\ell_0$  bound is sharp on its closure and strict for each  $\ell_0 < 0$ . For  $\ell_0 > 0$  the old comparison bound has a quantified strict gap, while the exactly flat supremum is  $2/(1 + v_0^2)$  and is approached by genuine analytic solutions. The exact positive- $\ell_0$ , positive-tolerance control optimum remains open.*
6. *If a nonzero constant- $\ell_0$  sheet itself reaches a standard asymptotically flat end and remains uniformly subluminal, then  $\eta = rv$  locks to  $q = -\ell_0^{-1} \log C$  and  $v = O(r^{-1})$ ; the retained-order logarithmic branch cannot supply such an end. If density is nonnegative in outer neighbourhoods, the end is eventually vacuum and locally flat. Its possible nonzero  $q$  is an affine helical holonomy, not a constructed string source, and it cannot extend within the same flat region to a regular axial fixed point.*

7. A compact connected body whose meridional image is a disk separated from the axis is a solid torus. Its vacuum orbit surface is not simply connected, and global Weyl and twist potentials require explicit period tests. At any finite boundary,  $\rho = 0$  is only a candidate condition: shell-free gluing is equivalent to equality of the full first and second fundamental forms. The linearized exterior Ernst data must additionally lie in a Calderón Lagrangian graph. None of these necessary conditions constructs a nonlinear isolated exterior.
8. The linear auxiliary equation possesses an exact weighted action, Noether currents, covariant Hamiltonian variables, and a multisymplectic current formula, but its Legendre map collapses at the density fold and it is not the Einstein–dust phase space. The physical theory is instead the Einstein–Hilbert–GHY action coupled to vortical Brown dust; equality of its boundary momenta is equivalent to the complete Darmois condition. In the different EMDA theory, neutral axidilaton rigidity holds only under the global hypotheses of Theorem 10; duality transformations do not by themselves produce a rotating-dust completion.
9. The local scalar  $v$  does not determine an emitter–observer frequency shift independently of the metric, the two observer frames, and the connecting null ray. A spectroscopic rotation curve therefore requires the completed spacetime and a specified forward problem.

Consequently, the constant-field, constant- $\ell$  edge construction does not by itself define a globally viable isolated rotating-dust galaxy. This conclusion does not exclude nonconstant- $\ell$  solutions, finite interiors with separately constructed exteriors, pressure-supported or kinetic matter, nonstandard asymptotics, or a spectroscopic curve obtained from a complete photon-propagation model.

*Proof.* Part 1 combines Proposition 1 with Theorem 7. Part 2 is Proposition 2 and Theorem 2. Part 3 follows from Theorems 11, 3, 4, and 6. Part 4 combines Theorems 12, 14, and 5. Part 5 is Theorem 16, Corollary 7, Theorem 17, and Propositions 11 and 12. Part 6 follows from Proposition 13, Theorem 18, and Theorem 19, together with Corollary 12. Part 7 is Theorem 20, Proposition 14, and Theorem 21. Part 8 combines Theorems 8 and 9, Propositions 5 and 7, and the conditional physical results of Sect. VIII. Part 9 is Propositions 15 and 16. The final scope statement records the non-implications stated alongside those results.  $\square$

#### IV. FAILURE OF THE CONSTANT-FIELD EDGE ANSATZ

This section examines whether a nonzero spatially constant ZAMO speed can satisfy the constant- $\ell$  velocity-field equation on a finite region. We first establish a pointwise obstruction on any open radial interval and then apply it to the specific edge calibration and frozen-coefficient approximation of Ref. [6]. Finally, we determine the scope of the obstruction by allowing  $\ell$  to vary, showing that the failure of the constant-field ansatz is a consequence of the constant- $\ell$  restriction rather than of the general  $(\eta, H)$  family.

##### A. Pointwise obstruction on an open region

We begin by showing that a nonzero spatially constant field cannot satisfy the constant- $\ell$  VFE on any open radial interval.

**Proposition 1** (Constant-field obstruction at fixed  $\ell$ ). *Let  $I \subset (0, \infty)$  and  $J \subset \mathbb{R}$  be nonempty open intervals, and let  $\ell_0 \in \mathbb{R}$  be constant. There is no constant  $v_0 \neq 0$  for which the field*

$$v(r, z) \equiv v_0 \quad \text{on } I \times J \quad (15)$$

*satisfies the retained-order VFE*

$$\mathcal{T}_{\ell_0}[v] = 0 \quad \text{pointwise on } I \times J. \quad (16)$$

*The same conclusion holds for the untruncated constant- $\ell$  VFE  $\mathcal{E}_{\ell_0}[v] = 0$ .*

*Proof.* For the field in Eq. (15), all first and second spatial derivatives vanish, i.e.

$$v_r = v_z = v_{rr} = v_{zz} = 0. \quad (17)$$

Substitution into Eq. (12) leads to

$$\mathcal{T}_{\ell_0}[v_0] = \frac{2}{r^2} (\ell_0 r - v_0). \quad (18)$$

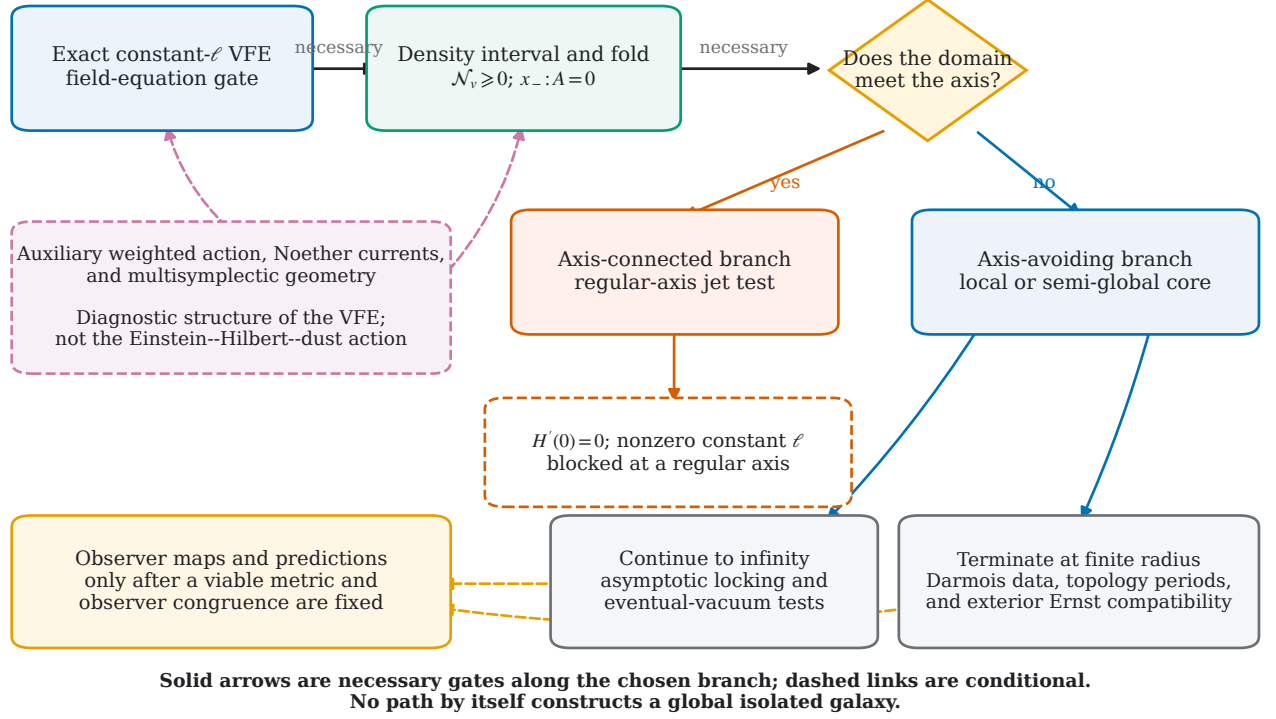


FIG. 1. Logical architecture of the constant- $\ell$  viability problem. Solid arrows are necessary gates along the chosen domain branch, while dashed links are conditional. A regular axis blocks nonzero constant  $\ell$ ; an axis-avoiding core still requires either a controlled asymptotic continuation or complete finite-boundary and exterior data before observable predictions can be assigned. Passing any gate is not a sufficiency theorem. The side panel records auxiliary variational and multisymplectic structure of the VFE and is explicitly distinct from the physical Einstein–Hilbert–Brown–dust action.

Thus, Eq. (16) would require

$$\ell_0 r = v_0 \quad \text{for every } r \in I. \quad (19)$$

Because  $I$  is an open interval, it contains two distinct radii  $r_1 \neq r_2$ . Evaluating Eq. (19) at these radii and subtracting gives  $\ell_0(r_1 - r_2) = 0$ , so  $\ell_0 = 0$ . Substitution back into Eq. (19) gives  $v_0 = 0$ , contradicting the hypothesis  $v_0 \neq 0$ . Hence, no nonzero spatially constant field can solve the retained-order VFE on  $I \times J$ . Finally, Eq. (14) and Eq. (17) imply  $\mathcal{E}_{\ell_0}[v_0] = \mathcal{T}_{\ell_0}[v_0]$ . Therefore, the same contradiction applies to the untruncated constant- $\ell$  VFE. This also shows that no term omitted in the low-velocity truncation can repair the constant-field ansatz.  $\square$

The proposition excludes a nonzero constant field on any open region. It is also useful to identify more precisely the set on which the corresponding residual may vanish.

**Corollary 1.** Fix  $v_0 \neq 0$ . The residual of a constant field is Eq. (18). Consequently, the following statements hold

1. if  $\ell_0 = 0$ , the residual is nonzero for every  $r > 0$ ;
2. if  $\ell_0 \neq 0$  but  $v_0/\ell_0 \leq 0$ , the residual is nonzero for every physical radius  $r > 0$ ;
3. if  $\ell_0 \neq 0$  and  $v_0/\ell_0 > 0$ , the residual vanishes only on the single cylinder

$$r = r_* := \frac{v_0}{\ell_0}. \quad (20)$$

In every case, the zero set has empty interior in the  $(r, z)$  domain.

*Proof.* Equation (18) vanishes if and only if  $\ell_0 r - v_0 = 0$ . The three cases follow immediately from  $r > 0$ . A set of the form  $\{r = r_*\}$  is codimension one and therefore contains no open subset of the  $(r, z)$  plane.  $\square$

### B. The proposed edge calibration and frozen coefficients

The parameter choice in Ref. [6] is

$$\ell_0 = \frac{av_c}{R_G}, \quad v_0 = av_c = \ell_0 R_G. \quad (21)$$

Inserting Eq. (21) into Eq. (18) gives the exact residual

$$\mathcal{T}_{\ell_0}[av_c] = \mathcal{E}_{\ell_0}[av_c] = \frac{2av_c}{r^2} \left( \frac{r}{R_G} - 1 \right). \quad (22)$$

For  $av_c \neq 0$ , Eq. (22) vanishes if and only if  $r = R_G$ . Thus, the value  $v = av_c$  is compatible with the constant- $\ell$  VFE at the single edge cylinder  $r = R_G$ , but not on any annulus containing more than one radius. To display precisely what is lost in the near-edge replacement, we write

$$r = R_G(1 + \delta), \quad \delta := \frac{r - R_G}{R_G}. \quad (23)$$

Then, Eq. (22) becomes

$$\mathcal{T}_{\ell_0}[av_c] = \frac{2av_c}{R_G^2} (\delta - 2\delta^2 + O(\delta^3)). \quad (24)$$

The near-edge Eq. (4.5) of Ref. [6] is obtained by freezing the dimensionless factors at  $r/R_G = 1$  and replacing  $1/R_G$  by  $1/r$  at that order. This operation sets the linear mismatch in Eq. (24) to zero. It therefore produces a frozen-coefficient equation for which  $v = av_c$  is a constant solution. Equation (24) shows that this is a pointwise or zeroth-order edge relation, not a solution of the parent retained-order equation on a finite radial interval. Nor is it consistent to promote the near-edge replacement  $R_G \simeq r$  to an identity throughout an interval while continuing to use Eq. (12). Such a promotion would change  $\ell_0 = av_c/R_G$  into a radially varying quantity  $av_c/r$ , whereas Eq. (12) was derived after imposing  $\ell'(\eta) = 0$ . Once  $\ell$  varies, the first term of the full VFE in Eq. (7) must be restored. A constant value may still serve as the leading term of a local expansion in a shrinking neighbourhood of  $r = R_G$ , for example, when  $\delta = O(\varepsilon)$  and a nonconstant correction is included at the next order. What Proposition 1 excludes is the stronger statement that an exactly constant field solves the constant- $\ell$  VFE throughout an open outer region. It does not, however, rule out a configuration that is nonconstant in the full  $(r, z)$  domain but has an exactly flat equatorial trace, since transverse curvature may enter the exact VFE through the vertical derivatives. Section IX addresses this remaining possibility without using the frozen-coefficient approximation.

### C. Role of the constant- $\ell$ assumption

The constant- $\ell$  assumption is essential to Proposition 1. To isolate the role of the constant- $\ell$  assumption, we return to the full VFE in Eq. (7) and impose a spatially constant field  $v \equiv v_0$  with  $0 < v_0 < 1$ , while allowing  $\ell = \ell(\eta)$  to vary. Since  $\eta = rv_0$ , Eq. (7) reduces to

$$v_0(1 - v_0^2)\ell'(\eta) + \frac{2\ell(\eta)}{r} - \frac{2v_0}{r^2} = 0. \quad (25)$$

Using  $r = \eta/v_0$  and dividing by  $v_0 \neq 0$  gives the linear ODE

$$(1 - v_0^2)\ell'(\eta) + \frac{2}{\eta}\ell(\eta) = \frac{2v_0^2}{\eta^2}. \quad (26)$$

After division by  $1 - v_0^2$ , an integrating factor is

$$\mu(\eta) = \exp \left( \int \frac{2}{(1 - v_0^2)\eta} d\eta \right) = \eta^{\frac{2}{1 - v_0^2}}, \quad \eta > 0. \quad (27)$$

Multiplying Eq. (26) by this factor yields

$$\frac{d}{d\eta} \left( \eta^{\frac{2}{1 - v_0^2}} \ell(\eta) \right) = \frac{2v_0^2}{1 - v_0^2} \eta^{\frac{2}{1 - v_0^2} - 2}. \quad (28)$$

Integration gives the complete local family

$$\ell(\eta) = \frac{2v_0^2}{1+v_0^2} \frac{1}{\eta} + C\eta^{-\frac{2}{1-v_0^2}}, \quad (29)$$

where  $C$  is an integration constant with the dimensions needed to make  $\ell$  an inverse length. Conversely, substitution into Eq. (26) verifies that every member of Eq. (29) supports the constant field at the level of the VFE. Since  $\ell = H_{,\eta}/H$ , each admissible  $\ell(\eta)$  determines, at least locally, a corresponding negative function  $H(\eta)$ . For any  $H_* > 0$  one may set  $H(\eta) = -H_* \exp(\int^\eta \ell(s) ds)$ . Equation (29) therefore shows that an exactly constant velocity field is not forbidden by the VFE in general, but is incompatible with a constant  $\ell$  on an open radial interval. The functions in Eq. (29) are generally singular as  $\eta \rightarrow 0$ , so their global physical viability remains a separate question.

## V. DENSITY-POSITIVITY BOUNDS AND RADIAL CUTOFFS

We next examine the restrictions imposed by the requirement of nonnegative dust density. The exact density formula allows its sign to be isolated in a simple algebraic factor, from which we derive a sharp admissible interval for the product  $r\ell$ . For constant- $\ell$  configurations with an exactly flat equatorial trace, this interval in turn yields finite radial cutoffs and permits a direct comparison with the parameter choice of Ref. [6]. Let us first recall that the exact density formula of Ref. [6] is

$$8\pi G_N \rho = \frac{1}{4g_{rr}} \left[ \frac{\eta^2}{r^2} (2 - \eta\ell)^2 - r^2 \ell^2 \right] \frac{\eta_r^2 + \eta_z^2}{\eta^2}. \quad (30)$$

Here,  $G_N$  denotes Newton's gravitational constant. Since  $g_{rr} = e^\mu$ , the metric coefficient  $g_{rr}$  is finite and strictly positive at every regular point. Consequently, it affects the magnitude of the density but not its sign. To isolate the remaining factors that control the sign of  $\rho$ , consider a regular point with  $r > 0$  and introduce  $\eta = rv$ , and  $x := r\ell$ . For  $0 < v < 1$ , one has  $\eta \neq 0$ . Substitution into the square bracket in Eq. (30) gives the dimensionless quadratic

$$\mathcal{N}_v(x) = 4v^2 - 4v^3x - (1 - v^4)x^2. \quad (31)$$

Thus, Eq. (30) can be written as

$$\rho = \frac{\eta_r^2 + \eta_z^2}{32\pi G_N g_{rr} \eta^2} \mathcal{N}_v(x). \quad (32)$$

The prefactor multiplying  $\mathcal{N}_v$  is nonnegative, and it is strictly positive precisely when  $\nabla_{rz}\eta \neq 0$ .

**Proposition 2** (Density-positivity domain). *Let  $p$  be a regular point with  $r > 0$ ,  $0 < v < 1$ , and finite  $\ell(\eta)$ . Set  $x = r\ell$  at  $p$ . Then, the algebraic density factor in Eq. (31) is nonnegative if and only if*

$$-\frac{2v}{1-v^2} \leq r\ell \leq \frac{2v}{1+v^2}. \quad (33)$$

*If, in addition,  $\nabla_{rz}\eta \neq 0$  at  $p$ , then Eq. (33) is equivalent to  $\rho \geq 0$  at that point. Under the same nondegeneracy hypothesis,  $\rho > 0$  at  $p$  if and only if both inequalities in Eq. (33) are strict. If either inequality is saturated, then  $\rho = 0$  at that point.*

*Proof.* To determine the sign of  $\mathcal{N}_v(x)$  in Eq. (31), we factor it as

$$\mathcal{N}_v(x) = [2v - (1 + v^2)x] [2v + (1 - v^2)x]. \quad (34)$$

Because  $0 < v < 1$ , both terms  $1 - v^2$  and  $1 + v^2$  are strictly positive. Moreover, the two factors in Eq. (34) vanish at

$$x_- := -\frac{2v}{1-v^2}, \quad x_+ := \frac{2v}{1+v^2}, \quad (35)$$

respectively, and  $x_- < 0 < x_+$ . Equivalently, Eq. (31) is a quadratic polynomial in  $x$  with leading coefficient  $-(1 - v^4) < 0$  for  $0 < v < 1$ . Since  $\mathcal{N}_v(x)$  is therefore strictly concave and has two distinct real roots  $x_- \leq x_+$ , it is nonnegative precisely for  $x \in [x_-, x_+]$  and negative outside this interval. Hence,  $\mathcal{N}_v(x) \geq 0$  if and only if  $x_- \leq x \leq x_+$ , which, after substituting  $x = r\ell$ , gives Eq. (33). Moreover,  $\mathcal{N}_v(x) > 0$  for  $x_- < x < x_+$ , while  $\mathcal{N}_v(x) = 0$  at the endpoints. At a regular noncritical point, where  $\nabla_{rz}\eta \neq 0$ , the term  $(\eta_r^2 + \eta_z^2)/(32\pi G_N g_{rr} \eta^2)$  is strictly positive. Indeed,  $G_N > 0$ ,  $g_{rr} > 0$ , and  $\eta^2 > 0$ , while the numerator is positive by the noncriticality assumption. It follows that  $\rho$  and  $\mathcal{N}_v$  have the same sign, which proves the proposition.  $\square$

At a critical point of  $\eta$ , where  $\nabla_{rz}\eta = 0$ , Eq. (32) gives  $\rho = 0$  independently of the sign of  $\mathcal{N}_v(r\ell)$ . Therefore, Eq. (33) is not a necessary condition for  $\rho \geq 0$  at such points. It is, however, necessary and sufficient for the nonnegativity of the algebraic density factor, and, at every regular noncritical point, it is equivalent to  $\rho \geq 0$ . For strictly positive density no qualification is needed, since  $\rho > 0$  itself implies  $\nabla_{rz}\eta \neq 0$ .

Under continuity and nonnegative density, a violation of the bound at a critical point necessarily extends to a local vacuum region.

**Corollary 2** (Failure of the bound forces a local vacuum plateau). *Let  $U$  be an open set of regular points on which  $0 < v < 1$  and  $\rho \geq 0$ , and assume that the functions entering Eq. (32) are continuous on  $U$ . If Eq. (33) fails at a point  $p \in U$ , then there exists a connected open neighbourhood  $V \subset U$  of  $p$  on which  $\nabla_{rz}\eta = 0$ , and  $\rho = 0$ . In particular, if  $\eta$  is not locally constant on any nonempty open subset of  $U$ , then Eq. (33) must hold throughout  $U$ , including at critical points.*

*Proof.* By Proposition 2, failure of Eq. (33) at  $p$  implies  $\mathcal{N}_v(r\ell) < 0$  there. By continuity, there exists a connected open neighbourhood  $V \subset U$  of  $p$  on which  $\mathcal{N}_v(r\ell) < 0$ . Since  $G_N > 0$ ,  $g_{rr} > 0$ , and  $\eta^2 > 0$  on  $V$ , Eq. (32) would give  $\rho < 0$  at any point where  $\nabla_{rz}\eta \neq 0$ , contradicting the assumption  $\rho \geq 0$ . Hence,  $\nabla_{rz}\eta = 0$  throughout  $V$ , and Eq. (32) then gives  $\rho = 0$  on  $V$ . Because  $V$  is connected, the vanishing of  $\nabla_{rz}\eta$  implies that  $\eta$  is constant on  $V$ . Therefore, if  $\eta$  is not locally constant on any nonempty open subset of  $U$ , the bound Eq. (33) cannot fail at any point of  $U$ .  $\square$

Having clarified the role of critical points, it is useful to rewrite the positivity bound in terms of the combination  $\eta\ell$ . Since  $\eta = rv$  and  $v > 0$ , one has  $\eta\ell = v(r\ell)$ . Multiplying Eq. (33) by  $v$  therefore gives

$$-\frac{2v^2}{1-v^2} \leq \eta\ell \leq \frac{2v^2}{1+v^2}. \quad (36)$$

This form makes the low-velocity behaviour of the positivity bound transparent. As  $v \rightarrow 0^+$ , the allowed interval for  $\eta\ell$  has width of order  $v^2$ , with endpoints  $-2v^2 - 2v^4 + O(v^6) \leq \eta\ell \leq 2v^2 - 2v^4 + O(v^6)$ , or equivalently  $-2v - 2v^3 + O(v^5) \leq r\ell \leq 2v - 2v^3 + O(v^5)$ . Thus, the asymmetry between the lower and upper bounds for  $r\ell$  first appears at cubic order in  $v$ . The preceding expressions use the positive-speed convention  $0 < v < 1$  adopted throughout this analysis. If  $v$  is instead regarded as an oriented quantity with  $-1 < v < 0$ , the same two algebraic roots are obtained, but their order is reversed. In that convention, the positivity domain can be written without reference to their ordering as

$$\min\left\{-\frac{2v}{1-v^2}, \frac{2v}{1+v^2}\right\} \leq r\ell \leq \max\left\{-\frac{2v}{1-v^2}, \frac{2v}{1+v^2}\right\}. \quad (37)$$

In what follows, we retain the positive-speed convention  $0 < v < 1$ . We now specialize the density-positivity bound to an exactly flat equatorial trace with constant  $\ell$ . In this setting, the bound translates directly into a finite radial range whenever  $\ell \neq 0$ .

**Corollary 3** (Finite positivity radius for an exactly flat equatorial trace). *Let  $I \subset (0, \infty)$  be an interval on which  $v(r, 0) \equiv v_0$  with  $0 < v_0 < 1$ , and let  $\ell \equiv \ell_0$  be constant. No assumption on  $v_z$  or  $v_{zz}$  is required. Nonnegative density along the trace is then equivalent to the following:*

(i) if  $\ell_0 > 0$ ,

$$0 < r \leq R_\rho^{(+)} := \frac{2v_0}{\ell_0(1+v_0^2)}; \quad (38)$$

(ii) If  $\ell_0 < 0$ ,

$$0 < r \leq R_\rho^{(-)} := \frac{2v_0}{|\ell_0|(1-v_0^2)}; \quad (39)$$

(iii) If  $\ell_0 = 0$ , the algebraic density factor is  $\mathcal{N}_{v_0}(0) = 4v_0^2 > 0$ , and remains positive throughout the trace.

In cases (i) and (ii), the density vanishes at the corresponding cutoff radius and is negative at every regular trace point beyond it.

This is a pointwise sign result for an exactly flat equatorial trace. It does not by itself establish that the trace extends to a global solution or that the zero-density radius defines a matchable material boundary.

*Proof.* Since  $\eta(r, 0) = rv_0$ , differentiation along the trace gives  $\eta_r(r, 0) = v_0$ . Therefore,  $\nabla_{rz}\eta \neq 0$  there, irrespective of the transverse derivative  $\eta_z$ . Proposition 2 applies without the critical-point exception. For  $\ell_0 > 0$ , the condition  $r\ell_0 > 0$  ensures that the lower bound in Eq. (33) is always satisfied, whereas the upper bound reduces to Eq. (38). For  $\ell_0 < 0$ , one has  $r\ell_0 < 0$ , so the upper bound in Eq. (33) is always satisfied, whereas the lower bound becomes

$$r|\ell_0| \leq \frac{2v_0}{1 - v_0^2}, \quad (40)$$

which is equivalent to Eq. (39). For  $\ell_0 = 0$ , direct substitution gives  $\mathcal{N}_{v_0}(0) = 4v_0^2 > 0$ . Finally, the endpoint and exterior sign statements follow from the strict concavity and root classification in Proposition 2.  $\square$

We next specialize the finite-radius bound to the parameter choice adopted in Ref. [6], which allows the exact cutoff to be compared directly with the retained-order prediction.

**Corollary 4.** *Let us assume  $a > 0$ ,  $v_c > 0$ ,  $0 < av_c < 1$ . Furthermore, we set  $\ell_0 = av_c/R_G$ , and suppose that, on a radial interval  $I \subset (0, \infty)$ , the equatorial trace satisfies  $v(r, 0) \equiv v_0 = av_c$ . Then, the exact algebraic density factor is nonnegative at precisely those radii  $r \in I$  for which*

$$0 < r \leq R_\rho^{\text{exact}} := \frac{2R_G}{1 + a^2v_c^2}. \quad (41)$$

*If  $R_\rho^{\text{exact}} \in I$ , the density vanishes at the cutoff radius, while at every regular point of the trace with  $r > R_\rho^{\text{exact}}$  it is negative. Moreover,*

$$R_\rho^{\text{exact}} = 2R_G [1 - a^2v_c^2 + O(a^4v_c^4)]. \quad (42)$$

*Hence,  $2R_G$  is the leading low-velocity approximation to the exact density-positivity cutoff.*

*Proof.* Since  $\ell_0 > 0$ , Eq. (38) applies. Substituting  $v_0 = av_c$  and  $\ell_0 = av_c/R_G$  into the expression for  $R_\rho^{(+)}$  gives

$$R_\rho^{(+)} = \frac{2R_G}{1 + a^2v_c^2}, \quad (43)$$

which is Eq. (41). The sign statements follow from Corollary 3. Finally, the geometric-series expansion

$$\frac{1}{1 + a^2v_c^2} = 1 - a^2v_c^2 + O(a^4v_c^4) \quad (44)$$

gives Eq. (42).  $\square$

The positivity bound should not be interpreted as implying that the algebraic density factor necessarily decreases when  $\ell$  departs from zero. At fixed  $v$ , the case  $\ell = 0$  corresponds to  $x = r\ell = 0$ , for which  $\mathcal{N}_v(0) = 4v^2$ . From Eq. (31),

$$\mathcal{N}_v(x) - \mathcal{N}_v(0) = -4v^3x - (1 - v^4)x^2. \quad (45)$$

Since  $\mathcal{N}_v$  is a concave quadratic in  $x$ , its maximum occurs at

$$x_* = -\frac{2v^3}{1 - v^4}, \quad \mathcal{N}_v(x_*) = \frac{4v^2}{1 - v^4} > \mathcal{N}_v(0). \quad (46)$$

More precisely, Eq. (45) shows that  $\mathcal{N}_v(x)$  exceeds its value at  $\ell = 0$  precisely for

$$-\frac{4v^3}{1 - v^4} < x < 0. \quad (47)$$

This interval lies strictly inside the positivity domain of Eq. (33). Hence, a sufficiently small negative value of  $r\ell$  increases the algebraic density factor relative to its value at  $\ell = 0$ . This comparison concerns only the algebraic factor  $\mathcal{N}_v$ . The full density also depends on  $g_{rr}$ , which generally changes when  $H$  is varied. Proposition 2 should therefore be understood as a sign criterion, rather than as a monotonicity statement for the density as a function of  $\ell$ .

## VI. FOLD GEOMETRY, AUTOMATIC REGULARITY, AND SEMI-GLOBAL REALIZATION

The auxiliary potential does more than shorten the velocity-field equation. It identifies the exact discriminant at which the physical inverse problem changes sheet, imports global tools from harmonic analysis, and makes the logical gap between analytic local construction and smooth classical solutions disappear. Throughout this section,  $D$  denotes a connected domain contained in  $\{r > 0, \eta > 0\}$  and

$$A = \partial_\eta \Phi_{\ell_0}(r, \eta) = 2 + \ell_0 r \left( \frac{1}{v} - v \right).$$

### A. The density boundary as a Whitney fold

Let  $x = r\ell_0$  and let  $\mathcal{N}_v(x)$  be the algebraic density polynomial in Eq. (31). Direct factorization gives

$$\mathcal{N}_v(x) = vA[2v - (1 + v^2)x]. \quad (48)$$

The two factors in Eq. (48) have different geometric meanings: the second is an ordinary zero of the density factor, whereas the first is the Jacobian of the point linearisation.

**Theorem 2** (Density fold and physical inverse sheet). *Fix  $0 < v_0 < 1$  and a point  $(r_0, \eta_0)$  with  $\eta_0 = r_0 v_0$ . At the lower density root*

$$r_0 \ell_0 = -\frac{2v_0}{1 - v_0^2},$$

*the map*

$$(r, \eta) \mapsto (r, \Phi_{\ell_0}(r, \eta))$$

*has a nondegenerate Whitney fold. Strictly positive density implies  $A > 0$  and therefore selects the orientation-preserving inverse sheet. A generic transverse auxiliary datum has a square-root inverse at the fold; a smooth crossing requires the pulled-back discriminant to admit a smooth square root compatible with one continuous choice of sheet.*

*Proof.* Equation (48) shows that the lower root is exactly  $A = 0$ . At that root  $\ell_0 < 0$ , and

$$\partial_{\eta\eta} \Phi_{\ell_0} = -\ell_0 \left( 1 + \frac{1}{v_0^2} \right) > 0.$$

Thus the derivative in the kernel direction vanishes simply while the second derivative in that direction does not. The first component of the map is  $r$  itself, so the parameter-dependent Morse lemma gives local coordinates  $(R, S)$  in source and  $(R, Y)$  in target for which the map is  $(R, S) \mapsto (R, S^2)$ . This is the Whitney-fold normal form. Between the two simple density roots, both factors on the right-hand side of Eq. (48) are positive, hence  $A > 0$ . Finally, inversion of  $Y = S^2$  is generically  $S = \pm\sqrt{Y}$ . A smooth inverse exists only when the pulled-back discriminant is a square in the local smooth ring, with a coherent choice of sign. Along a one-dimensional transverse slice, finite even vanishing order with positive leading coefficient is sufficient for such a smooth signed square root. In the analytic category the corresponding condition is local analytic factorisation as a square; contact parity alone is not asserted to be sufficient in several variables.  $\square$

*Remark 1.* The theorem is local. It does not say that every even-contact datum produces a globally admissible metric, and it does not identify the fold with an optical caustic. In particular, metric regularity must be checked after the inverse is reconstructed; it cannot be inferred from the square-root model alone.

### B. Automatic analyticity and plateau propagation

**Theorem 3** (Automatic analyticity and connected-sheet rigidity). *Let  $\eta \in C^2(D)$  solve the exact constant- $\ell$  VFE and suppose  $A \neq 0$  on  $D$ . Then  $\eta$ ,  $v$ , and the locally reconstructed metric are real analytic. If  $\eta$  is constant on a nonempty open subset of  $D$ , then it is constant throughout  $D$ .*

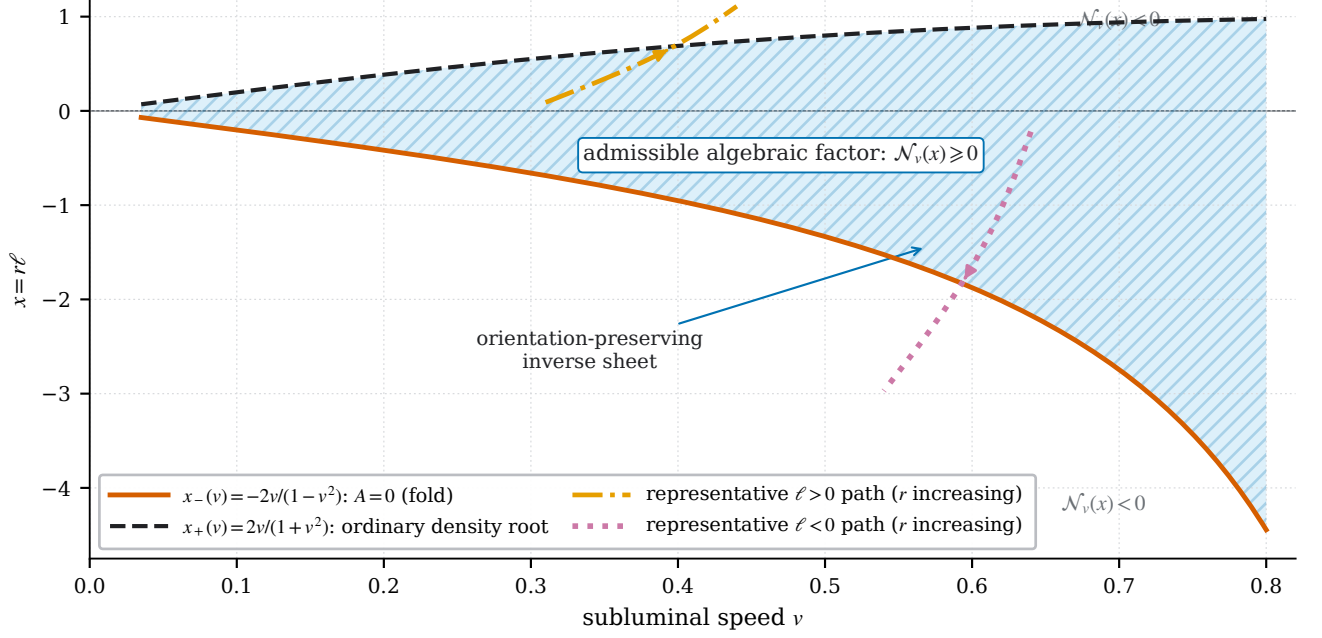


FIG. 2. Exact density-admissibility diagram in the variables  $0 < v \leq 0.8$  and  $x = r\ell$ . At a regular noncritical point, the hatched strip  $x_-(v) \leq x \leq x_+(v)$  is equivalent to  $\rho \geq 0$ . The lower boundary is simultaneously  $A = 0$ , the Whitney fold of the point-linearising map, whereas the upper boundary is an ordinary zero of the algebraic density factor. The dash-dotted and dotted curves illustrate only the kinematics of  $x = r\ell$  for fixed-sign  $\ell$  as  $r$  increases; they are not solutions of the VFE. Thus the diagram selects a local inverse sheet but does not establish a global spacetime.

*Proof.* Set  $\mathcal{F} = \Phi_{\ell_0}(r, \eta)$ . The exact chain-rule identity behind Eq. (138) gives  $\mathcal{L}\mathcal{F} = 0$  in the classical, hence distributional, sense. The operator  $\mathcal{L}$  is uniformly elliptic on every relatively compact subdomain of  $D$  and has analytic coefficients. Analytic elliptic regularity therefore makes  $\mathcal{F}$  real analytic [34]. Since  $A$  never vanishes, the analytic implicit-function theorem makes  $\eta$  and  $v$  analytic. The algebraic Killing block and the closed reconstruction quadrature are then analytic as well.

Suppose  $\eta = \eta_0$  on a nonempty open set. The comparison function

$$\mathcal{F}_0(r) = \Phi_{\ell_0}(r, \eta_0) = c_0 + c_2 r^2$$

is  $\mathcal{L}$ -harmonic. Unique continuation applied to  $\mathcal{F} - \mathcal{F}_0$  gives  $\mathcal{F} = \mathcal{F}_0$  on  $D$ . Local inverse uniqueness makes the set  $\{\eta = \eta_0\}$  both open and closed in  $D$ , and connectedness completes the proof.  $\square$

Strict positive density forces  $A > 0$  by Eq. (48). Thus no analyticity assumption is needed anywhere inside a strictly positive-density regular sheet. Plateau propagation can stop only at a genuine boundary, the axis,  $\eta = 0$ , the fold, a change of equations, or loss of classical regularity.

### C. Exact elliptic Cauchy instability

The preceding theorem is a regularity statement, not a stability theorem. The following sequence quantifies the distinction for the precise operator used here.

**Theorem 4** (Bessel–Hadamard instability). *For  $k > 0$ , define*

$$\mathcal{F}_k(r, z) = e^{-\sqrt{k}r} J_1(kr) \cosh(kz).$$

*Then  $\mathcal{L}\mathcal{F}_k = 0$ . On every compact annulus  $I = [r_1, r_2] \Subset (0, \infty)$ , the Cauchy data at  $z = 0$  converge to zero in every fixed  $C^m$  and Sobolev norm, while at every fixed  $z_0 > 0$  their  $L^2(I)$  norm diverges. Consequently, continuation from smooth Cauchy data to any collar of positive depth is discontinuous.*

*Proof.* The Bessel equation for  $J_1$  gives

$$\left(\partial_r^2 - \frac{1}{r}\partial_r\right)[rJ_1(kr)] = -k^2 r J_1(kr),$$

which cancels  $\partial_z^2 \cosh(kz) = k^2 \cosh(kz)$ . At  $z = 0$  the normal datum vanishes, while every fixed radial derivative of the Dirichlet datum is bounded by a polynomial in  $k$  times  $e^{-\sqrt{k}}$  on  $I$ ; hence all fixed smooth and Sobolev norms tend to zero. Standard large-argument Bessel asymptotics [35] give an  $L^2(I)$  norm bounded below by a fixed negative power of  $k$  along all sufficiently large  $k$ . Multiplication by  $e^{-\sqrt{k}} \cosh(kz_0)$  therefore makes that norm diverge exponentially. If a continuous Cauchy solution map to the collar existed, this null sequence of data would have a bounded image, a contradiction.  $\square$

Analytic data with Fourier decay  $Me^{-a|\xi|}$  can be continued only to depth  $d < a$  with a conditional model rate  $O(M^{d/a} \varepsilon^{1-d/a})$ . This observation describes the mechanism; regularisation and numerical conditioning are reserved for Part II.

#### D. Five-dimensional lift and a global rigidity theorem

**Proposition 3** ( $SO(5)$  harmonic lift). *Writing  $\mathcal{F} = r^2 U$  gives*

$$\mathcal{L}\mathcal{F} = r^2 \left( U_{rr} + \frac{3}{r} U_r + U_{zz} \right). \quad (49)$$

*Thus  $U(|y|, z)$  is harmonic on  $\mathbb{R}_y^4 \times \mathbb{R}_z$  wherever  $\mathcal{F}$  is  $\mathcal{L}$ -harmonic. With  $R = (r^2 + z^2)^{1/2}$ , Kelvin inversion is*

$$\mathcal{F}^K(r, z) = R\mathcal{F}\left(\frac{r}{R^2}, \frac{z}{R^2}\right).$$

*The regular and decaying  $SO(4)$ -invariant multipoles are*

$$\mathcal{F}_n^{\text{reg}} = r^2 R^n C_n^{3/2}(z/R), \quad \mathcal{F}_n^{\text{dec}} = r^2 R^{-n-3} C_n^{3/2}(z/R).$$

*Proof.* Equation (49) follows by differentiating  $r^2 U$ . The remaining statements are the five-dimensional Kelvin transform and the zonal harmonic expansion, multiplied by  $r^2$ .  $\square$

**Theorem 5** (Global regular-axis rigidity for  $\ell_0 = 0$ ). *Suppose the  $\ell_0 = 0$  branch occupies the complete orbit half-plane, has a regular axis and no finite boundary, puncture, additional end, horizon, or singular axis segment. Assume, locally uniformly in  $z$ ,  $|\eta(r, z)| \leq C_K r^2$  near the axis and, for sufficiently large  $R = (r^2 + z^2)^{1/2}$ ,*

$$|\eta(r, z)| \leq A \frac{r}{R}.$$

*Then  $\eta \equiv 0$ . After stationary and meridional normalisation, the spacetime is Minkowski and the dust density vanishes.*

*Proof.* For  $\ell_0 = 0$ , set  $U = 2\eta/r^2$ . The local axis estimate makes  $U$  bounded near the lifted axis. That axis has removable capacity for bounded harmonic functions in five dimensions, so  $U$  extends harmonically across it. On a lifted sphere of radius  $S$ , the asymptotic estimate and the  $\sin^3 \theta$  angular measure give an  $O(S^{-2})$  spherical  $L^1$  mean. The Poisson formula bounds  $|U(X)|$  at each fixed interior point by  $O(S^{-2})$ . Letting  $S \rightarrow \infty$  gives  $U(X) = 0$ . Hence  $\eta = 0$ ; the remaining algebraic relations and quadratures give the stated normalised metric and density.  $\square$

The domain hypotheses are essential. The theorem does not exclude a bounded dust body stopped at a finite surface and matched to vacuum.

#### E. Runge realization and formal integrability

**Theorem 6** (Semi-global realization on a protected physical core). *Let  $K \Subset K' \Subset D_0 \Subset D_1 \Subset \{r > 0\}$ , and let  $\mathcal{F}_0$  be  $\mathcal{L}$ -harmonic near  $K'$ . Assume the elliptic Runge no-compact-hole condition for the pair  $(K', D_1)$ . Then, for every*

integer  $m \geq 0$  there are  $\mathcal{L}$ -harmonic functions  $\mathcal{F}_j$  on  $D_1$  converging to  $\mathcal{F}_0$  in  $C^m(K)$ . If the inverse sheet on  $K$  has quantitative margins

$$A \geq a_0 > 0, \quad 0 < v \leq 1 - \delta, \quad \rho \geq \rho_0 > 0,$$

then, for  $m \geq 2$  and a fixed local normalization of the  $\mu$  quadrature on each connected core, sufficiently accurate approximants invert on the same sheet and retain these inequalities, with smaller positive margins, on  $K$ .

*Proof.* The first assertion is the Lax–Malgrange Runge theorem for elliptic operators with unique continuation [36], applied on the larger compact set  $K'$ . Elliptic interior estimates then promote uniform approximation near  $K'$  to  $C^m(K)$ . For  $m \geq 2$ , the implicit inverse, its required first derivatives, the normalized local quadrature, and the density depend continuously on the relevant finite jet on  $K$  as long as  $A \geq a_0$ . Openness of the three strict inequalities proves the second assertion.  $\square$

No physical conclusion is asserted on  $D_1 \setminus K$ . In particular, Runge approximation neither preserves an exactly flat trace nor supplies an asymptotically flat exterior or a matching surface.

**Theorem 7** (Regular-sheet involutivity and fold rank jump). *Regard the exact VFE as a hypersurface in  $J^2(D, \mathbb{R}_v)$ . On  $A \neq 0$ , its order-two symbol is*

$$\sigma_2(\xi) = A(\xi_r^2 + \xi_z^2),$$

its Cartan characters are  $(2, 0)$ , and the prolonged point transformation  $\mathcal{F} = \Phi_{\ell_0}(r, rv)$  identifies it with the prolonged linear equation  $\mathcal{L}\mathcal{F} = 0$ . At  $A = 0$ , the symbol rank falls from one to zero and  $\dim g_2$  jumps from two to three. Appending  $v_r = v_z = 0$  makes every positive constant-field branch inconsistent; the first prolongation leaves only  $\ell_0 = v = 0$ .

*Proof.* The principal part of the VFE is  $A(v_{rr} + v_{zz})$ . Its kernel in  $S^2T^*D$  is the two-dimensional space of trace-free quadratic forms. Its successive prolongations are the homogeneous harmonic polynomials in two variables and remain two-dimensional, giving characters  $(2, 0)$ . On  $A \neq 0$  the point map is a jet diffeomorphism, so formal integrability and involutivity follow constructively from the linear equation. When  $A = 0$  no second derivative remains in the symbol, so  $g_2 = S^2T^*D$  has dimension three. Finally,  $v_r = v_z = 0$  kills their first prolongations, while the VFE reduces to  $\ell_0 r - v = 0$ ; differentiation in  $r$  gives  $\ell_0 = 0$  and then  $v = 0$ .  $\square$

This last calculation is also recorded as a branch-sensitive ‘rfsimp()’/Thomas certificate. Its differential-algebra output is evidence for the encoded local system; it does not establish positivity, global existence, or boundary regularity.

## VII. VARIATIONAL AND MULTISYMPLECTIC GEOMETRY OF THE AUXILIARY FIELD

The linearizing potential does more than simplify the velocity-field equation. It places the regular constant- $\ell_0$  sheets inside an exact covariant Hamiltonian system, identifies the lower density root as a degeneracy of the Legendre map, and turns the Green identities into a multisymplectic conservation law. This structure is useful only if its logical level is kept clear. It belongs to the auxiliary meridional equation; it is not, by itself, the physical phase space or energy of the complete Einstein–dust system. We use the standard variational-bicomplex, multisymplectic, and covariant-polymomentum conventions of Refs. [37–39].

Throughout this section,  $D$  is an oriented, connected, bounded domain with piecewise  $C^2$  boundary whose closure is contained in  $\{r > 0\}$ , and  $d^2x := dr \wedge dz$ . Repeated meridional indices  $i, j \in \{r, z\}$  are summed with the Euclidean metric. The outward unit normal and arclength measure on  $\partial D$  are denoted by  $n$  and  $ds$ , respectively.

### A. Weighted action and its singular pullback

Recall the auxiliary operator

$$\mathcal{L}\mathcal{F} := \mathcal{F}_{rr} - \frac{1}{r}\mathcal{F}_r + \mathcal{F}_{zz} = r \partial_i \left( \frac{\partial_i \mathcal{F}}{r} \right). \quad (50)$$

Its natural action is

$$\mathcal{A}_D[\mathcal{F}] := \frac{1}{2} \int_D \frac{|\nabla \mathcal{F}|^2}{r} dr dz. \quad (51)$$

**Proposition 4** (Weighted variational principle). *Let  $\mathcal{F} \in H^1(D)$ , and assume sufficient regularity whenever a classical trace or derivative is used. For every smooth variation  $f$ ,*

$$\delta \mathcal{A}_D[\mathcal{F}](f) = - \int_D \frac{\mathcal{L}\mathcal{F}}{r} f \, dr \, dz + \int_{\partial D} \frac{\partial_n \mathcal{F}}{r} f \, ds. \quad (52)$$

*Consequently, the critical points for fixed Dirichlet data are precisely the weak solutions of  $\mathcal{L}\mathcal{F} = 0$ . The action is strictly convex on each affine Dirichlet class and is strictly convex modulo constants when no Dirichlet trace is fixed.*

*Proof.* Equation (52) is the weighted divergence theorem. The second variation is  $\int_D |\nabla f|^2 / r \, dr \, dz$ , which proves the convexity statements.  $\square$

We next pull back the action through the constant- $\ell_0$  linearizing map. On  $\eta > 0$ , set

$$\Phi_{\ell_0}(r, \eta) := 2\eta + \ell_0 r^2 \log\left(\frac{\eta}{\eta_*}\right) - \frac{\ell_0}{2} \eta^2, \quad \eta_* > 0, \quad (53)$$

$$A(r, \eta) := \partial_\eta \Phi_{\ell_0} = 2 + \frac{\ell_0 r^2}{\eta} - \ell_0 \eta, \quad B(r, \eta) := \partial_r \Phi_{\ell_0}|_\eta = 2\ell_0 r \log\left(\frac{\eta}{\eta_*}\right). \quad (54)$$

Thus

$$\mathcal{F}_r = B + A\eta_r, \quad \mathcal{F}_z = A\eta_z. \quad (55)$$

**Theorem 8** (Variational equivalence and fold degeneracy). *The pullback of Eq. (51) is the first-order action with density*

$$L_\eta = \frac{(B + A\eta_r)^2 + A^2 \eta_z^2}{2r}. \quad (56)$$

*Its Euler–Lagrange derivative and fibre Hessian are*

$$\frac{\delta L_\eta}{\delta \eta} = -\frac{A}{r} \mathcal{L}\mathcal{F}, \quad (57)$$

$$\frac{\partial^2 L_\eta}{\partial \eta_i \partial \eta_j} = \frac{A^2}{r} \delta_{ij}. \quad (58)$$

*Since  $\mathcal{L}\Phi_{\ell_0}(r, v) = r\mathcal{E}_{\ell_0}[v]$ , the pulled-back Euler–Lagrange equation is exactly  $A\mathcal{E}_{\ell_0}[v] = 0$ . It is equivalent to the VFE on every sheet on which  $A \neq 0$ , but not on the fold  $A = 0$ .*

*More precisely, for  $\eta = rv$  with  $0 < v < 1$ ,*

$$A = 0 \iff r\ell_0 = -\frac{2v}{1-v^2}. \quad (59)$$

*If  $\ell_0 \neq 0$ , then at such a point*

$$\partial_{\eta\eta} \Phi_{\ell_0} = -\ell_0 \left(1 + \frac{1}{v^2}\right) \neq 0. \quad (60)$$

*Hence the map  $(r, \eta) \mapsto (r, \mathcal{F})$  has a nondegenerate fold. The Legendre rank of Eq. (56) is two for  $A \neq 0$  and zero for  $A = 0$ .*

*Proof.* The first identity is the Euler–Lagrange chain rule for a base-preserving point transformation:  $\delta \mathcal{F} = A \delta \eta$ . Direct differentiation of Eq. (56) gives Eq. (58). Equation (59) follows by substituting  $\eta = rv$  into  $A = 0$ , and Eq. (60) follows by differentiating  $A$  at fixed  $r$ .  $\square$

**Remark 2** (What the singular action does not prove). The factor  $A$  in Eq. (57) creates a spurious variational stratum: a smooth section whose image lies in  $A = 0$  is stationary for the pulled-back action even when  $\mathcal{L}\mathcal{F} \neq 0$ . Therefore, one must impose the inverse-sheet condition or return to the original VFE at the fold. The rank loss is not a gauge symmetry and does not, by itself, exclude a specially compatible smooth crossing. Since the rank is not constant across the fold, a conventional unstratified Dirac–Bergmann analysis is inapplicable there.

### B. De Donder–Weyl Hamiltonian and Poincaré–Cartan form

Let  $Y = D \times \mathbb{R}_{\mathcal{F}} \rightarrow D$  be the auxiliary configuration bundle. Its first jet has coordinates  $(r, z, \mathcal{F}, \mathcal{F}_r, \mathcal{F}_z)$ , while the restricted multiphase bundle has coordinates  $(r, z, \mathcal{F}, p^r, p^z)$ . With  $d^1 x_r = dz$  and  $d^1 x_z = -dr$ , define

$$p^r := \frac{\mathcal{F}_r}{r}, \quad p^z := \frac{\mathcal{F}_z}{r}, \quad H_{\mathcal{F}} := \frac{r}{2}[(p^r)^2 + (p^z)^2]. \quad (61)$$

Our sign convention for the Hamiltonian Poincaré–Cartan and multisymplectic forms is

$$\Theta_{\mathcal{F}} := p^r d\mathcal{F} \wedge dz - p^z d\mathcal{F} \wedge dr - H_{\mathcal{F}} dr \wedge dz, \quad (62)$$

$$\Omega_{\mathcal{F}} := -d\Theta_{\mathcal{F}}. \quad (63)$$

**Theorem 9** (Exact covariant canonical transformation). *The De Donder–Weyl equations determined by Eqs. (61)–(63) are*

$$\mathcal{F}_r = rp^r, \quad \mathcal{F}_z = rp^z, \quad \partial_r p^r + \partial_z p^z = 0, \quad (64)$$

and are equivalent to  $\mathcal{L}\mathcal{F} = 0$ .

On  $A \neq 0$ , the  $\eta$  polymomenta and Hamiltonian are

$$\pi^r = \frac{A}{r}(B + A\eta_r) = Ap^r, \quad \pi^z = \frac{A^2}{r}\eta_z = Ap^z, \quad (65)$$

$$H_{\eta} = \frac{r}{2A^2}[(\pi^r)^2 + (\pi^z)^2] - \frac{B}{A}\pi^r. \quad (66)$$

The map

$$\Psi : (r, z, \eta, \pi^r, \pi^z) \longmapsto \left( r, z, \Phi_{\ell_0}(r, \eta), \frac{\pi^r}{A}, \frac{\pi^z}{A} \right) \quad (67)$$

satisfies

$$\Psi^* \Theta_{\mathcal{F}} = \Theta_{\eta}, \quad \Psi^* \Omega_{\mathcal{F}} = \Omega_{\eta}, \quad (68)$$

where

$$\Theta_{\eta} := \pi^r d\eta \wedge dz - \pi^z d\eta \wedge dr - H_{\eta} dr \wedge dz. \quad (69)$$

Thus the point linearization is an exact local multisymplectic transformation on every invertible sheet.

*Proof.* The first two equations in Eq. (64) are the polymomentum definitions, and the third is the weighted field equation. Solving Eq. (65) for the first derivatives gives

$$\eta_r = \frac{r\pi^r}{A^2} - \frac{B}{A}, \quad \eta_z = \frac{r\pi^z}{A^2}, \quad (70)$$

from which Eq. (66) follows by Legendre transform. Finally,  $d\mathcal{F} = A d\eta + B dr$ . Substitution into Eq. (62) gives Eq. (69) coefficient by coefficient. Exterior differentiation proves the second identity in Eq. (68).  $\square$

At  $A = 0$ , the map in Eq. (67) ceases to exist, the velocity Hessian vanishes, and the fibre directions of the pulled-back Cartan form acquire a kernel. The appropriate object is therefore a stratified premultisymplectic system, not a single smooth hyperregular Hamiltonian theory crossing the density fold.

### C. Multisymplectic conservation and Noether currents

Let  $f$  and  $g$  be two solutions of the linearized equation, which here is again  $\mathcal{L}h = 0$ . Define the one-form

$$\omega(f, g) := \frac{fg_r - gfr}{r} dz - \frac{fg_z - gfr}{r} dr. \quad (71)$$

Direct differentiation gives the off-shell identity

$$d\omega(f, g) = \frac{f\mathcal{L}g - g\mathcal{L}f}{r} dr \wedge dz. \quad (72)$$

Consequently,

$$d\omega(f, g) = 0 \quad (73)$$

for every pair of Jacobi fields. Equation (73) is the multisymplectic form formula for the present theory. It is also the local Green identity and hence makes precise why the same conserved quantity controls solution transformations, boundary data, and numerical residuals.

Several exact Noether currents follow.

**Proposition 5** (Noether and stress currents). *Let  $\mathcal{L}\mathcal{F} = 0$  on  $D$ .*

*First, for any fixed  $h$  satisfying  $\mathcal{L}h = 0$ , the vertical transformation  $\delta_h\mathcal{F} = h$  is a divergence variational symmetry. Its Noether current is*

$$\mathbf{J}_h := \frac{h\mathcal{F}_r - \mathcal{F}h_r}{r} dz - \frac{h\mathcal{F}_z - \mathcal{F}h_z}{r} dr, \quad d\mathbf{J}_h = 0. \quad (74)$$

*Second, translation in  $z$  gives the divergence-free vector current*

$$J_z^r = \frac{\mathcal{F}_r\mathcal{F}_z}{r}, \quad J_z^z = \frac{\mathcal{F}_z^2 - \mathcal{F}_r^2}{2r}, \quad \partial_r J_z^r + \partial_z J_z^z = 0. \quad (75)$$

*Third, define the weighted stress tensor*

$$T_{ij} := \frac{\mathcal{F}_i\mathcal{F}_j}{r} - \delta_{ij} \frac{|\nabla\mathcal{F}|^2}{2r}. \quad (76)$$

*Then*

$$\partial_i T_{iz} = 0, \quad \partial_i T_{ir} = \frac{|\nabla\mathcal{F}|^2}{2r^2}. \quad (77)$$

*The bundle dilation  $(r, z, \mathcal{F}) \mapsto (e^s r, e^s z, e^{s/2}\mathcal{F})$  gives, up to an overall sign, the improved current*

$$D_i := x^j T_{ij} - \frac{\mathcal{F}\mathcal{F}_i}{2r}, \quad \partial_i D_i = 0, \quad (x^r, x^z) = (r, z). \quad (78)$$

*In particular, the unimproved stress identity gives the exact Pohozaev formula*

$$\frac{1}{2} \int_D \frac{|\nabla\mathcal{F}|^2}{r} dr dz = \int_{\partial D} x^j T_{ij} n_i ds. \quad (79)$$

*Proof.* For the first statement,

$$\left. \frac{d}{d\epsilon} L_{\mathcal{F}+\epsilon h} \right|_{\epsilon=0} = \partial_i \left( \frac{\mathcal{F}h_i}{r} \right) \quad (80)$$

because  $\mathcal{L}h = 0$ . Equation (74) is the resulting Noether current. The remaining identities follow by direct differentiation and Eq. (50). Integrating  $\partial_i(x^j T_{ij}) = |\nabla\mathcal{F}|^2/(2r)$  proves Eq. (79).  $\square$

Pure amplitude scaling  $\mathcal{F} \mapsto a\mathcal{F}$  preserves the linear field equation but multiplies the action by  $a^2$ ; it is therefore a solution symmetry, not a Noether symmetry. Translation in  $r$  is broken by the explicit weight  $r^{-1}$ . None of the currents above is a sign-definite physical mass current.

### D. A formal Hamiltonian slice and its instability

The covariant theory permits a canonical split, but no meridional coordinate is physical time. To make this distinction explicit, let  $I = (r_-, r_+) \subseteq (0, \infty)$  and use  $z$  as a formal evolution variable. Set

$$\pi(r) := \frac{\mathcal{F}_z(r, z)}{r}, \quad \Omega_I := \int_I \delta \mathcal{F} \wedge \delta \pi \, dr, \quad (81)$$

and

$$H_z[\mathcal{F}, \pi] := \int_I \left( \frac{r}{2} \pi^2 - \frac{\mathcal{F}_r^2}{2r} \right) dr. \quad (82)$$

For variations obeying boundary conditions that remove the endpoint term, the Hamilton equations are

$$\mathcal{F}_z = \frac{\delta H_z}{\delta \pi} = r\pi, \quad \pi_z = -\frac{\delta H_z}{\delta \mathcal{F}} = -\partial_r \left( \frac{\mathcal{F}_r}{r} \right), \quad (83)$$

and hence reproduce  $\mathcal{L}\mathcal{F} = 0$ . The corresponding formal Poisson bracket is

$$\{P, Q\}_I := \int_I \left( \frac{\delta P}{\delta \mathcal{F}} \frac{\delta Q}{\delta \pi} - \frac{\delta P}{\delta \pi} \frac{\delta Q}{\delta \mathcal{F}} \right) dr. \quad (84)$$

For two solution translations  $f$  and  $g$ , the bilinear form

$$c_I(f, g) := \int_I \frac{f g_z - g f_z}{r} dr \quad (85)$$

is independent of  $z$  when the radial endpoint flux vanishes. It is an antisymmetric two-cocycle on the abelian solution-translation algebra. It may be degenerate for a chosen boundary condition and is not a quantum central extension.

The Hamiltonian in Eq. (82) is indefinite. The exact high-frequency Bessel solutions used in the Hadamard-instability theorem therefore have a complementary interpretation: the formal  $z$ -Hamiltonian flow amplifies arbitrarily small smooth Cauchy data exponentially. This is an elliptic Cauchy evolution, not a unitary or causal dynamics. In particular, no causal Peierls bracket follows from Eq. (84).

### E. Boundary Lagrangians, gluing, and the physical interface

For a fixed smooth boundary component  $\Sigma$ , define the conormal momentum

$$p_n := \frac{\partial_n \mathcal{F}}{r}. \quad (86)$$

The trace space

$$\mathcal{H}_\Sigma := H^{1/2}(\Sigma) \oplus H^{-1/2}(\Sigma) \quad (87)$$

has the canonical symplectic form

$$\omega_\Sigma((f, p), (g, q)) := \int_\Sigma (fq - gp) \, ds. \quad (88)$$

Integration of Eq. (73) proves that the Cauchy data of  $\mathcal{L}$ -harmonic fields are isotropic. Under the standard elliptic trace and unique-continuation hypotheses, the Calderón projector identifies their completed Cauchy-data space as a Lagrangian subspace of  $\mathcal{H}_{\partial D}$  [40, 41]. This functional-analytic conclusion is stronger than the formal Green identity and uses the stated trace and closed-range hypotheses.

On a fixed boundary and an invertible sheet,

$$\delta \mathcal{F} = A \delta \eta, \quad \pi_n := A p_n, \quad \delta \mathcal{F} \wedge \delta p_n = \delta \eta \wedge \delta \pi_n. \quad (89)$$

Thus the auxiliary trace transformation is symplectic for  $A \neq 0$  and collapses at  $A = 0$ .

Suppose  $D = D_- \cup_\Sigma D_+$  and orient both copies of  $\Sigma$  by their outward normals. The two variational boundary terms cancel precisely when  $\mathcal{F}$  and its conormal momentum match. Hence the solution spaces of the two pieces define Lagrangian correspondences, and source-free gluing is their composition along the diagonal. If  $\mathcal{F}$  is continuous but  $p_n$  jumps, the uncanceled term is exactly the auxiliary delta-layer source. This statement concerns the linear auxiliary equation: its jump is not an Israel surface stress.

**Proposition 6** (Exact distributional interface dictionary). *Let  $\Sigma \Subset \{r > 0\}$  be a smooth oriented curve with unit normal  $n$  pointing from  $D_-$  to  $D_+$ , and let  $\mathcal{F}$  be piecewise  $C^2$  with one-sided traces. In the sense of Schwartz distributions,*

$$\mathcal{L}\mathcal{F} = (\mathcal{L}\mathcal{F})_{\text{reg}} + [\partial_n \mathcal{F}] \delta_\Sigma + \text{div}([\mathcal{F}] n \delta_\Sigma) - \frac{n_r}{r} [\mathcal{F}] \delta_\Sigma. \quad (90)$$

If  $[\mathcal{F}] = 0$ , only the single-layer term remains. Within one constant- $\ell_0$  chart, continuity of  $\eta$  and  $A \neq 0$  give

$$[\partial_n \mathcal{F}] = A[\partial_n \eta]. \quad (91)$$

The exact reflected mode

$$\mathcal{F}(r, z) = r J_1(kr) e^{-k|z|} \quad (92)$$

satisfies

$$\mathcal{L}\mathcal{F} = -2kr J_1(kr) \delta(z). \quad (93)$$

It is a razor-thin auxiliary layer, not a smooth volume-dust solution.

*Proof.* Write the piecewise field with a Heaviside function of signed distance to  $\Sigma$  and differentiate distributionally. The regular gradients produce  $[\partial_n \mathcal{F}] \delta_\Sigma$ , the jump of the field produces the divergence term, and  $-r^{-1} \partial_r$  produces the last term in Eq. (90). If the field is continuous, the last two terms vanish. On one inverse chart,  $\partial_n \mathcal{F} = B n_r + A \partial_n \eta$ ; continuity of  $r$ ,  $\eta$ ,  $A$ , and  $B$  proves Eq. (91). Finally,  $\partial_z^2 e^{-k|z|} = k^2 e^{-k|z|} - 2k \delta(z)$ , while the Bessel equation gives

$$(\partial_r^2 - r^{-1} \partial_r + k^2) [r J_1(kr)] = 0,$$

which proves Eq. (93).  $\square$

The wavefront of  $\delta_\Sigma$  is the nonzero conormal bundle of  $\Sigma$ . Ellipticity therefore localizes the singular support of the auxiliary response to that of its source, rather than producing propagation along real characteristics [42]. If  $\eta$  or the metric itself jumps, expressions such as  $|\nabla \eta|^2$ ,  $\log \eta$ ,  $g^{-1}$ , and the nonlinear Einstein tensor contain products that ordinary distribution theory need not define. They must not be formed without a separately specified generalized-function framework or regularization. Moreover, the coefficient in Eq. (90) is not an Israel stress tensor: that tensor is determined only after the full metric has been reconstructed and its extrinsic-curvature jump evaluated. The Geroch–Traschen regular-metric class accommodates hypersurface shells but not generic codimension-two strings [43].

For a fixed piecewise smooth boundary, the first-order scalar action creates no additional codimension-two charge. Shape variations, moving interfaces, or independent boundary actions enlarge the phase space and can generate corner terms. A Maslov index or spectral-flow statement requires a specified self-adjoint Fredholm family; it is not a consequence of the word “Lagrangian” alone.

The vacuum exterior has a different variational system. Where a global twist potential  $\chi$  exists, the gauge-fixed Ernst subsystem may be written, up to an overall conventional constant, as

$$\mathcal{A}_E[f, \chi] = \frac{1}{2} \int_D \frac{r}{f^2} (|\nabla f|^2 + |\nabla \chi|^2) dr dz, \quad f > 0. \quad (94)$$

It has polymomenta

$$p_f^i = \frac{r}{f^2} \partial_i f, \quad p_\chi^i = \frac{r}{f^2} \partial_i \chi, \quad (95)$$

and an analogous conserved multisymplectic current for pairs of Jacobi fields. This structure belongs to the nonlinear Ernst harmonic map, not to  $\mathcal{F}$ . On a multiply connected exterior, Eq. (94) is global only after the twist periods vanish; otherwise one must use local potentials or the universal cover together with the cohomology sector.

Finally, the physical variational problem starts from the Einstein–Hilbert action with its gravitational boundary terms and a dust action capable of vorticity. For example, a material-coordinate representation uses

$$\mathcal{A}_{\text{dust}} = -\frac{1}{2} \int_{\mathcal{M}} \sqrt{-g} M (g^{ab} U_a U_b + 1) d^4 x, \quad U_a = -\partial_a T + W_A \partial_a Z^A. \quad (96)$$

Such matter actions and the required gravitational boundary term are reviewed in Refs. [44, 45]. The full covariant presymplectic form has diffeomorphism degeneracies and must be reduced only after the constraints, asymptotic

conditions, and boundary gauge have been fixed [46, 47]. Variation at an internal timelike boundary relates continuity of the induced metric and gravitational canonical momentum to the Darmois conditions; the matter variables additionally require a no-flux or prescribed-flux condition. No equality between that physical form and  $\Omega_{\mathcal{F}}$  has been established here.

This separation prevents three incorrect conclusions. The convex auxiliary energy is not the ADM mass, the formal  $z$ -Hamiltonian is not physical time evolution, and the auxiliary Lagrangian matching relation is only one compatibility layer within the complete dust–vacuum boundary problem.

### VIII. PHYSICAL COVARIANT VARIATION AND CONDITIONAL STRING EXTENSIONS

The auxiliary action of Sect. VII is an exact and useful structure, but it is not an action for the spacetime and its matter. This section supplies the physical variational comparison. We first formulate Einstein gravity coupled to vortical dust and identify its covariant presymplectic and boundary data. We then ask a deliberately separate question: what survives if the exterior theory is enlarged to Einstein–Maxwell–dilaton–axion gravity? The answer includes a rigidity theorem, but not a string-theoretic explanation of the rotating-dust interior.

Throughout this section the spacetime signature is  $(-, +, +, +)$ ,  $G_N$  is Newton’s constant, and all orientations and normal directions are fixed before boundary quantities are compared.

#### A. Einstein gravity with vortical dust

A pressureless fluid is not adequately represented by one velocity potential when vorticity is present. Brown’s material-coordinate action keeps the particle current and three comoving labels as independent fields [44]. Let  $\mathcal{J}^\mu$  be a future-directed vector density, let  $\alpha^A$  be material labels, and let  $\vartheta$  and  $\beta_A$  be Clebsch variables, where  $A = 1, 2, 3$ . Define

$$|\mathcal{J}|_g := \sqrt{-g_{\mu\nu}\mathcal{J}^\mu\mathcal{J}^\nu}, \quad n := \frac{|\mathcal{J}|_g}{\sqrt{-g}}, \quad u^\mu := \frac{\mathcal{J}^\mu}{|\mathcal{J}|_g}. \quad (97)$$

For particles of constant rest mass  $m > 0$ , the dust action is

$$\mathcal{S}_B[g, \mathcal{J}, \vartheta, \alpha, \beta] = \int_{\mathcal{M}} [-m|\mathcal{J}|_g + \mathcal{J}^\mu (\partial_\mu \vartheta + \beta_A \partial_\mu \alpha^A)] d^4x. \quad (98)$$

This is the  $\rho(n) = mn$  specialization of the perfect-fluid action. Its independent variations give

$$\begin{aligned} \partial_\mu \mathcal{J}^\mu &= 0, & \mathcal{J}^\mu \partial_\mu \alpha^A &= 0, \\ \partial_\mu (\beta_A \mathcal{J}^\mu) &= 0, & m u_\mu + \partial_\mu \vartheta + \beta_A \partial_\mu \alpha^A &= 0. \end{aligned} \quad (99)$$

Metric variation produces

$$T_{\mu\nu}^d = mn u_\mu u_\nu. \quad (100)$$

Moreover, Eq. (99) implies

$$m du^b = -d\beta_A \wedge d\alpha^A, \quad (101)$$

so the action admits genuine vortical flow. In particular, the stationary rotating dust considered in this paper need not be forced into an irrotational scalar ansatz. The constant shift  $\vartheta \mapsto \vartheta + c$  has Noether current  $\mathcal{J}^\mu$ , while material-label transformations preserving  $d\vartheta + \beta_A d\alpha^A$  encode the circulation symmetries. These are matter symmetries, not replacements for spacetime diffeomorphism charges.

On a spacetime region with smooth non-null boundary components, the physical Dirichlet action is

$$\mathcal{S}_{\text{phys}} = \frac{1}{16\pi G_N} \int_{\mathcal{M}} R \epsilon_g + \frac{1}{8\pi G_N} \int_{\partial\mathcal{M}} \epsilon_n K \epsilon_h + \mathcal{S}_B, \quad (102)$$

where  $\epsilon_n = g(n, n) = \pm 1$ ,  $h$  is the induced metric, and  $K = h^{ab} K_{ab}$  in the convention of Eq. (279). Corner terms must be added when boundary components meet non-smoothly. The second term in Eq. (102) is the Gibbons–Hawking–York term in these conventions [45].

For the bulk Lagrangian four-form  $\mathbf{L}_{\text{phys}}$ , write its first variation as

$$\delta \mathbf{L}_{\text{phys}} = \mathbf{E}(\Psi) \cdot \delta \Psi + d\boldsymbol{\theta}(\Psi; \delta \Psi), \quad \Psi = (g, \mathcal{J}, \vartheta, \alpha, \beta). \quad (103)$$

The Lee–Wald presymplectic current and the diffeomorphism Noether current are, respectively,

$$\boldsymbol{\omega}(\Psi; \delta_1 \Psi, \delta_2 \Psi) := \delta_1 \boldsymbol{\theta}(\Psi; \delta_2 \Psi) - \delta_2 \boldsymbol{\theta}(\Psi; \delta_1 \Psi), \quad (104)$$

$$\mathbf{J}_\xi := \boldsymbol{\theta}(\Psi; \mathcal{L}_\xi \Psi) - \iota_\xi \mathbf{L}_{\text{phys}}. \quad (105)$$

For solutions and linearized solutions,  $d\boldsymbol{\omega} = 0$ . Diffeomorphism covariance gives  $\mathbf{J}_\xi = \mathbf{C}_\xi + d\mathbf{Q}_\xi$ , with the constraint term  $\mathbf{C}_\xi$  vanishing on shell. Subject to the chosen boundary conditions and integrability of the right-hand side, the Hamiltonian variation is

$$\delta H_\xi = \int_{\partial C} (\delta \mathbf{Q}_\xi - \iota_\xi \boldsymbol{\theta}) \quad (106)$$

on a Cauchy surface  $C$  [46, 47]. Boundary counterterms and corners shift the representative of  $\boldsymbol{\theta}$  and must be included before a charge is declared finite or integrable. Equations (104)–(106) refer to the coupled metric–dust theory. They cannot be replaced by the weighted meridional current of Eq. (71). In particular, choosing  $\xi$  to be the stationary or axial Killing field defines physical energy or angular-momentum variations only after the asymptotic normalization, boundary conditions, and integrability test have been supplied.

The GHY term also makes the relation between variational and Darms data precise. For a non-null three-boundary define the gravitational momentum density conjugate to  $h^{ab}$  by

$$\Pi_{ab} := \frac{\epsilon_n \sqrt{|h|}}{16\pi G_N} (K_{ab} - K h_{ab}). \quad (107)$$

**Proposition 7** (Darms data as equality of boundary momenta). *Let two  $C^2$  spacetime regions meet across a non-null hypersurface  $\Sigma$ , and express both second fundamental forms using one common normal orientation. Once the first fundamental forms have been identified,*

$$[\Pi_{ab}] = 0 \quad \Longleftrightarrow \quad [K_{ab}] = 0. \quad (108)$$

*Thus a shell-free stationary point of the joined Dirichlet action has the complete Darms data. Equality only of selected Killing-block derivatives does not suffice.*

*Proof.* With  $h_{ab}$  common,  $[\Pi_{ab}] = 0$  is equivalent to  $[K_{ab}] - h_{ab}[K] = 0$ . Contracting with the inverse three-metric gives  $[K] - 3[K] = -2[K] = 0$ . Hence  $[K] = 0$ , and substitution gives  $[K_{ab}] = 0$ . The converse is immediate. With the outward normals used in the two separate actions, one first reverses one normal to obtain the common-orientation convention. A nonzero momentum jump is the variational counterpart of an Israel surface stress [48–50].  $\square$

There is one further reduction caveat. Substituting a stationary, axisymmetric ansatz directly into Eq. (102) and only then varying is valid only when the relevant principle of symmetric criticality holds. Its group cohomology and isotropy hypotheses are nontrivial in general relativity [51]. The safe logical route in the present paper is to derive the covariant equations and boundary variation first and then impose the two Killing symmetries. No auxiliary scalar action establishes this missing equivalence.

## B. Einstein-frame EMDA and neutral axidilaton rigidity

We now change the field content. In a standard four-dimensional Einstein-frame normalization, the bosonic EMDA bulk action is

$$\begin{aligned} \mathcal{S}_{\text{EMDA}}^E = \frac{1}{16\pi G_N} \int_{\mathcal{M}} \epsilon_E \left[ R_E - 2(\nabla\varphi)^2 - \frac{1}{2}e^{4\varphi}(\nabla\kappa)^2 - e^{-2\varphi}F_{\mu\nu}F^{\mu\nu} \right. \\ \left. - \kappa F_{\mu\nu}^* F^{\mu\nu} \right] + \mathcal{S}_B[g_E]. \end{aligned} \quad (109)$$

Here  $\varphi$  is the dilaton,  $\kappa$  is the axion,  $F = dA$ , and the dust is neutral and minimally coupled only to  $g_E$ . This convention is equivalent, after the signature translation, to the primary EMDA actions used in Refs. [52, 53]. The gravitational boundary term has the form already displayed in Eq. (102) with  $g = g_E$ .

When  $F = 0$ , the scalar equations are

$$\nabla_\mu (e^{4\varphi} \nabla^\mu \kappa) = 0, \quad \square_E \varphi = \frac{1}{2} e^{4\varphi} (\nabla \kappa)^2. \quad (110)$$

There is no dust source in either equation because  $\mathcal{S}_B[g_E]$  is independent of  $\varphi$  and  $\kappa$ . For invariant scalars in the circular metric

$$ds_E^2 = G_{AB} dx^A dx^B + e^\mu (dr^2 + dz^2), \quad \det G = -r^2, \quad (111)$$

Eq. (110) becomes

$$\partial_a (r e^{4\varphi} \partial_a \kappa) = 0, \quad \partial_a (r \partial_a \varphi) = \frac{r}{2} e^{4\varphi} |\nabla_{rz} \kappa|^2, \quad a \in \{r, z\}. \quad (112)$$

**Theorem 10** (Neutral axidilaton no-hair theorem). *Let a connected, horizonless, stationary and axisymmetric solution of Eq. (109) satisfy  $F = 0$ . Assume all of the following.*

1. *The spacetime is globally regular and asymptotically flat with one end, the axial action has a regular fixed axis, and the complete orbit space is a half-plane with no excised inner boundary. It admits one global circular Weyl chart  $(r, z)$  in which Eq. (111) holds with  $\det G = -r^2$ , the axis is  $r = 0$ , and the orbit measure in the reduced scalar equations is  $r dr dz$ .*
2. *The neutral dust is minimally coupled to  $g_E$ . Across every internal matter–vacuum interface,  $\varphi$ ,  $\kappa$ , and their canonical normal fluxes are continuous; in particular there is no scalar surface source.*
3. *The invariant scalars are  $C^2$  away from interfaces,  $C^1$  in the global weak sense, and regular at the axis. The axion has a globally single-valued real lift, so there is no axionic monodromy.*
4. *With  $R = (r^2 + z^2)^{1/2}$ , constants  $\varphi_\infty$  and  $\kappa_\infty$  exist such that*

$$\begin{aligned} \varphi - \varphi_\infty &= O(R^{-1}), & |\nabla_{rz} \varphi| &= O(R^{-2}), \\ \kappa - \kappa_\infty &= O(R^{-1}), & |\nabla_{rz} \kappa| &= O(R^{-2}). \end{aligned} \quad (113)$$

*The weighted scalar energies*

$$\int r |\nabla_{rz} \varphi|^2 dr dz, \quad \int r e^{4\varphi} |\nabla_{rz} \kappa|^2 dr dz \quad (114)$$

*are finite over the orbit space.*

*Then*

$$\kappa \equiv \kappa_\infty, \quad \varphi \equiv \varphi_\infty. \quad (115)$$

*Consequently, appending a neutral, source-free axidilaton sector cannot dress a globally regular galaxy satisfying these hypotheses.*

*Proof.* Let  $D_R$  be the intersection of the orbit half-plane with a Euclidean half-disk of radius  $R$ , with small half-disks around any intermediate corners used only during the exhaustion. Multiply the first equation in Eq. (112) by  $\kappa - \kappa_\infty$  and integrate by parts. The interface terms cancel because both the scalar and its canonical normal flux are continuous. Regularity makes  $\kappa$  and its first derivatives bounded at the axis, where the factor  $r$  makes the boundary term vanish. On the outer semicircle, Eq. (113) gives

$$\int_{\partial D_R} r e^{4\varphi} (\kappa - \kappa_\infty) \partial_n \kappa ds = O(R^{-1}). \quad (116)$$

There is no horizon or other inner boundary. Passing to the limit, using Eq. (114), yields

$$\int_D r e^{4\varphi} |\nabla_{rz} \kappa|^2 dr dz = 0. \quad (117)$$

The integrand is nonnegative and  $e^{4\varphi} > 0$ , so connectedness gives  $\kappa \equiv \kappa_\infty$ .

The second equation in Eq. (112) now reduces to  $\partial_a (r \partial_a \varphi) = 0$ . Multiplication by  $\varphi - \varphi_\infty$  and the identical exhaustion argument give

$$\int_D r |\nabla_{rz} \varphi|^2 dr dz = 0. \quad (118)$$

Thus  $\varphi \equiv \varphi_\infty$ , proving Eq. (115).  $\square$

*Remark 3* (Sharp logical content of the rigidity theorem). The proof fails, for identifiable reasons, in the presence of gauge flux, direct matter–scalar coupling, a scalar-charged shell, a horizon or another inner boundary, a second asymptotic end, axionic monodromy, singularities, or insufficient decay. These are possible mechanisms for nonconstant scalars; the theorem does not assert that any of them produces an acceptable galaxy.

### C. Exact Einstein–string-frame dust dictionary

In four dimensions and in the normalization of Eq. (109), fix the conformal relation

$$g_E = e^{-2\varphi} g_s, \quad \sqrt{-g_E} = e^{-4\varphi} \sqrt{-g_s}. \quad (119)$$

This is a field redefinition. Which frame is minimally coupled is an additional physical choice about the matter sector.

**Proposition 8** (Conformal transformation of the Brown dust action). *Keep the vector density  $\mathcal{J}^\mu$  and the Clebsch variables fixed under Eq. (119). Then*

$$|\mathcal{J}|_E = e^{-\varphi} |\mathcal{J}|_s, \quad n_E = e^{3\varphi} n_s, \quad u_E^\mu = e^\varphi u_s^\mu, \quad (120)$$

and the Einstein-minimal action Eq. (98) becomes exactly

$$\mathcal{S}_B^E = \int_{\mathcal{M}} [-me^{-\varphi} |\mathcal{J}|_s + \mathcal{J}^\mu (\partial_\mu \vartheta + \beta_A \partial_\mu \alpha^A)] d^4x. \quad (121)$$

Thus the same particles have string-frame mass  $m_s(\varphi) = me^{-\varphi}$ ; they are not minimally coupled to  $g_s$ .  
Let

$$T_{\mu\nu}^s := -\frac{2}{\sqrt{-g_s}} \frac{\delta \mathcal{S}_B^E}{\delta g_s^{\mu\nu}}, \quad \rho_s := me^{-\varphi} n_s. \quad (122)$$

On the dust equations,

$$T_s^{\mu\nu} = \rho_s u_s^\mu u_s^\nu, \quad (123)$$

$$\nabla_\mu^s T_s^\mu{}_\nu = \rho_s \partial_\nu \varphi, \quad (124)$$

$$u_s^\mu \nabla_\mu^s u_s^\nu = (g_s^{\nu\lambda} + u_s^\nu u_s^\lambda) \partial_\lambda \varphi. \quad (125)$$

Conversely, if constant-mass dust is postulated to be minimal in the string frame, its Einstein-frame mass is  $m_E(\varphi) = me^\varphi$ , and the right-hand sides of Eqs. (124) and (125) acquire the opposite sign after interchanging  $s$  and  $E$ .

*Proof.* Equation (120) follows directly from the definition  $|\mathcal{J}|^2 = -g_{\mu\nu} \mathcal{J}^\mu \mathcal{J}^\nu$  and the volume scaling in Eq. (119). The current–potential term is a coordinate density and is conformally unchanged, which proves Eq. (121).

The scalar source of this action is

$$\frac{1}{\sqrt{-g_s}} \frac{\delta \mathcal{S}_B^E}{\delta \varphi} = \rho_s. \quad (126)$$

Diffeomorphism invariance therefore gives the Ward identity Eq. (124). Particle-number conservation remains  $\nabla_\mu^s (n_s u_s^\mu) = 0$ . Projecting Eq. (124) orthogonally to  $u_s$  proves Eq. (125). Starting instead from  $-m|\mathcal{J}|_s$  gives  $-me^\varphi |\mathcal{J}|_E$ , which proves the converse statement.  $\square$

The fifth force in Eq. (125) is not a coordinate artifact: it records which metric defines constant particle masses and freely falling matter. A conformal rewriting neither creates a new solution nor preserves the geodesic-dust interpretation in both frames. In particular, the neutral no-hair theorem above assumes Einstein-minimal dust; changing the matter coupling changes its scalar equation and its hypotheses.

### D. Conditional hidden symmetries, Buscher duality, and junctions

Source-free stationary EMDA admits a finite-dimensional hidden  $Sp(4, \mathbb{R})$  symmetry, denoted  $Sp(2, \mathbb{R})$  in part of the physics literature, and the two-Killing reduction admits a Lax representation and an affine extension [53, 54]. These statements concern the source-free reduced equations. With dust present, the matrix equation has the schematic form

$$\partial_r (r \mathcal{M}_r \mathcal{M}^{-1}) + \partial_z (r \mathcal{M}_z \mathcal{M}^{-1}) = \mathcal{J}_d. \quad (127)$$

No zero-curvature representation has been established for a generic nonzero source  $\mathcal{J}_d$ . A hidden-symmetry transformation may therefore be applied to a valid vacuum exterior seed, but not presumed to map the rotating-dust interior to another solution with the same matter.

Buscher duality is a different operation: it acts locally on a string-frame sigma-model background  $(g_s, B, \Phi_s)$  with an Abelian isometry [55, 56]. In units in which  $\alpha' = 1$  and the sigma-model metric components are dimensionless, let  $y$  be the isometry coordinate and let  $i, j$  denote the other coordinates. The Abelian Buscher rules are

$$\tilde{g}_{yy} = \frac{1}{g_{yy}}, \quad \tilde{g}_{yi} = \frac{B_{yi}}{g_{yy}}, \quad \tilde{B}_{yi} = \frac{g_{yi}}{g_{yy}}, \quad (128)$$

$$\tilde{g}_{ij} = g_{ij} - \frac{g_{yi}g_{yj} - B_{yi}B_{yj}}{g_{yy}}, \quad \tilde{B}_{ij} = B_{ij} - \frac{g_{yi}B_{yj} - B_{yi}g_{yj}}{g_{yy}}, \quad (129)$$

$$\tilde{\Phi}_s = \Phi_s - \frac{1}{2} \log g_{yy}. \quad (130)$$

For the spacelike duality used below they require  $g_{yy} > 0$  and a well-defined local isometry quotient. Restoring dimensions replaces the logarithm by the corresponding dimensionless ratio in string units.

For illustration only, suppose that the helical plateau Eq. (270) is used as a string-frame seed, with constant dilaton, vanishing initial  $B$ -field, and the constant Einstein–string conformal factor absorbed into the normalization of the displayed metric. Taking  $y = \phi$  then gives  $g_{\phi\phi} = r^2 - q_0^2$  and  $g_{t\phi} = q_0$ , and hence

$$\tilde{B}_{\phi t} = \frac{q_0}{r^2 - q_0^2}, \quad \tilde{g}_{\phi\phi} = \frac{1}{r^2 - q_0^2}. \quad (131)$$

This conditional dual is singular at the chronology cylinder. More generally, axial duality at a regular fixed axis has  $g_{\phi\phi} \sim r^2$ , so  $\tilde{g}_{\phi\phi} \sim r^{-2}$  and  $\tilde{\Phi}_s \sim \Phi_s - \log r$ . It is therefore neither a regular axis nor a global galactic completion.

The field enlargement also enlarges the junction problem. In the conventions of Eq. (109), introduce the antisymmetric Maxwell excitation tensor

$$\mathcal{G}^{\mu\nu} := e^{-2\varphi} F^{\mu\nu} + \kappa^* F^{\mu\nu}, \quad \nabla_\mu^* F^{\mu\nu} = 0, \quad \nabla_\mu \mathcal{G}^{\mu\nu} = 0 \quad (132)$$

in a source-free neighbourhood of the matching surface.

**Proposition 9** (Necessary shell-free EMDA junction data). *Let a non-null hypersurface  $\Sigma$  separate two piecewise  $C^2$  EMDA solutions. Assume there is no gravitational shell, scalar surface source, electric surface current, or magnetic surface current. In a common normal orientation, the distributional field equations require*

$$[h_{ab}] = 0, \quad [K_{ab}] = 0, \quad (133)$$

$$[\varphi] = 0, \quad [n(\varphi)] = 0, \quad (134)$$

$$[\kappa] = 0, \quad [e^{4\varphi} n(\kappa)] = 0, \quad (135)$$

$$[i^* F] = 0, \quad [n_\mu \mathcal{G}^{\mu a}] = 0. \quad (136)$$

Here  $i : \Sigma \hookrightarrow \mathcal{M}$  is the inclusion. These are necessary local data; they do not prove that a global exterior realizing them exists.

*Proof.* The first line is Proposition 7. A jump of either scalar itself would produce an inadmissible derivative of a delta distribution in its second-order equation, while integration of the scalar equations across a Gaussian pillbox gives the two normal-flux conditions. Writing a piecewise smooth two-form as  $F = F^+ \mathbf{1}_+ + F^- \mathbf{1}_-$  shows that the singular part of  $dF$  is the normal one-form wedged with  $[F]$ ; its vanishing is equivalent to  $[i^* F] = 0$ . Integrating  $\nabla_\mu \mathcal{G}^{\mu\nu} = 0$  across the same pillbox proves the last condition. This is the non-null, source-free specialization of the general distributional scalar-junction framework [57].  $\square$

Applying a duality only to the exterior generally violates one or more of Eqs. (133)–(136) and creates a shell or surface charge. Even when the local junction data pass, asymptotic normalization, NUT charge, gauge charge, scalar charge, axis regularity, conicity, topology, and causality must all be rechecked.

*Remark 4* (Non-implications). Nothing in this section implies that the dust interior is an EMDA solution with nonconstant scalars, that an EMDA hidden symmetry survives a dust source, or that Buscher duality preserves pressureless geodesic dust. A pressureless dust element is not a fundamental-string worldsheet or a string gas; those sources have additional tension data. The helical flat quotient is not evidence for a string source. A transformed finite-parameter black-hole exterior is not a generic galactic exterior. Finally, calling the retained  $U(1)$  field electromagnetic or heterotic requires a specified compactification and matter-coupling prescription. The only unconditional new result in the string-effective direction is the neutral rigidity theorem under its stated hypotheses.

## IX. EXACTLY FLAT EQUATORIAL TRACES: LOCAL EXISTENCE AND GLOBAL OBSTRUCTIONS

Proposition 1 does not exclude a nonconstant solution whose equatorial trace is exactly flat. We first show that such traces can be realized locally on patches separated from the rotation axis. We then examine whether they can be extended beyond this local setting. Regularity at the axis excludes every nontrivial branch with  $\ell_0 \neq 0$ , while an analytic-continuation argument rules out an axis-connected exactly flat trace for the remaining case  $\ell_0 = 0$ . Finally, for branches confined to annular regions, we combine density positivity with vertical localization to derive a finite bound on the radial extent of an exactly flat trace.

### A. Local analytic realization of an exactly flat trace

For constant  $\ell = \ell_0$ , it is useful to recast Eq. (2.10) of Ref. [6] in terms of an auxiliary potential  $\mathcal{F}(r, z)$ . On any patch where  $\eta > 0$ , the relation between  $\mathcal{F}$  and  $\eta$  can be written as

$$\mathcal{F}(r, z) = \Phi_{\ell_0}(r, \eta(r, z)), \quad \Phi_{\ell_0}(r, \eta) := 2\eta + \ell_0 r^2 \log\left(\frac{\eta}{\eta_*}\right) - \frac{\ell_0}{2}\eta^2, \quad (137)$$

where  $\eta_* > 0$  is an arbitrary reference scale. The auxiliary potential satisfies the linear equation

$$\mathcal{L}\mathcal{F} = 0, \quad \mathcal{L} := \partial_r^2 - \frac{1}{r}\partial_r + \partial_z^2. \quad (138)$$

Replacing  $\eta_*$  by another positive reference scale changes  $\mathcal{F}$  only by a constant multiple of  $r^2$ . Since  $\mathcal{L}(r^2) = 0$ , this freedom leaves the field equation  $\mathcal{L}\mathcal{F} = 0$  unchanged. The derivative of the transformation in Eq. (137) with respect to  $\eta$  is

$$\partial_\eta \Phi_{\ell_0}(r, \eta) = 2 + \frac{\ell_0 r^2}{\eta} - \ell_0 \eta = 2 + \ell_0 r \left( \frac{1}{v} - v \right) =: A_{\ell_0}(r, v). \quad (139)$$

Thus, at fixed  $r$ , the Jacobian relating  $\eta$  to the auxiliary potential  $\mathcal{F}$  is precisely the same factor that multiplies the second-order derivatives in the exact VFE. The nonvanishing of  $A_{\ell_0}$  is precisely the condition that allows the transformation in Eq. (137) to be inverted locally with respect to  $\eta$ . This observation leads to the following local existence result for an exactly flat equatorial trace.

**Theorem 11** (Local analytic realization of an exactly flat trace). *Let  $r_* > 0$ ,  $v_0 \in (0, 1)$ , and  $\ell_0 \in \mathbb{R}$ , and assume that*

$$A_* := A_{\ell_0}(r_*, v_0) = 2 + \ell_0 r_* \left( \frac{1}{v_0} - v_0 \right) \neq 0. \quad (140)$$

*Then, there exist  $\delta, \varepsilon > 0$ , with  $\delta < r_*$ , and a unique real-analytic solution of the exact constant- $\ell$  VFE on  $(r_* - \delta, r_* + \delta) \times (-\varepsilon, \varepsilon)$  satisfying the Cauchy data  $v(r, 0) = v_0$ ,  $v_z(r, 0) = 0$ , and  $r \in (r_* - \delta, r_* + \delta)$ . The solution is even in  $z$  and need not be spatially constant. Its leading transverse correction is*

$$v(r, z) = v_0 + \frac{v_0 - \ell_0 r}{r^2 [2 + \ell_0 r (1/v_0 - v_0)]} z^2 + O(z^4). \quad (141)$$

*For every choice of  $C > 0$ , of the additive integration constant in  $\Omega$ , and of the value  $\mu(r_*, 0)$ , the field  $\eta = rv$  lifts uniquely to a real-analytic local metric of the exact  $(\eta, H)$  rotating-dust family, with  $H(\eta) = -Ce^{\ell_0 \eta}$ . If, in addition,*

$$-\frac{2v_0}{1 - v_0^2} < \ell_0 r_* < \frac{2v_0}{1 + v_0^2}, \quad (142)$$

*then the neighbourhood can be chosen sufficiently small that  $0 < v < 1$  and  $\rho > 0$  hold throughout it.*

*Proof.* We first solve the linear equation for the auxiliary potential  $\mathcal{F}$ . Along  $z = 0$ , we prescribe the analytic Cauchy data

$$\mathcal{F}(r, 0) := \Phi_{\ell_0}(r, rv_0) = 2rv_0 + \ell_0 r^2 \log\left(\frac{rv_0}{\eta_*}\right) - \frac{\ell_0}{2}r^2 v_0^2, \quad (143)$$

$$\mathcal{F}_z(r, 0) := 0. \quad (144)$$

Since  $r_* > 0$  and  $v_0 > 0$ , these data are analytic in a neighbourhood of  $r = r_*$ . Moreover, Eq. (138) can be written as

$$\mathcal{F}_{zz} = -\mathcal{F}_{rr} + \frac{1}{r}\mathcal{F}_r, \quad (145)$$

whose coefficients are analytic near  $(r_*, 0)$ . The surface  $z = 0$  is therefore noncharacteristic, and the Cauchy–Kowalevski theorem yields a unique real-analytic solution  $\mathcal{F}$  in a neighbourhood of  $(r_*, 0)$  (see, for example, Ref. [58]). Because both the equation and the Cauchy data are invariant under  $z \mapsto -z$ , uniqueness implies  $\mathcal{F}(r, z) = \mathcal{F}(r, -z)$ . At  $(r_*, 0)$ , the condition Eq. (140) is precisely  $\partial_\eta \Phi_{\ell_0}(r_*, r_* v_0) = A_{\ell_0}(r_*, v_0) \neq 0$ . By continuity, after restricting to a sufficiently small neighbourhood,  $\partial_\eta \Phi_{\ell_0}$  remains nonzero. The analytic implicit-function theorem then provides a unique real-analytic field  $\eta(r, z)$  satisfying  $\mathcal{F}(r, z) = \Phi_{\ell_0}(r, \eta(r, z))$ . On  $z = 0$ , the prescribed data and local uniqueness give  $\eta(r, 0) = r v_0$ , and hence,  $v(r, 0) = \eta(r, 0)/r = v_0$ . The evenness of  $\mathcal{F}$  and uniqueness of the implicit solution imply that  $\eta$ , and therefore  $v = \eta/r$ , is even in  $z$ . In particular,  $v_z(r, 0) = 0$ . Since the transformation Eq. (137) recasts the constant- $\ell$  field equation as  $\mathcal{L}\mathcal{F} = 0$ , the resulting field  $v$  satisfies the exact constant- $\ell$  VFE. We now verify that this VFE solution determines a full local Einstein–dust metric. Let us fix  $C > 0$  and the additive integration constant in  $\Omega$ . Moreover, we prescribe a value  $\mu_*$  at  $(r_*, 0)$ . After shrinking the neighbourhood if necessary, one has  $\eta > 0$  and  $0 < v < 1$ . Let us set  $H(\eta) = -C e^{\ell_0 \eta}$  and  $d\Omega/d\eta = H_\eta/(2\eta)$ . Further, define the Killing block  $G = (g_{AB})$  by Eq. (4). Because  $\eta$  stays away from zero on this patch,  $H$ ,  $\Omega$ , and all entries of  $G$  are real analytic. Moreover,  $g_{\phi\phi} = (-H)^{-1}(r^2 - \eta^2) > 0$  and  $\det G = -r^2 < 0$ , so the Killing block has Lorentzian signature. It remains to reconstruct the meridional conformal factor. Let  $M_r$  and  $M_z$  denote the right-hand sides of the exact quadratures

$$M_r := \frac{1}{2r} \left[ (g_{tt})_r (g_{\phi\phi})_r - (g_{tt})_z (g_{\phi\phi})_z - (g_{t\phi})_r^2 + (g_{t\phi})_z^2 \right], \quad (146)$$

$$M_z := \frac{1}{2r} \left[ (g_{tt})_z (g_{\phi\phi})_r + (g_{tt})_r (g_{\phi\phi})_z - 2(g_{t\phi})_r (g_{t\phi})_z \right]. \quad (147)$$

The Winicour construction summarized in Eqs. (2.5), (2.9), and (2.10) of Ref. [6] shows that, after the Killing block is expressed through  $(\eta, H, \Omega)$ , the compatibility condition  $\partial_z M_r = \partial_r M_z$  is precisely the remaining Einstein-equation integrability condition, and it follows from  $\mathcal{L}\mathcal{F} = 0$ . Thus, the analytic one-form  $M_r dr + M_z dz$  is closed. On a sufficiently small rectangular neighbourhood, the analytic Poincaré lemma therefore gives a unique real-analytic function  $\mu$  satisfying  $\mu_r = M_r$ ,  $\mu_z = M_z$ , and  $\mu(r_*, 0) = \mu_*$ . Hence, the metric associated to the line element  $ds^2 = G_{AB} dx^A dx^B + e^\mu (dr^2 + dz^2)$  is a real-analytic Lorentzian metric of the exact rotating-dust family. Once the constants just specified are fixed, the reconstruction is unique. We may now choose  $\delta, \varepsilon > 0$ , with  $\delta < r_*$ , so that the construction is valid on  $(r_* - \delta, r_* + \delta) \times (-\varepsilon, \varepsilon)$ . Along the equatorial trace, we have  $v_r(r, 0) = v_{rr}(r, 0) = v_z(r, 0) = 0$ . Substitution into the exact constant- $\ell$  VFE, Eq. (10), gives

$$\left[ 2 + \ell_0 r \left( \frac{1}{v_0} - v_0 \right) \right] v_{zz}(r, 0) + \frac{2\ell_0}{r} - \frac{2v_0}{r^2} = 0. \quad (148)$$

Since the coefficient of  $v_{zz}$  remains nonzero on a sufficiently small radial interval around  $r_*$ , this yields

$$v_{zz}(r, 0) = \frac{2(v_0 - \ell_0 r)}{r^2 [2 + \ell_0 r(1/v_0 - v_0)]}. \quad (149)$$

Because  $v$  is even in  $z$ , its Taylor expansion about the equatorial plane therefore gives

$$v(r, z) = v_0 + \frac{v_0 - \ell_0 r}{r^2 [2 + \ell_0 r(1/v_0 - v_0)]} z^2 + O(z^4), \quad (150)$$

which is Eq. (141). Finally, suppose that the strict inequalities Eq. (142) hold. At  $(r_*, 0)$  we have  $\eta_r = v_0 \neq 0$  and  $\eta_z = 0$ , so the point is noncritical and the algebraic density factor is strictly positive by Proposition 2. The reconstructed metric has  $g_{rr} = e^\mu > 0$ , with  $\mu$  finite and analytic, while  $\eta > 0$  on the chosen patch. The exact density formula therefore gives  $\rho(r_*, 0) > 0$ . Since every factor entering that formula is continuous, the neighbourhood can be reduced once more so that  $0 < v < 1$  and  $\rho > 0$  hold throughout it.  $\square$

## B. Regular-axis obstruction for nonzero constant- $\ell$ branches

The preceding construction establishes local analytic solutions on patches separated from the rotation axis. We now ask whether a nontrivial constant- $\ell$  branch can be extended regularly to the axis when  $\ell_0 \neq 0$ . Recall that  $k = \partial_t$  and

$m = \partial_\phi$  denote the stationary and axial Killing fields. A regular segment of the rotation axis lies in the fixed-point set of the  $U(1)$  action generated by  $m$ . In local Cartesian coordinates  $(x, y, z)$  adapted to this action, the axial Killing field takes the form

$$m = -y \partial_x + x \partial_y, \quad \mathcal{A} = \{x = y = 0\}. \quad (151)$$

The obstruction is already contained in the metric jets; regular extension of the material congruence is not needed.

**Theorem 12** (Metric-only axis-jet obstruction). *Let a  $C^2$  stationary Lorentz metric extend across a fixed component  $\mathcal{A}$  of an effective isometric  $U(1)$  action of period  $2\pi$ , and let the stationary Killing field  $k$  commute with its axial generator  $m$ . Assume that a nontrivial  $(\eta, H)$  branch is connected to  $\mathcal{A}$ , that  $k$  is timelike on  $\mathcal{A}$ , that  $0 < |\eta| < r$  on the punctured branch with  $\eta \rightarrow 0$  at  $\mathcal{A}$ , that  $H$  is  $C^1$  at  $\eta = 0$  with  $H(0) = -C < 0$ , and that the exact relations in Eqs. (2)–(4) hold off the axis. Then*

$$H'(0) = 0.$$

Consequently, a constant logarithmic derivative  $H'/H = \ell_0$  on the connected branch must satisfy  $\ell_0 = 0$ .

*Proof.* Let  $\varrho$  be a smooth transverse distance to the axis. In a smooth invariant frame the axis jets are

$$g(m, m) = a\varrho^2 + o(\varrho^2), \quad g(k, m) = O(\varrho^2), \quad g(k, k) = -D + O(\varrho^2), \quad (152)$$

with  $a > 0$  and  $D > 0$ . The determinant identity then gives

$$r^2 = Da\varrho^2 + o(\varrho^2), \quad (153)$$

so  $r \asymp \varrho$  and  $g(m, m)/r^2 \rightarrow 1/D$ .

Suppose, for contradiction, that  $h_1 := H'(0) \neq 0$ . The exact relation  $\Omega, \eta = H, \eta/(2\eta)$  gives

$$\Omega(\eta) = \frac{h_1}{2} \log |\eta| + o(|\log |\eta||),$$

and therefore  $|\Omega| \rightarrow \infty$ . From  $\eta = g(k, m) + \Omega g(m, m)$  and Eqs. (152)–(153),

$$\frac{\eta}{r^2 \Omega} \longrightarrow \frac{1}{D}.$$

Writing  $y = |\log |\eta||$ , the resulting lower comparison  $|\eta| \geq cr^2 |\Omega| \geq c'r^2 y$  gives  $y + \log y \leq 2|\log r| + O(1)$ . Thus  $|\Omega| = O(|\log r|)$ ,  $r^2 \Omega^2 \rightarrow 0$ , and  $\Omega \eta \rightarrow 0$ . The exact formula for  $g(k, k)$  now gives  $g(k, k)|_{\mathcal{A}} = H(0) = -C$ , so  $D = C$ . Moreover,  $H + C = O(|\eta|) = O(r^2 |\Omega|)$ . Hence

$$\frac{H + C}{r^2 \Omega^2} \longrightarrow 0, \quad \frac{g(k, m)}{r^2 \Omega} \longrightarrow 0.$$

On the one hand, metric regularity gives  $(g(k, k) + C)/(r^2 \Omega^2) \rightarrow 0$ . On the other hand, substituting  $g(k, k) = H - 2\Omega \eta + \Omega^2 g(m, m)$  and then  $\eta = g(k, m) + \Omega g(m, m)$  yields

$$\frac{g(k, k) + C}{r^2 \Omega^2} = \frac{H + C}{r^2 \Omega^2} - \frac{2g(k, m)}{r^2 \Omega} - \frac{g(m, m)}{r^2} \longrightarrow -\frac{1}{C}, \quad (154)$$

a contradiction. Thus  $H'(0) = 0$ . For  $H = -Ce^{\ell_0 \eta}$ , one has  $H'(0) = -C\ell_0$ , proving the final assertion.  $\square$

The proof uses the full quadratic axis jets rather than elementary flatness alone. This distinction is sharp: elementary-flat punctured metrics of lower regularity need not satisfy the conclusion.

The following lemma records the local regularity property of azimuthal vector fields that will be needed in the axis analysis.

**Lemma 1** (Regularity of an azimuthal coefficient at the axis). *Let  $X = \omega(x, y, z)m$  be an axisymmetric vector field defined on a punctured neighbourhood of a regular axis segment. If  $X$  extends as a  $C^1$  vector field across the axis, then  $\omega$  has a finite limit as the axis is approached.*

*Proof.* Let us consider first the half-line  $y = 0$ ,  $x > 0$ . From Eq. (151), we have  $X^y(x, 0, z) = x\omega(x, 0, z)$ . The continuous axisymmetric extension of the purely azimuthal field  $X$  must vanish on the axis. In particular, we find  $X^y(0, 0, z) = 0$ . For  $x > 0$ , we have

$$\omega(x, 0, z) = \frac{X^y(x, 0, z) - X^y(0, 0, z)}{x}. \quad (155)$$

Since  $X^y$  is differentiable at the axis, taking  $x \downarrow 0$  gives

$$\lim_{x \downarrow 0} \omega(x, 0, z) = \partial_x X^y(0, 0, z), \quad (156)$$

which is finite. Axisymmetry implies that  $\omega$  depends only on the radial distance from the axis and on  $z$ , so the same limit is obtained from any radial direction.  $\square$

The preceding lemma provides the regularity condition needed to test whether a constant- $\ell$  branch can approach the rotation axis. The potential obstruction comes from the angular velocity  $\Omega(\eta)$ . For real  $x \neq 0$ , the principal-value exponential integral has the small-argument expansion

$$\text{Ei}(x) = \gamma_E + \log|x| + O(x), \quad x \rightarrow 0, \quad (157)$$

where  $\gamma_E$  is the Euler–Mascheroni constant (see Eqs. 6.2.6–6.2.7 in [35]). Accordingly, for  $\ell_0 \neq 0$ , Eq. (9) gives

$$\Omega(\eta) = -\frac{C\ell_0}{2} \log|\eta| + O(1) \quad (158)$$

as  $\eta \rightarrow 0$  with  $\eta \in \mathbb{R} \setminus \{0\}$ . Thus,  $\Omega$  diverges logarithmically as a nonzero constant- $\ell$  branch approaches  $\eta = 0$  from either side. The following theorem shows that this behaviour is incompatible with a regular extension to the rotation axis.

**Theorem 13** (Regular-axis no-go for nonzero constant- $\ell$  branches). *Let an exact stationary, axisymmetric rotating-dust solution of the  $(\eta, H)$  family be defined on a connected invariant region  $\mathcal{U}$  whose closure meets a regular segment  $\mathcal{A}$  of the rotation axis. Assume that*

1. *the metric extends across  $\mathcal{A}$  as a  $C^2$  Lorentzian metric, the stationary Killing field is nonzero on  $\mathcal{A}$ , and the axial Killing field generates closed orbits with angular period  $2\pi$ ;*
2. *the dust four-velocity extends as a  $C^1$  timelike vector field to every regular limit point of the branch in  $\overline{\mathcal{U}}$ ;*
3.  *$|v| = |\eta|/r < 1$  away from the axis and  $H < 0$ ;*
4. *the logarithmic derivative  $H_{,\eta}/H$  is constant on the connected range of  $\eta$  attained in  $\mathcal{U}$ , with  $H_{,\eta}/H \equiv \ell_0$ ;*
5. *the branch is nontrivial, so that  $\eta \neq 0$  on  $\mathcal{U}$ .*

*Then necessarily  $\ell_0 = 0$ . This conclusion does not depend on the spatial form of  $v(r, z)$ .*

*Proof.* The subluminal condition  $|v| < 1$ , together with  $\eta = rv$ , gives  $|\eta| < r$  away from the axis. Since the orbit-area radius tends to zero at a regular axis segment,  $\eta$  tends to zero along every approach to  $\mathcal{A}$  from  $\mathcal{U}$ . Let us set  $\mathcal{Z} := \{\eta \neq 0\} \cap \mathcal{U}$ . The branch is nontrivial, so  $\mathcal{Z}$  is nonempty. Moreover, let us choose a connected component  $\mathcal{U}_*$  of  $\mathcal{Z}$ . If  $\mathcal{U}_* = \mathcal{U}$ , then its closure meets  $\mathcal{A}$ , and there is a sequence  $p_j \in \mathcal{U}_*$  approaching a regular axis point. If  $\mathcal{U}_*$  is a proper subset of  $\mathcal{U}$ , then the connectedness of  $\mathcal{U}$  implies that its relative boundary  $\partial_{\mathcal{U}}\mathcal{U}_*$  is nonempty. Choose  $q \in \partial_{\mathcal{U}}\mathcal{U}_*$ . Continuity forces  $\eta(q) = 0$  because otherwise  $q$  would lie in the open set  $\mathcal{Z}$  and in the same connected component as nearby points of  $\mathcal{U}_*$ . Thus, in either case, there is a sequence  $p_j \in \mathcal{U}_*$  such that

$$\eta(p_j) \longrightarrow 0, \quad (159)$$

where  $p_j$  approaches either a regular axis point or a regular interior point  $q \in \mathcal{U}$  with  $\eta(q) = 0$ . On  $\mathcal{U}_*$  the field  $\eta$  has a fixed sign, and the dust four-velocity may be written as

$$u = N(k + \Omega m), \quad N := (-H)^{-1/2}. \quad (160)$$

For constant  $\ell = \ell_0$ , Eq. (8) and Eq. (159) give

$$N(p_j) \longrightarrow N_0 := C^{-1/2} \in (0, \infty). \quad (161)$$

If  $\ell_0 \neq 0$ , Eq. (158) gives  $|\Omega(p_j)| \rightarrow \infty$ . Suppose first that  $p_j \rightarrow q$  for a regular interior point with  $r(q) > 0$ . The identity  $\det G = -r^2 < 0$  implies that  $k$  and  $m$  are linearly independent near  $q$ . After shrinking to a neighbourhood of  $q$ , we choose a smooth one-form  $\vartheta$  dual to the axial direction in the sense that  $\vartheta(k) = 0$ , and  $\vartheta(m) = 1$ . Then,  $\vartheta(u) = N\Omega$  diverges along  $p_j$ , whereas both  $u$  and  $\vartheta$  extend continuously to  $q$ . This contradiction excludes an interior limit point with  $\eta = 0$  when  $\ell_0 \neq 0$ . It remains to consider a sequence approaching the regular axis. The local product structure of the commuting stationary and axial actions, together with the nonvanishing of  $k$  on  $\mathcal{A}$ , provides a smooth one-form  $\tau$  in a neighbourhood of the axis satisfying  $\tau(k) = 1$  and  $\tau(m) = 0$ . Equivalently, in a smooth chart adapted to both actions we may take  $\tau = dt$ . Unlike  $d\phi$ , this one-form is regular on the axis. Applying  $\tau$  to Eq. (160) gives  $N = \tau(u)$  on the punctured branch. Because  $u$  extends as a  $C^1$  vector field and  $\tau$  is smooth,  $N$  extends as a  $C^1$  scalar to the axis. Therefore,  $Nk$  is  $C^1$ , and the azimuthal part  $X := u - Nk = N\Omega m$  extends as a  $C^1$  axisymmetric vector field. By Lemma 1, the coefficient  $N\Omega$  has a finite axis limit. Since  $N \rightarrow N_0 \neq 0$ , the angular velocity  $\Omega$  must also have a finite limit, in contradiction with Eq. (158). Both possible limits of the sequence in Eq. (159) are therefore incompatible with  $\ell_0 \neq 0$ . Hence,  $\ell_0 = 0$ .  $\square$

The regular-axis condition also has a simple interpretation at the level of  $H(\eta)$ . If the stationary time coordinate is normalized so that  $H(0) = -1$ , regularity requires the linear term in an analytic expansion of  $H$  about  $\eta = 0$  to vanish, i.e.  $H(\eta) = -1 + O(\eta^2)$ . The same coefficient condition is obtained independently in Ref. [28] by imposing regularity of the axis together with vanishing matter rotation there. By contrast, a nonzero constant  $\ell_0$  gives

$$H(\eta) = -e^{\ell_0 \eta} = -1 - \ell_0 \eta + O(\eta^2), \quad (162)$$

after the normalization  $H(0) = -1$ , and therefore produces precisely the excluded linear term. Theorem 13 provides an independent congruence-level proof without assuming analyticity of  $H$ , while Theorem 12 removes the congruence hypothesis altogether. Thus, unless one of the hypotheses of Theorem 12 fails, a nonzero constant- $\ell$  branch can represent only an annular or piecewise region. The remaining value  $\ell_0 = 0$  is compatible with a regular axis, and its axis behaviour is determined next.

**Lemma 2** (Axis behaviour of the  $\ell_0 = 0$  branch). *Under the regular-axis hypotheses of Theorem 13, let  $\ell_0 = 0$ . Then, in the orbit-area radial coordinate  $r$ ,*

$$\eta = O(r^2), \quad v = \frac{\eta}{r} = O(r) \quad (163)$$

as  $r \rightarrow 0$ . In particular,  $v \rightarrow 0$  as any regular segment of the rotation axis is approached.

*Proof.* For  $\ell_0 = 0$ , Eq. (8) gives  $H = -C < 0$ , while Eq. (3) implies that the coordinate angular velocity is a finite constant,  $\Omega = \Omega_0$ . The metric relation for the mixed Killing component therefore reduces to

$$\eta = g_{t\phi} + \Omega_0 g_{\phi\phi}. \quad (164)$$

Let  $\varrho = \sqrt{x^2 + y^2}$  denote the local cylindrical distance from the axis in the Cartesian coordinates introduced above. Smoothness and axisymmetry imply

$$g_{\phi\phi} = O(\varrho^2), \quad g_{t\phi} = O(\varrho^2) \quad (165)$$

as  $\varrho \rightarrow 0$ . Indeed, the axial Killing field  $m = -y\partial_x + x\partial_y$  vanishes linearly at the axis, and the regular axisymmetric metric coefficients involving  $m$  therefore vanish quadratically. It remains to relate  $\varrho$  to the orbit-area coordinate  $r$ . The exact identity  $g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -r^2$  together with regularity and nondegeneracy of the stationary direction at the axis shows that  $r$  vanishes linearly with the local cylindrical distance  $\varrho$ . Hence,  $r = O(\varrho)$ , and  $\varrho = O(r)$  for  $\varrho \rightarrow 0$ . Combining this relation with Eqs. (164) and (165) gives  $\eta = O(\varrho^2) = O(r^2)$ . Since  $v = \eta/r$ , it follows immediately that  $v = O(r)$ , proving Eq. (163).  $\square$

### C. Global obstruction to axis-connected exactly flat traces

The regular-axis theorem excludes every nonzero constant- $\ell$  branch from extending regularly to the rotation axis. It remains to examine the only admissible constant value,  $\ell_0 = 0$ . In this case, analyticity away from the axis implies that an exactly flat equatorial trace on any open radial interval propagates across the entire connected equatorial domain. The resulting behaviour is incompatible with the axis decay established in Lemma 2. This gives the following global obstruction, without requiring any assumption on the sign of  $v_{zz}$ .

**Theorem 14** (No axis-connected exactly flat constant- $\ell$  trace). *Let a nontrivial exact constant- $\ell$  solution satisfy the regular-axis hypotheses of Theorem 13. Define its equatorial radial domain by  $\mathcal{I}_0 := \{r > 0 : (r, 0) \in \mathcal{U}\}$ . Assume that  $\mathcal{I}_0$  is a connected open interval with  $0 \in \overline{\mathcal{I}_0}$ , so that the equatorial part of the branch extends continuously toward the regular axis without an intervening junction, singularity, or gap. If there exist a nonempty open interval  $I \subset \mathcal{I}_0$  and a constant  $v_0 \in (0, 1)$  such that*

$$v(r, 0) = v_0 \quad (166)$$

for all  $r \in I$ , then no such solution exists.

*Proof.* Theorem 13 first implies  $\ell_0 = 0$ . For this value, Eq. (137) reduces to  $\mathcal{F} = 2\eta$ , up to an irrelevant additive solution of the linear equation. Equation (138) therefore becomes

$$\eta_{rr} - \frac{1}{r}\eta_r + \eta_{zz} = 0 \quad (167)$$

with  $r > 0$ . Away from the axis, Eq. (167) is a linear elliptic equation with real-analytic coefficients. Interior analytic regularity therefore implies that  $\eta$  is real analytic on every neighbourhood contained in  $r > 0$  [34]. Since  $v = \eta/r$ , the function  $r \mapsto v(r, 0)$  is consequently real analytic on the connected interval  $\mathcal{I}_0$ . By assumption,  $v(r, 0) - v_0 = 0$  for all  $r \in I$ , on a nonempty open subinterval  $I \subset \mathcal{I}_0$ . The one-dimensional identity theorem for real-analytic functions therefore extends this equality to the whole connected interval, i.e.

$$v(r, 0) \equiv v_0 \quad (168)$$

for all  $r \in \mathcal{I}_0$ . Because 0 is in the closure of  $\mathcal{I}_0$ , Eq. (168) implies  $v(r, 0) \rightarrow v_0 > 0$  as  $r \downarrow 0$ . On the other hand, Lemma 2 requires  $v(r, 0) = O(r) \rightarrow 0$  as  $r \downarrow 0$  at a regular axis. The two limits are incompatible, and the assumed axis-connected solution therefore cannot exist.  $\square$

The connectedness assumption in Theorem 14 ensures that the equatorial domain belongs to a single axis-connected branch. In particular, it excludes an intervening vacuum region, junction, singular ring, or other gap between the rotation axis and the flat annulus. The result does not rely on reflection symmetry, density positivity, or any assumption on the sign of the vertical curvature. Moreover, there is no conflict with Theorem 11, which establishes the existence of local solutions on annular patches away from the axis. The global theorem shows instead that such a patch cannot be continued as a single regular constant- $\ell$  branch to the rotation axis while preserving an exactly flat equatorial trace on any open radial interval. For the remaining case  $\ell_0 = 0$ , the local continuation of flat, reflection-symmetric Cauchy data can be obtained explicitly. This also reveals the transverse behaviour responsible for the failure of a globally regular flat trace.

**Corollary 5** (Explicit  $\ell_0 = 0$  solution with an exactly flat trace). *Let  $\ell_0 = 0$  and  $r_* > 0$ . Consider the reflection-symmetric Cauchy data  $v(r, 0) = v_0$  and  $v_z(r, 0) = 0$ , with  $0 < v_0 < 1$ , for  $r$  in a neighbourhood of  $r_*$ . The unique local analytic solution of the exact constant- $\ell$  VFE is*

$$v(r, z) = v_0 \frac{\sqrt{r^2 + z^2}}{r}. \quad (169)$$

Along the equatorial plane,

$$v_{zz}(r, 0) = \frac{v_0}{r^2} > 0, \quad (170)$$

so  $z = 0$  is a strict local minimum of  $v$  in the vertical direction. If the explicit solution is continued at fixed  $r > 0$  along the vertical direction, it reaches  $v = 1$  at

$$|z| = r\sqrt{v_0^{-2} - 1}, \quad (171)$$

and becomes superluminal for larger  $|z|$ .

*Proof.* For  $\ell_0 = 0$ , the field  $\eta = rv$  satisfies Eq. (167). Consider  $\eta_0(r, z) := v_0\sqrt{r^2 + z^2}$ . To verify that  $\eta_0$  is a solution, let  $R = \sqrt{r^2 + z^2}$  and suppose that a function of  $(r, z)$  depends only on  $R$ , so that it can be written as  $f = f(R)$ . A direct calculation gives

$$f_{rr} - \frac{1}{r}f_r + f_{zz} = f''(R). \quad (172)$$

For  $f(R) = v_0 R$ , one has  $f''(R) = 0$ , and therefore  $\eta_0$  satisfies Eq. (167) wherever  $r > 0$ . On the equatorial plane,  $\eta_0(r, 0) = rv_0$  and  $(\eta_0)_z(r, 0) = 0$ , which are precisely the Cauchy data in the statement of the corollary. Analytic uniqueness near  $(r_*, 0)$  therefore gives  $\eta = \eta_0$ , and hence

$$v(r, z) = \frac{\eta(r, z)}{r} = v_0 \frac{\sqrt{r^2 + z^2}}{r}, \quad (173)$$

which proves Eq. (169). Differentiation with respect to  $z$  gives Eq. (170). Finally, solving  $v(r, z) = 1$  at fixed  $r > 0$  yields Eq. (171). Since  $v$  increases with  $|z|$  away from the equatorial plane, it exceeds unity beyond that height.  $\square$

#### D. Finite radial bounds for annular flat traces

We now ask how far an exactly flat trace can extend in such an annular branch under nonnegative density and a local vertical-maximum condition, without otherwise restricting the off-plane profile.

**Theorem 15** (Finite radial window for an exactly flat annular trace). *Let an exact constant- $\ell$  solution be  $C^2$  on  $I \times (-\varepsilon, \varepsilon)$ , where  $I \subset (0, \infty)$  is a nonempty open interval. Further assume that*

1. *the equatorial trace is exactly flat,  $v(r, 0) = v_0$  with  $0 < v_0 < 1$  for all  $r \in I$ ;*
2. *the density is nonnegative along the trace,  $\rho(r, 0) \geq 0$  for all  $r \in I$ ;*
3. *for each  $r \in I$ , the equatorial plane is a local maximum of  $v$  in the vertical direction.*

*Then,  $\ell_0 > 0$ , and every  $r \in I$  satisfies*

$$\frac{v_0}{\ell_0} \leq r \leq \frac{2v_0}{\ell_0(1 + v_0^2)}. \quad (174)$$

*If, in addition,  $\rho(r, 0) > 0$ , and  $v_{zz}(r, 0) < 0$  for all  $r \in I$ , then both inequalities in Eq. (174) are strict.*

*Proof.* Along the exactly flat trace,  $\eta(r, 0) = rv_0$  and  $\eta_r(r, 0) = v_0$ . Since  $z = 0$  is a local vertical maximum and  $v$  is  $C^2$ ,  $v_z(r, 0) = 0$ , and hence  $\eta_z(r, 0) = 0$ . Therefore,  $\nabla_{rz}\eta \neq 0$  throughout the trace, and Proposition 2 applies directly. Let us set  $x := \ell_0 r$ . Nonnegative density then requires

$$x_- := -\frac{2v_0}{1 - v_0^2} \leq x \leq x_+ := \frac{2v_0}{1 + v_0^2}. \quad (175)$$

The coefficient of  $v_{zz}$  in Eq. (148) can be written as

$$A(x) := 2 + \frac{1 - v_0^2}{v_0} x. \quad (176)$$

At the lower density endpoint  $x = x_-$ , one has  $A(x_-) = 0$ . Equation (148) would then reduce to  $2(x_- - v_0)/r^2 = 0$ , which is impossible because  $x_- < 0 < v_0$ . Hence,  $x = x_-$  cannot occur on a  $C^2$  flat trace. It follows that  $x > x_-$  and  $A(x) > 0$  hold throughout  $I$ . Equation (149) may therefore be written as

$$v_{zz}(r, 0) = \frac{2(v_0 - x)}{r^2 A(x)}. \quad (177)$$

Because  $A(x) > 0$ , the condition  $v_{zz}(r, 0) \leq 0$  implies  $x \geq v_0$ . Since  $v_0 > 0$  and  $r > 0$ , this gives  $\ell_0 > 0$ . Combining  $x \geq v_0$  with the upper density bound  $x \leq 2v_0/(1 + v_0^2)$  and using  $x = \ell_0 r$  yields Eq. (174). Finally, if  $\rho(r, 0) > 0$ , the upper density bound is strict, so  $x < x_+$ . If  $v_{zz}(r, 0) < 0$ , Eq. (177) gives  $x > v_0$ . The two radial inequalities are therefore strict.  $\square$

An immediate consequence of Theorem 15 is a bound on the radial span of the flat trace. For any two radii  $r_1 < r_2$  in  $I$ , Eq. (174) gives

$$\frac{r_2}{r_1} \leq \frac{2}{1 + v_0^2} < 2, \quad (178)$$

where the final inequality follows from  $0 < v_0 < 1$ . Thus, irrespective of its off-plane structure, a constant- $\ell$  solution satisfying the hypotheses of Theorem 15 cannot support an exactly flat equatorial trace over a radial factor of two or more.

## X. QUANTITATIVE BOUNDS FOR APPROXIMATELY FLAT EQUATORIAL TRACES

The preceding results concern equatorial traces that are exactly flat. We now relax this condition and ask how slowly the equatorial speed may vary while remaining compatible with the exact constant- $\ell$  field equation, nonnegative density, and vertical localization. The strategy is to measure deviations from flatness by the logarithmic slope of the equatorial trace, reduce the exact VFE on the equatorial plane to a one-dimensional equation, and use the resulting differential inequalities to constrain the radial interval over which approximate flatness can persist.

### A. Exact reduction on the equatorial plane

We begin by introducing a dimensionless measure of approximate flatness. Because logarithmic radial derivatives are invariant under a rescaling of the radial coordinate, they provide a natural measure of the variation of a rotation curve across a finite radial interval.

**Definition 1** ( $\varepsilon$ -flat equatorial trace). Let  $0 < R_1 < R_2$ . We define the logarithmic radial span as

$$\Lambda := \log\left(\frac{R_2}{R_1}\right). \quad (179)$$

Furthermore, let  $u(r) := v(r, 0)$  be a positive  $C^1$  equatorial trace on  $[R_1, R_2]$ . For  $0 \leq \varepsilon < 1$ , we say that  $u$  is  $\varepsilon$ -flat if

$$\left| \frac{d \log u}{d \log r} \right| \leq \varepsilon \quad (180)$$

throughout  $[R_1, R_2]$ .

The case  $\varepsilon = 0$  corresponds to exact flatness, whereas  $\varepsilon > 0$  permits a slowly rising or slowly falling trace. The restriction  $\varepsilon < 1$  will also be useful below. Indeed, if  $s := d \log u / d \log r$ , then

$$\frac{d \log(ru)}{d \log r} = 1 + s \geq 1 - \varepsilon > 0. \quad (181)$$

Hence,  $\eta(r, 0) = ru(r)$  is strictly increasing along an  $\varepsilon$ -flat trace, so the critical-point degeneracy of the density formula cannot occur there. To reduce the equatorial field equation, we introduce the dimensionless logarithmic radial coordinate

$$t := \log\left(\frac{r}{R_1}\right), \quad 0 \leq t \leq \Lambda. \quad (182)$$

For a sufficiently smooth trace, we define

$$s := \frac{d \log u}{dt}, \quad q := \frac{ds}{dt}, \quad y := \frac{\ell_0 r}{u}, \quad Z := \frac{r^2 v_{zz}(r, 0)}{u}. \quad (183)$$

Since  $dt = d \log r$ , the  $\varepsilon$ -flatness condition is simply  $|s| \leq \varepsilon$ . Using  $d/dr = r^{-1} d/dt$ , one obtains

$$u_r = \frac{us}{r}, \quad u_{rr} = \frac{u}{r^2} (q + s^2 - s), \quad \frac{dy}{dt} = (1 - s)y. \quad (184)$$

The logarithmic variables introduced above allow the exact constant- $\ell$  VFE to be reduced directly on the equatorial plane. The resulting identity also makes it possible to incorporate the density-positivity and vertical-localization conditions in a form suited to quantitative estimates.

**Lemma 3** (Exact logarithmic equatorial identity). *Let  $v$  be a  $C^2$  reflection-symmetric solution of the exact constant- $\ell$  VFE in a neighbourhood of  $[R_1, R_2] \times \{0\}$ , and assume  $0 < u(r) < 1$  for  $R_1 \leq r \leq R_2$ . In terms of the variables defined in Eq. (183), the exact VFE restricted to the equatorial plane becomes*

$$[2 + y(1 - u^2)](Z + q) + 2(1 - yu^2)s^2 + 2y(1 - u^2)s + 2y - 2 = 0. \quad (185)$$

*If, in addition, the trace is  $\varepsilon$ -flat and  $\rho(r, 0) \geq 0$  throughout  $[R_1, R_2]$ , then*

$$-\frac{2}{1 - u^2} \leq y \leq \frac{2}{1 + u^2}. \quad (186)$$

Finally, define

$$A := 2 + y(1 - u^2). \quad (187)$$

At every point where  $A > 0$  and the equatorial plane is a local maximum of  $v$  in the vertical direction, we have

$$Aq \geq 2(1 - y) - 2(1 - yu^2)s^2 - 2y(1 - u^2)s. \quad (188)$$

*Proof.* Reflection symmetry implies  $v_z(r, 0) = 0$ . Substituting the derivative identities in Eq. (184) into the exact constant- $\ell$  VFE, evaluating at  $z = 0$ , multiplying by  $r^2/u$ , and using  $\ell_0 r = yu$  gives Eq. (185) after collecting terms. Now suppose that the trace is  $\varepsilon$ -flat and that  $\rho(r, 0) \geq 0$ . Along the equatorial plane, we find  $\eta_r(r, 0) = u + ru_r = u(1 + s)$ . Since  $|s| \leq \varepsilon < 1$  and  $u > 0$ , we have  $\eta_r(r, 0) > 0$ . Together with reflection symmetry, which gives  $\eta_z(r, 0) = 0$ , this shows that  $\nabla_{rz}\eta \neq 0$  throughout the trace. Hence, Proposition 2 applies without the critical-point qualification. Dividing its bounds for  $x = r\ell_0$  by  $u > 0$  and using  $y = x/u$  yields Eq. (186). Finally, Eq. (185) can be rearranged as

$$Aq = 2(1 - y) - 2(1 - yu^2)s^2 - 2y(1 - u^2)s - AZ. \quad (189)$$

If  $A > 0$  and the equatorial plane is a local vertical maximum, then  $v_{zz}(r, 0) \leq 0$ . Since  $u > 0$ , this is equivalent to  $Z \leq 0$ . Therefore,  $-AZ \geq 0$ , which gives Eq. (188).  $\square$

### B. Calibrated density-only span bound

Before imposing any condition on the vertical curvature, the density-positivity bound alone already constrains the radial span once the strength of the constant- $\ell$  branch is fixed at the inner edge. The following corollary makes this dependence explicit.

**Corollary 6** (Density-only span bound for a calibrated branch). *Under the hypotheses of Lemma 3, let us suppose that the equatorial trace is  $\varepsilon$ -flat,  $\rho(r, 0) \geq 0$  throughout  $[R_1, R_2]$ , and  $\ell_0 > 0$ . No condition on the sign of  $v_{zz}$  is imposed. Further, we define the dimensionless inner-edge parameter*

$$\gamma_1 := \frac{\ell_0 R_1}{u(R_1)} > 0 \quad (190)$$

and let  $\underline{v} := \min_{[R_1, R_2]} u$ . Then the following inequality holds

$$\frac{R_2}{R_1} \leq \left[ \frac{2}{\gamma_1(1 + \underline{v}^2)} \right]^{1/(1-\varepsilon)}. \quad (191)$$

For the edge calibration of Ref. [6], that is

$$R_1 = R_G, \quad \ell_0 = \frac{av_c}{R_G}, \quad u(R_G) = av_c, \quad (192)$$

one has  $\gamma_1 = 1$ , and therefore

$$\frac{R_2}{R_G} \leq \left( \frac{2}{1 + \underline{v}^2} \right)^{1/(1-\varepsilon)} < 2^{1/(1-\varepsilon)}. \quad (193)$$

For  $\varepsilon = 0$ , this reduces to the exact density-positivity cutoff of Corollary 4.

*Proof.* Since  $\ell_0 > 0$  and  $u > 0$  for all  $r \in [R_1, R_2]$ , the variable  $y = (\ell_0 r)/u$  is positive throughout  $[R_1, R_2]$ . From Eq. (184), we have

$$\frac{1}{y} \frac{dy}{dt} = 1 - s. \quad (194)$$

Integrating from  $t = 0$ , corresponding to  $r = R_1$ , to  $t = \Lambda$ , corresponding to  $r = R_2$ , gives

$$\log \left( \frac{y(R_2)}{y(R_1)} \right) = \int_0^\Lambda (1 - s) dt. \quad (195)$$

Because the trace is  $\varepsilon$ -flat,  $|s| \leq \varepsilon$ , and hence

$$\log\left(\frac{y(R_2)}{y(R_1)}\right) \geq (1 - \varepsilon)\Lambda. \quad (196)$$

The density bound in Eq. (186) gives

$$y(R_2) \leq \frac{2}{1 + u(R_2)^2} \leq \frac{2}{1 + \underline{v}^2}, \quad (197)$$

while  $y(R_1) = \gamma_1$ . It follows that

$$(1 - \varepsilon)\Lambda \leq \log\left[\frac{2}{(1 + \underline{v}^2)\gamma_1}\right]. \quad (198)$$

Using  $\Lambda = \log(R_2/R_1)$  and exponentiating yields Eq. (191). For the calibration in Eq. (192),

$$\gamma_1 = \frac{(av_c/R_G)R_G}{av_c} = 1, \quad (199)$$

which gives Eq. (193). When  $\varepsilon = 0$ , the trace is constant, so  $\underline{v} = av_c$ , and the bound becomes

$$\frac{R_2}{R_G} \leq \frac{2}{1 + a^2 v_c^2}, \quad (200)$$

in agreement with Eq. (41).  $\square$

The dependence on  $\gamma_1$  is essential. Approximate flatness and nonnegative density alone do not yield a parameter-free upper bound on the radial span, because  $\gamma_1$  may be chosen arbitrarily small. This limitation is also visible in the explicit  $\ell_0 = 0$  solution of Corollary 5, for which  $s \equiv 0$  and the density is positive on every compact annulus on which the solution remains subluminal. In that example, however, the equatorial plane is a vertical minimum. A condition controlling the transverse behaviour is therefore needed to obtain a universal finite-span estimate. In what follows, we impose only the weak pointwise localization condition  $v_{zz}(r, 0) \leq 0$ , without assuming any bound on the logarithmic slope derivative  $q = ds/dt$ .

### C. Universal radial-span bounds under vertical localization

The localization condition has an invariant formulation. Let  $q = e^\mu(dr^2 + dz^2)$  be the orbit metric and let  $r = \sqrt{-\det G}$  be the orbit-area scalar. On a reflection surface on which  $dr \neq 0$ , choose the unit transverse normal

$$N = \frac{J \operatorname{grad}_q r}{|\operatorname{grad}_q r|_q},$$

where  $J$  is the orientation-compatible quarter turn. The relative dust–ZAMO rapidity is the scalar  $\psi = \operatorname{artanh} v$ .

**Proposition 10** (Invariant vertical-shape condition). *On a reflection surface, with the orientation of  $N$  chosen as above,*

$$N(N\psi) = e^{-\mu} \frac{v_{zz}}{1 - v^2}. \quad (201)$$

Consequently,  $v_{zz} \leq 0$  is equivalent to the invariant statement  $N(N\psi) \leq 0$ .

*Proof.* In the Weyl chart one has  $N = e^{-\mu/2} \partial_z$  on the reflection surface. Reflection symmetry gives  $v_z = \psi_z = 0$ , and hence

$$N(N\psi) = e^{-\mu} \psi_{zz} = e^{-\mu} \frac{v_{zz}}{1 - v^2}.$$

The factor multiplying  $v_{zz}$  is positive, which proves the equivalence.  $\square$

This condition says that the relative rapidity has a transverse maximum. It is not a statement that the density or total mass is localized there.

**Theorem 16** (Universal span bounds for approximately flat, vertically localized traces). *Let an exact constant- $\ell$  solution be  $C^2$  in a neighbourhood of  $[R_1, R_2] \times \{0\}$ , and set  $u(r) := v(r, 0)$ . Assume that*

1. *the solution is reflection symmetric with respect to the equatorial plane, and  $0 < u(r) < 1$  for all  $r \in [R_1, R_2]$ ;*
2. *the density is nonnegative along the trace,  $\rho(r, 0) \geq 0$  for all  $r \in [R_1, R_2]$ ;*
3. *the reflection surface is vertically localized in the invariant sense  $N(N\psi) \leq 0$ , equivalently  $v_{zz}(r, 0) \leq 0$ , for all  $r \in [R_1, R_2]$ ;*
4. *the equatorial trace is  $\varepsilon$ -flat for some  $0 \leq \varepsilon < 1$ .*

Let us define

$$\underline{v} := \min_{R_1 \leq r \leq R_2} u(r) > 0. \quad (202)$$

Then, the logarithmic radial span  $\Lambda = \log(R_2/R_1)$  satisfies the following bounds.

- If  $\ell_0 \leq 0$ , then

$$\Lambda \leq 2 \operatorname{artanh} \varepsilon = \log \left( \frac{1 + \varepsilon}{1 - \varepsilon} \right). \quad (203)$$

- If  $\ell_0 > 0$ , then for every  $\alpha \in (0, 1 - \varepsilon)$ ,

$$\Lambda \leq \mathcal{B}_+(\varepsilon, \alpha; \underline{v}) := \frac{\varepsilon(2 + \alpha)}{(1 + \varepsilon)(1 - \varepsilon - \alpha)} + \frac{1}{1 - \varepsilon} \log \left[ \frac{2}{(1 + \underline{v}^2)\alpha} \right]. \quad (204)$$

Thus, an annulus whose logarithmic radial span exceeds the corresponding bound cannot support a solution satisfying all four hypotheses.

*Proof.* Let us first recall that Lemma 3 gives the density bound (186). We first show that the coefficient  $A = 2 + y(1 - u^2)$  is strictly positive throughout  $[R_1, R_2]$ . If  $\ell_0 = 0$ , then  $y = 0$  and hence,  $A = 2$ . If  $\ell_0 > 0$ , then  $y > 0$ , so  $A > 2$ . It remains to consider  $\ell_0 < 0$ . In this case, the lower density bound implies  $A \geq 0$ . Suppose that  $A = 0$  at some point. Then necessarily  $y = -2/(1 - u^2)$ . At that point, the term  $A(Z + q)$  in Eq. (185) vanishes, and the remaining terms reduce to  $(s + 1)[(1 + u^2)s - (3 - u^2)] = 0$ . The two possible values are  $s = -1$ , and  $s = (3 - u^2)/(1 + u^2) > 1$ , both of which contradict  $|s| \leq \varepsilon < 1$ . Therefore,  $A > 0$  throughout the trace, and Eq. (188) applies everywhere.

- **Case  $\ell_0 \leq 0$ :** Subtract  $A(1 - s^2)$  from the right-hand side of Eq. (188). Exact factorization gives

$$\begin{aligned} 2(1 - y) - 2(1 - yu^2)s^2 - 2y(1 - u^2)s - A(1 - s^2) \\ = -y(1 + s)[3 - u^2 - (1 + u^2)s]. \end{aligned} \quad (205)$$

Here  $y \leq 0$ ,  $1 + s > 0$ , and the final factor is strictly positive because  $0 < u < 1$  and  $s < 1$ . Therefore  $q \geq 1 - s^2$ . Dividing by  $1 - s^2$  and integrating gives

$$\Lambda \leq \operatorname{artanh} s(\Lambda) - \operatorname{artanh} s(0) \leq 2 \operatorname{artanh} \varepsilon,$$

which is Eq. (203).

- **Case  $\ell_0 > 0$ :** In this case,  $y > 0$ . Equation (184) gives

$$\frac{d \log y}{dt} = 1 - s \geq 1 - \varepsilon > 0, \quad (206)$$

so  $y$  is strictly increasing with  $t$ . The upper density bound also implies

$$0 < y(r) \leq \frac{2}{1 + u(r)^2} \leq Y_{\max} := \frac{2}{1 + \underline{v}^2} < 2. \quad (207)$$

Let us fix  $\alpha \in (0, 1 - \varepsilon)$ . At every point where  $y \leq \alpha$ , we have  $0 < 1 - yu^2 \leq 1$  and  $A \leq 2 + \alpha$ . Using  $|s| \leq \varepsilon$  in Eq. (188), its right-hand side is bounded below by  $2(1 + \varepsilon)(1 - \varepsilon - \alpha)$ . Therefore, it follows that

$$q \geq c_\alpha := \frac{2(1 + \varepsilon)(1 - \varepsilon - \alpha)}{2 + \alpha} > 0 \quad (208)$$

whenever  $y \leq \alpha$ . Because  $y$  is strictly increasing, the set on which  $y \leq \alpha$  is, if nonempty, an initial portion of the logarithmic radial interval. Integrating  $q = ds/dt \geq c_\alpha$  over this portion and using  $-\varepsilon \leq s \leq \varepsilon$  shows that its length is at most

$$\frac{2\varepsilon}{c_\alpha} = \frac{\varepsilon(2 + \alpha)}{(1 + \varepsilon)(1 - \varepsilon - \alpha)}. \quad (209)$$

This estimate also applies if  $y$  remains below  $\alpha$  throughout the entire interval. Once  $y \geq \alpha$ , Eq. (206) shows that  $\log y$  increases at least at rate  $1 - \varepsilon$ . Since  $y \leq Y_{\max}$ , the remaining logarithmic radial length is at most

$$\frac{1}{1 - \varepsilon} \log \left( \frac{Y_{\max}}{\alpha} \right) = \frac{1}{1 - \varepsilon} \log \left[ \frac{2}{(1 + \underline{v}^2)\alpha} \right]. \quad (210)$$

Adding the bounds in Eqs. (209) and (210) yields Eq. (204).

This completes the proof.  $\square$

**Corollary 7** (Sharpness of the nonpositive- $\ell$  span bound). *For every  $0 < \varepsilon < 1$ , the first bound in Theorem 16 is sharp over the class  $\ell_0 \leq 0$ . Equality is attained by the rigid branch  $\ell_0 = 0$  with*

$$u(t) = a \cosh(t - t_0), \quad s(t) = \tanh(t - t_0), \quad (211)$$

*restricted to  $t - t_0 \in [-\operatorname{artanh} \varepsilon, \operatorname{artanh} \varepsilon]$  and with  $a > 0$  sufficiently small.*

*Proof.* For  $\ell_0 = 0$ ,  $\mathcal{F} = 2\eta$ . The trace in Eq. (211) gives

$$\eta = ru = \frac{a}{2} \left( \frac{r^2}{r_0} + r_0 \right),$$

so  $\mathcal{F} = c_2 r^2 + c_0$  is  $\mathcal{L}$ -harmonic and independent of  $z$ . Thus  $v_{zz} = 0$ , and the exact VFE, density, and vertical-localisation hypotheses hold on every compact subinterval on which  $0 < u < 1$  and  $\nabla \eta \neq 0$ . Moreover  $s$  runs from  $-\varepsilon$  to  $+\varepsilon$  in logarithmic time  $2 \operatorname{artanh} \varepsilon$ . The two inequalities used in the proof are equalities, establishing sharpness.  $\square$

The positive- $\ell$  estimate in Theorem 16 holds for every  $\alpha \in (0, 1 - \varepsilon)$ . We now optimize over this auxiliary parameter and examine the limit in which approximate flatness approaches exact flatness.

**Corollary 8** (Optimized positive- $\ell$  span bound and exact-flat limit). *For  $0 < \varepsilon < 1$ , let us define*

$$K_\varepsilon := \frac{\varepsilon(3 - \varepsilon)(1 - \varepsilon)}{1 + \varepsilon} \quad (212)$$

*and*

$$\alpha_\varepsilon := 1 - \varepsilon + \frac{K_\varepsilon}{2} - \frac{1}{2} \sqrt{K_\varepsilon [K_\varepsilon + 4(1 - \varepsilon)]}. \quad (213)$$

*Then,  $0 < \alpha_\varepsilon < 1 - \varepsilon$ , and  $\alpha_\varepsilon$  is the unique value of  $\alpha$  in  $(0, 1 - \varepsilon)$  that minimizes the function  $\mathcal{B}_+(\varepsilon, \alpha; \underline{v})$  defined in Eq. (204). Consequently, for  $\ell_0 > 0$ , we have*

$$\frac{R_2}{R_1} \leq \exp[\mathcal{B}_+(\varepsilon, \alpha_\varepsilon; \underline{v})]. \quad (214)$$

*In the limit  $\varepsilon \downarrow 0$ , the optimized bound becomes*

$$\frac{R_2}{R_1} \leq \frac{2}{1 + \underline{v}^2}. \quad (215)$$

*For an exactly flat trace,  $\underline{v} = v_0$ , so Eq. (215) agrees with the exact-flat span bound in Eq. (178). For fixed  $\underline{v}$ ,*

$$\inf_{0 < \alpha < 1 - \varepsilon} \mathcal{B}_+(\varepsilon, \alpha; \underline{v}) = \log \left( \frac{2}{1 + \underline{v}^2} \right) + 2\sqrt{3\varepsilon} + O(\varepsilon) \quad (216)$$

*as  $\varepsilon \downarrow 0$ .*

*Proof.* For fixed  $\varepsilon \in (0, 1)$  and  $\underline{v}$ , let us consider the function  $\mathcal{B}_+(\varepsilon, \alpha; \underline{v})$  defined in Eq. (204). It diverges as  $\alpha \downarrow 0$  and as  $\alpha \uparrow 1 - \varepsilon$ . Its derivative with respect to  $\alpha$  vanishes precisely when

$$(1 - \varepsilon - \alpha)^2 = K_\varepsilon \alpha. \quad (217)$$

The function  $\alpha \mapsto (1 - \varepsilon - \alpha)^2/\alpha$  is strictly decreasing on  $(0, 1 - \varepsilon)$ , from  $+\infty$  to 0. Hence, Eq. (217) has a unique solution in this interval. Solving the quadratic equation and selecting that root gives Eq. (213). Since  $\mathcal{B}_+$  diverges at both endpoints, this critical point is the unique global minimizer. Equation (214) then follows directly from Theorem 16. As  $\varepsilon \downarrow 0$ ,  $K_\varepsilon = 3\varepsilon + O(\varepsilon^2)$ , and  $\alpha_\varepsilon = 1 - \sqrt{3\varepsilon} + O(\varepsilon)$ . Thus,  $\alpha_\varepsilon \rightarrow 1$ , and substitution into the optimized bound gives Eq. (215). For the small- $\varepsilon$  expansion, the rational term in  $\mathcal{B}_+$  satisfies

$$\frac{\varepsilon(2 + \alpha_\varepsilon)}{(1 + \varepsilon)(1 - \varepsilon - \alpha_\varepsilon)} = \sqrt{3\varepsilon} + O(\varepsilon), \quad (218)$$

while  $-\log \alpha_\varepsilon = \sqrt{3\varepsilon} + O(\varepsilon)$ . Since  $(1 - \varepsilon)^{-1} = 1 + O(\varepsilon)$ , combining these expansions yields Eq. (216).  $\square$

#### D. Sharpness audit via a state-constrained control problem

The preceding estimates are upper bounds, but their logical status is not the same in the two signs of  $\ell_0$ . We now formulate the trace problem as an exact one-dimensional control system. This formulation proves sharpness for  $\ell_0 \leq 0$ , detects a strict gap in the positive- $\ell_0$  estimate of Corollary 8, and identifies the residual optimization problem without confusing a formal trace with a solution of the two-dimensional field equation.

For  $\ell_0 > 0$ , define

$$\mathfrak{g}(u, y, s) := \frac{2(1 + s)[(1 - s) - y(1 - u^2 s)]}{2 + y(1 - u^2)}. \quad (219)$$

The exact equatorial identity in Lemma 3 is equivalent to

$$\dot{u} = su, \quad \dot{y} = (1 - s)y, \quad \dot{s} = \mathfrak{g}(u, y, s) + w, \quad w := -Z \geq 0, \quad (220)$$

where a dot denotes  $d/dt$ . The state constraints are

$$0 < u < 1, \quad 0 < y \leq \frac{2}{1 + u^2}, \quad |s| \leq \varepsilon. \quad (221)$$

Thus the logarithmic span is precisely the exit time of a state-constrained system with the one-sided transverse-curvature control  $w \geq 0$ . In particular, the control can increase  $s$  relative to the zero-curvature drift, but it cannot decrease it.

There is also a geometric version of the same reduction. Set  $X = r^2$  and let  $f(X) = \Phi_{\ell_0}(r, ru(r))$  be the auxiliary trace. A direct calculation gives

$$\frac{d^2 f}{dX^2} = \frac{uA}{4r^3}(\dot{s} - \mathfrak{g}(u, y, s)) = \frac{uA}{4r^3}w, \quad A = 2 + y(1 - u^2) > 0. \quad (222)$$

Consequently, vertical localization is exactly convexity of the auxiliary equatorial datum as a function of  $r^2$ . This is a useful structural interpretation of the control; it is not an additional inequality.

**Lemma 4** (Analytic lifting of strict trace controls). *Let  $u$  be positive and real analytic near a compact radial interval away from the axis. Suppose that the inequalities in Eq. (221) are strict, that  $A > 0$ , and that  $\dot{s} > \mathfrak{g}(u, y, s)$ . Then the trace lifts to a reflection-symmetric real-analytic solution of the exact constant- $\ell$  VFE in a neighbourhood of the interval, with  $v(r, 0) = u(r)$ ,  $v_z(r, 0) = 0$ , positive density, and  $v_{zz}(r, 0) < 0$ .*

*Proof.* Prescribe analytic Cauchy data for the linear auxiliary equation by

$$\mathcal{F}(r, 0) = \Phi_{\ell_0}(r, ru(r)), \quad \mathcal{F}_z(r, 0) = 0.$$

The Cauchy–Kowalevski theorem applied to  $\mathcal{F}_{zz} = -\mathcal{F}_{rr} + r^{-1}\mathcal{F}_r$  gives a unique analytic solution near each point of the interval. Uniqueness makes the local solutions agree on overlaps, and compactness gives a common collar. Since  $A = \partial_\eta \Phi_{\ell_0} > 0$ , the analytic implicit-function theorem recovers a unique analytic field  $\eta$  on the same inverse sheet. The chain-rule identity underlying Eq. (138) then gives the exact VFE. On the trace,  $Z = \mathfrak{g} - \dot{s} < 0$ , so  $v_{zz} = uZ/r^2 < 0$ . Strict density and subluminality persist in a smaller collar by continuity.  $\square$

The lemma shows that strict analytic controls are genuine local PDE solutions. It does not make the elliptic Cauchy problem stable, nor does it provide an axis, a vacuum exterior, or a global completion.

We next improve the positive- $\ell_0$  comparison estimate. The result is deliberately explicit because it proves that the previously optimized bound is not the true optimum.

**Theorem 17** (Strict positive- $\ell$  gap and a refined span bound). *Assume the hypotheses of Theorem 16, let  $0 < \varepsilon < 1$ , suppose  $\ell_0 > 0$ , and write  $v_{\min} := \underline{v} \in (0, 1)$ . For  $\alpha \in (0, 1 - \varepsilon)$ , define*

$$c_+(\varepsilon, \alpha; v_{\min}) := \frac{2(1 + \varepsilon)[1 - \varepsilon - \alpha(1 - v_{\min}^2 \varepsilon)]}{2 + \alpha(1 - v_{\min}^2)}, \quad (223)$$

$$c_-(\varepsilon, \alpha; v_{\min}) := \begin{cases} \frac{2(1 - \varepsilon)[1 + \varepsilon - \alpha(1 + v_{\min}^2 \varepsilon)]}{2 + \alpha(1 - v_{\min}^2)}, & \alpha \leq \frac{1 - \varepsilon}{1 + \varepsilon}, \\ (1 - \varepsilon^2)(1 - \alpha), & \alpha > \frac{1 - \varepsilon}{1 + \varepsilon}, \end{cases} \quad (224)$$

$$c_*(\varepsilon, \alpha; v_{\min}) := \min\{c_+, c_-\}. \quad (225)$$

Then  $c_* > 0$  and

$$\Lambda \leq \mathcal{B}_*(\varepsilon, \alpha; v_{\min}) := \frac{2\varepsilon}{c_*(\varepsilon, \alpha; v_{\min})} + \frac{1}{1 - \varepsilon} \log \left[ \frac{2}{(1 + v_{\min}^2)\alpha} \right]. \quad (226)$$

Moreover, if

$$c_0(\varepsilon, \alpha) := \frac{2(1 + \varepsilon)(1 - \varepsilon - \alpha)}{2 + \alpha}, \quad (227)$$

then

$$c_*(\varepsilon, \alpha; v_{\min}) > c_0(\varepsilon, \alpha), \quad \mathcal{B}_*(\varepsilon, \alpha; v_{\min}) < \mathcal{B}_+(\varepsilon, \alpha; v_{\min}). \quad (228)$$

In particular, at the optimizer  $\alpha_\varepsilon$  in Eq. (213),

$$\begin{aligned} \Lambda &\leq \inf_{0 < \alpha < 1 - \varepsilon} \mathcal{B}_+(\varepsilon, \alpha; v_{\min}) \\ &\quad - \delta(\varepsilon, v_{\min}), \\ \delta(\varepsilon, v_{\min}) &:= 2\varepsilon \left[ \frac{1}{c_0(\varepsilon, \alpha_\varepsilon)} \right. \\ &\quad \left. - \frac{1}{c_*(\varepsilon, \alpha_\varepsilon; v_{\min})} \right] > 0. \end{aligned} \quad (229)$$

Thus the earlier positive- $\ell$  estimate is separated from the true trace optimum for every fixed  $\varepsilon > 0$  and  $v_{\min} > 0$ ; it is neither attained nor asymptotically attained with those parameters fixed.

*Proof.* On the portion of the trace where  $y \leq \alpha$ , the master inequality reads  $\dot{s} \geq \mathfrak{g}(u, y, s)$ . Put  $x = u^2$ . Differentiation of the drift gives

$$\partial_y \mathfrak{g} = \frac{2(1 + s)(sx + s + x - 3)}{[2 + y(1 - x)]^2}, \quad (230)$$

$$\partial_x \mathfrak{g} = \frac{2y(1 + s)[(1 + s) - y(1 - s)]}{[2 + y(1 - x)]^2}. \quad (231)$$

For  $|s| \leq \varepsilon < 1$  and  $0 \leq x < 1$ , the numerator in Eq. (230) is negative. Hence the minimum over  $0 < y \leq \alpha$  occurs at  $y = \alpha$ . For fixed  $x$ , the derivative with respect to  $s$  has the sign of

$$2s(x\alpha - 1) + \alpha(x - 1),$$

which is strictly decreasing in  $s$ . Any interior stationary point is therefore a maximum, and the minimum over  $[-\varepsilon, \varepsilon]$  occurs at an endpoint. At  $s = \varepsilon$ , Eq. (231) is positive, so the minimum over  $v_{\min}^2 \leq x < 1$  is  $c_+$  at  $x = v_{\min}^2$ . At

$s = -\varepsilon$ , its sign is that of  $(1 - \varepsilon) - \alpha(1 + \varepsilon)$ . This gives the two cases in Eq. (224); in the second case the infimum is the limit  $x \uparrow 1$ . It follows that

$$\dot{s} \geq c_*(\varepsilon, \alpha; v_{\min}) > 0$$

whenever  $y \leq \alpha$ . Since  $s$  remains in an interval of width  $2\varepsilon$ , this initial portion has length at most  $2\varepsilon/c_*$ . On the remaining portion,  $d \log y/dt = 1 - s \geq 1 - \varepsilon$ , while density positivity gives  $y \leq 2/(1 + u^2) \leq 2/(1 + v_{\min}^2)$ . Its length is therefore at most the logarithmic term in Eq. (226).

It remains to prove that the improvement is strict. Direct subtraction gives

$$c_+ - c_0 = \frac{2\alpha v_{\min}^2(1 + \varepsilon)[1 + \varepsilon - \alpha(1 - \varepsilon)]}{(2 + \alpha)[2 + \alpha(1 - v_{\min}^2)]} > 0. \quad (232)$$

If  $\alpha \leq (1 - \varepsilon)/(1 + \varepsilon)$ , monotonicity in  $x$  at  $s = -\varepsilon$  gives  $c_- \geq \mathbf{g}(0, \alpha, -\varepsilon)$ , and

$$\mathbf{g}(0, \alpha, -\varepsilon) - c_0 = \frac{4\alpha\varepsilon}{2 + \alpha} > 0. \quad (233)$$

In the other case,

$$c_- - c_0 = \frac{\alpha(1 + \varepsilon)[1 + \varepsilon - \alpha(1 - \varepsilon)]}{2 + \alpha} > 0. \quad (234)$$

Thus  $c_* > c_0$ . The first term in the old bound is  $2\varepsilon/c_0$ , whereas its logarithmic term is unchanged, proving Eqs. (228) and (229).  $\square$

The refined comparison is still not asserted to be optimal. It minimizes the drift over a rectangular state enclosure and then adds two exit-time estimates. The exact positive- $\ell_0$ , positive- $\varepsilon$  supremum requires the coupled evolution of  $(u, y, s)$ , including possible arcs on  $s = \varepsilon$ , and remains open. Equation (220) is the precise problem to be solved; any proposed extremal must also obey the density boundary and the tangency condition  $\mathbf{g}(u, y, \varepsilon) \leq 0$  on a sustained upper-slope arc.

The limiting exactly flat problem is more rigid and can be settled.

**Proposition 11** (Sharp exact-flat limit for positive  $\ell$ ). *Fix  $0 < v_0 < 1$  and  $\ell_0 > 0$ . Among real-analytic exactly flat traces with strict positive density and  $v_{zz} < 0$ , the supremum of the radial ratio is*

$$\sup \frac{R_2}{R_1} = \frac{2}{1 + v_0^2}. \quad (235)$$

*The supremum is approached by genuine reflection-symmetric analytic solutions of the two-dimensional VFE. It is not attained within the strict-margin class.*

*Proof.* The upper bound is Eq. (178). Conversely, choose closed intervals compactly contained in

$$\frac{v_0}{\ell_0} < r < \frac{2v_0}{\ell_0(1 + v_0^2)}$$

whose endpoint ratio tends to  $2/(1 + v_0^2)$ . The analytic trace  $u \equiv v_0$  has strict density because  $\ell_0 r < 2v_0/(1 + v_0^2)$ , while Eq. (177) gives  $v_{zz} < 0$  because  $\ell_0 r > v_0$ . Lemma 4, or equivalently the analytic auxiliary Cauchy construction used in Theorem 11, supplies a genuine analytic solution near each interval. The two strict inequalities prevent an endpoint from attaining the limiting ratio.  $\square$

Finally, the nonpositive- $\ell_0$  result admits a stronger sharpness statement than attainment by the endpoint  $\ell_0 = 0$  alone.

**Proposition 12** (Sharpness throughout the nonpositive- $\ell$  closure). *For every  $0 < \varepsilon < 1$ , the bound  $\Lambda \leq 2 \operatorname{artanh} \varepsilon$  is attained at  $\ell_0 = 0$  by Eq. (211). If  $\ell_0 < 0$ , every nondegenerate trace satisfying the same hypotheses obeys the strict inequality*

$$\Lambda < 2 \operatorname{artanh} \varepsilon. \quad (236)$$

*Nevertheless, the right-hand side is the supremum over the strict class  $\ell_0 < 0$ : there is a sequence of genuine analytic, reflection-symmetric, vertically neutral solutions with  $\ell_0 \uparrow 0$  whose spans converge to  $2 \operatorname{artanh} \varepsilon$ .*

*Proof.* For  $\ell_0 = 0$ , let  $u(t) = a \cosh(t - t_0)$  and  $s(t) = \tanh(t - t_0)$ . Then  $\dot{s} = 1 - s^2$ , and

$$\eta = ru = \frac{a}{2} \left( \frac{r^2}{r_0} + r_0 \right)$$

satisfies  $\mathcal{L}(2\eta) = 0$ . Hence the solution is independent of  $z$ , has  $v_{zz} = 0$ , and traverses from  $s = -\varepsilon$  to  $s = \varepsilon$  in exactly  $2 \operatorname{artanh} \varepsilon$ . Taking  $a$  sufficiently small gives strict subluminality and positive density on the compact annulus.

For  $\ell_0 < 0$ , the exact factorization (205) is strictly positive because  $y < 0$ ,  $1 + s > 0$ , and  $3 - u^2 - (1 + u^2)s > 0$ . Since  $A > 0$ , it follows that  $\dot{s} > 1 - s^2$  pointwise. Integration proves Eq. (236).

To prove asymptotic sharpness, keep the preceding radial auxiliary datum  $\mathcal{F}_0 = 2\eta_0 = c_2 r^2 + c_0$  fixed. For all sufficiently small negative  $\ell_0$ , the analytic implicit-function theorem solves

$$\Phi_{\ell_0}(r, \eta_{\ell_0}(r)) = \mathcal{F}_0(r)$$

on a common annulus, because  $\partial_\eta \Phi_0 = 2$ . The resulting  $\eta_{\ell_0}$  is independent of  $z$  and solves the exact VFE. Its speed, slope, and density converge uniformly with their required derivatives to the strict-margin  $\ell_0 = 0$  solution. The two transverse crossings  $s = -\varepsilon$  and  $s = \varepsilon$  persist by the implicit-function theorem because  $\dot{s} = 1 - \varepsilon^2 > 0$  at the limiting crossings. Their time separation therefore converges to  $2 \operatorname{artanh} \varepsilon$ . This construction proves sharpness within genuine analytic PDE solutions, not merely within formal trace controls.  $\square$

As an immediate consequence, the preceding bounds also exclude a semi-infinite approximately flat trace. Suppose that the hypotheses of Theorem 16 hold on every finite subinterval  $[R_1, R_2]$  of  $[R_1, \infty)$ . If  $\ell_0 \leq 0$ , Eq. (203) gives

$$\log \left( \frac{R_2}{R_1} \right) \leq 2 \operatorname{artanh} \varepsilon, \quad (237)$$

which is incompatible with  $R_2 \rightarrow \infty$ . For  $\ell_0 > 0$ , Eq. (214) likewise provides a finite upper bound on  $R_2/R_1$  when  $0 < \varepsilon < 1$ . Indeed, since  $\underline{v}^2 \geq 0$ , the right-hand side is bounded above by the same expression evaluated at  $\underline{v} = 0$ . At  $\varepsilon = 0$ , the positive- $\ell_0$  exact-flat bound Eq. (178) gives the finite ratio directly. Hence, an  $\varepsilon$ -flat trace with  $0 \leq \varepsilon < 1$  cannot persist on  $[R_1, \infty)$  under the stated density and vertical-localization conditions. More generally, any proposed approximately flat interval is excluded whenever its logarithmic radial span exceeds the bound in Eq. (203) for  $\ell_0 \leq 0$ , or its radial ratio exceeds the optimized bound in Eq. (214) for  $\ell_0 > 0$ .

## XI. ASYMPTOTIC FLATNESS AND THE CONSTANT- $\ell$ LOGARITHMIC BRANCH

The preceding analysis constrains constant- $\ell$  configurations on finite radial domains. We now ask whether such a branch can extend to an asymptotically flat isolated end. We first test the logarithmic retained-order solution of Ref. [6] against the falloff required in an asymptotically nonrotating frame. We then examine the range of validity of the approximation that produces this logarithmic behaviour and derive the corresponding obstruction directly from the exact constant- $\ell$  metric. This distinction separates a failure of the retained-order asymptotic ansatz from the stronger constraints imposed by the exact solution. Before carrying out the asymptotic test, it is important to distinguish the angular velocity of the dust from that of the ZAMO congruence. Equation (4.7) of Ref. [6] gives the coordinate angular velocity  $\Omega = d\phi/dt$  of the dust, whereas the angular velocity of the ZAMO congruence is

$$\chi := -\frac{g_{t\phi}}{g_{\phi\phi}}. \quad (238)$$

Up to the conventional sign,  $-\chi$  is the azimuthal component of the shift vector in coordinates adapted to stationarity and axisymmetry. Thus, Eq. (4.7) is not, by itself, a formula for the metric shift. The same logarithmic dependence appears in the shift only after Eq. (4.7) is combined with the metric components in Eq. (4.8). Under the constant-field and constant- $\ell$  approximation adopted in Sect. 4 of Ref. [6], and after absorbing the additive integration constant into a positive scale  $r_0$ , we find

$$\Omega^{(1)}(r) = -\frac{\ell_0}{2} \log \left( \frac{r}{r_0} \right), \quad (239)$$

together with

$$g_{tt}^{(1)} = -1, \quad g_{t\phi}^{(1)} = \frac{\ell_0}{2} r^2 \log \left( \frac{r}{r_0} \right) + v_0 r, \quad g_{\phi\phi}^{(1)} = r^2. \quad (240)$$

The superscript (1) indicates that these expressions belong to the retained-order approximation. They are not exact large-radius asymptotic expansions. Equations (238) and (240) then give

$$\chi^{(1)}(r) = -\frac{\ell_0}{2} \log\left(\frac{r}{r_0}\right) - \frac{v_0}{r}. \quad (241)$$

The first asymptotic obstruction derived below depends only on the growth order of these terms and is therefore insensitive to a possible factor-of-two convention for the mixed term in the line element.

### A. Asymptotic-flatness test of the retained-order branch

To test the retained-order branch against asymptotic flatness, we first translate the standard falloff conditions in asymptotically Cartesian coordinates into the azimuthal metric components relevant to Eq. (240). Let  $\mathcal{R} := \sqrt{x^2 + y^2 + z^2}$ , and  $\varrho := \sqrt{x^2 + y^2}$ , where  $(t, x, y, z)$  are asymptotically Cartesian coordinates on the spatial end. We assume the weak asymptotic falloff

$$g_{tt} = -1 + O(\mathcal{R}^{-1}), \quad g_{ti} = O(\mathcal{R}^{-1}), \quad g_{ij} = \delta_{ij} + O(\mathcal{R}^{-1}), \quad (242)$$

with  $i, j \in \{x, y, z\}$ , in an asymptotically nonrotating frame. The usual derivative falloffs may also be imposed, but they are not needed for the argument below. Conditions of this form are standard in the asymptotically flat isolated-system framework (see, for example, Refs. [59–61]). No use will be made of the exterior field equations or of the stronger stationary-vacuum multipole expansion.

**Lemma 5** (Azimuthal falloff implied by asymptotic flatness). *Let us assume Eq. (242), and choose the asymptotic rotation axis to be the  $z$ -axis. Let  $k = \partial_t$  and  $m = \partial_\phi = -y\partial_x + x\partial_y$  denote the stationary and axial coordinate vector fields. Along the equatorial end  $z = 0$ , where  $\mathcal{R} = \varrho$ , we have*

$$g_{t\phi} = g(k, m) = O(1), \quad g_{\phi\phi} = g(m, m) = \varrho^2 + O(\varrho) \quad (243)$$

as  $\varrho \rightarrow \infty$ . Consequently,

$$\chi = -\frac{g_{t\phi}}{g_{\phi\phi}} = O(\varrho^{-2}). \quad (244)$$

The same estimates hold on any fixed cone away from the rotation axis for which  $\varrho/\mathcal{R}$  is bounded below by a positive constant.

*Proof.* From  $m = -y\partial_x + x\partial_y$ , we obtain  $g_{t\phi} = g(k, m) = -y g_{tx} + x g_{ty}$ . Equation (242) gives  $g_{tx}, g_{ty} = O(\mathcal{R}^{-1})$ , while  $|x| + |y| = O(\varrho)$ . Hence,  $g_{t\phi} = O(\varrho/\mathcal{R})$ . Along the equatorial end, where  $\mathcal{R} = \varrho$ , this reduces to  $g_{t\phi} = O(1)$ . For the azimuthal component, the Euclidean norm of the axial vector is  $\delta_{ij} m^i m^j = x^2 + y^2 = \varrho^2$ . Using  $g_{ij} = \delta_{ij} + O(\mathcal{R}^{-1})$ , we therefore find

$$g_{\phi\phi} = g_{ij} m^i m^j = \delta_{ij} m^i m^j + (g_{ij} - \delta_{ij}) m^i m^j = \varrho^2 + O\left(\frac{\varrho^2}{\mathcal{R}}\right). \quad (245)$$

On the equatorial end,  $\mathcal{R} = \varrho$ , and hence,  $g_{\phi\phi} = \varrho^2 + O(\varrho) = \varrho^2 [1 + O(\varrho^{-1})]$ . Combining this estimate with  $g_{t\phi} = O(1)$  gives

$$\chi = -\frac{g_{t\phi}}{g_{\phi\phi}} = O(\varrho^{-2}), \quad (246)$$

which proves Eq. (244). Finally, on any fixed cone for which  $\varrho/\mathcal{R}$  is bounded below by a positive constant,  $\varrho$  and  $\mathcal{R}$  have the same asymptotic order. The preceding estimates therefore give  $g_{t\phi} = O(1)$ ,  $g_{\phi\phi} = \varrho^2 + O(\varrho)$ , and  $\chi = O(\varrho^{-2})$  there as well.  $\square$

The preceding lemma expresses the asymptotic-flatness conditions in terms of the ordinary cylindrical distance  $\varrho$ . The constant- $\ell$  metric, however, is written in terms of the usual orbit-area coordinate  $r$ . These two radial variables agree to leading order on an asymptotically flat end. Indeed, using the exact identity  $g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -r^2$  together with Eq. (242) and Lemma 5 gives, along the equatorial end,

$$r^2 = \varrho^2 + O(\varrho), \quad \frac{r}{\varrho} = 1 + O(\varrho^{-1}) \quad (247)$$

as  $\varrho \rightarrow \infty$ . Thus, the asymptotic estimates of Lemma 5 may equivalently be expressed in terms of  $r$ .

**Proposition 13** (Asymptotic obstruction for the retained-order logarithmic branch). *Let a stationary, axisymmetric metric possess an asymptotically flat end satisfying Eq. (242) in coordinates adapted to  $k = \partial_t$  and  $m = \partial_\phi$ . Suppose that, along the equatorial part of that end, the retained-order logarithmic branch of Eqs. (239)–(240) gives the leading asymptotic behaviour of the metric in the form*

$$g_{t\phi}(r, 0) = \frac{\ell_0}{2} r^2 \log\left(\frac{r}{r_0}\right) + v_0 r + \mathcal{E}_{t\phi}(r), \quad (248)$$

$$g_{\phi\phi}(r, 0) = r^2 [1 + \mathcal{E}_\phi(r)], \quad (249)$$

where  $r$  is the usual orbit-area radius,  $r_0 > 0$ ,  $v_0$  is finite,  $\ell_0$  is constant, and

$$\mathcal{E}_{t\phi}(r) = o(r^2 \log r), \quad \mathcal{E}_\phi(r) = o(1) \quad (250)$$

for  $r \rightarrow \infty$ . Then, asymptotic flatness requires  $\ell_0 = 0$ . Consequently, a nonzero constant  $\ell_0$  and the logarithmically growing retained-order branch obtained from Eqs. (4.7)–(4.8) of Ref. [6] cannot describe an asymptotically flat isolated end.

*Proof.* Since  $r_0 > 0$  is fixed,  $\log(r/r_0)/\log r \rightarrow 1$  as  $r \rightarrow \infty$ . Dividing Eq. (248) by  $r^2 \log r$  gives

$$\frac{g_{t\phi}(r, 0)}{r^2 \log r} = \frac{\ell_0}{2} \frac{\log(r/r_0)}{\log r} + \frac{v_0}{r \log r} + \frac{\mathcal{E}_{t\phi}(r)}{r^2 \log r}. \quad (251)$$

The second term tends to zero because  $v_0$  is finite, while the third tends to zero by Eq. (250). Therefore,

$$\lim_{r \rightarrow \infty} \frac{g_{t\phi}(r, 0)}{r^2 \log r} = \frac{\ell_0}{2}. \quad (252)$$

If  $\ell_0 \neq 0$ , Eq. (252) shows that  $g_{t\phi}$  grows in magnitude as  $r^2 \log r$ . On the other hand, Lemma 5 requires  $g_{t\phi} = O(1)$  in terms of  $\varrho$ , and Eq. (247) gives  $r/\varrho \rightarrow 1$ . Hence, the same  $O(1)$  bound holds when the end is parametrized by  $r$ . This contradiction excludes  $\ell_0 \neq 0$ . The same obstruction can be expressed in terms of the ZAMO angular velocity. Equations (248) and (249) give

$$\lim_{r \rightarrow \infty} \frac{\chi(r, 0)}{\log r} = -\frac{\ell_0}{2}, \quad (253)$$

whereas Lemma 5, together with Eq. (247), requires  $\chi = O(r^{-2})$ . Thus, asymptotic flatness necessarily requires  $\ell_0 = 0$ .  $\square$

The preceding proposition tests the logarithmic branch through the asymptotic behaviour of the metric. The same logarithmic growth produces an independent obstruction at the level of the dust congruence itself. Indeed, a circular trajectory whose coordinate angular velocity grows logarithmically cannot remain timelike at arbitrarily large radius on an asymptotically flat end.

**Corollary 9** (No timelike circular dust motion with logarithmic angular velocity). *Under the asymptotic-flatness hypotheses of Lemma 5, let us consider a circular dust congruence extending to arbitrarily large equatorial values of the orbit-area radius  $r$ . Suppose that its coordinate angular velocity satisfies*

$$\Omega(r) = -\frac{\ell_0}{2} \log\left(\frac{r}{r_0}\right) + o(\log r) \quad (254)$$

with  $\ell_0 \neq 0$  and  $r_0 > 0$ . Then, the circular tangent direction becomes spacelike for all sufficiently large  $r$ . Consequently, such a congruence cannot remain a timelike dust flow on an asymptotically flat end.

*Proof.* A circular trajectory with coordinate angular velocity  $\Omega$  has tangent direction  $K := k + \Omega m$ , where  $k = \partial_t$  and  $m = \partial_\phi$ . Such a direction can be normalized to a timelike four-velocity only where  $g(K, K) < 0$ . Its squared norm is

$$g(K, K) = g(k, k) + 2\Omega g(k, m) + \Omega^2 g(m, m). \quad (255)$$

Along the asymptotically flat equatorial end, Lemma 5 gives  $g(k, k) = -1 + O(\varrho^{-1})$ ,  $g(k, m) = O(1)$ , and  $g(m, m) = \varrho^2 + O(\varrho)$ . Using Eq. (247), these estimates may equivalently be written in terms of the radius  $r$  as  $g(k, k) = -1 + O(r^{-1})$ ,

$g(k, m) = O(1)$ , and  $g(m, m) = r^2[1 + o(1)]$ . Equation (254) with  $\ell_0 \neq 0$  implies  $|\Omega(r)|r \rightarrow \infty$ . Dividing Eq. (255) by the positive quantity  $\Omega^2 r^2$  therefore gives

$$\frac{g(K, K)}{\Omega^2 r^2} = \frac{g(k, k)}{\Omega^2 r^2} + \frac{2g(k, m)}{\Omega r^2} + \frac{g(m, m)}{r^2} \rightarrow 1. \quad (256)$$

Hence,  $g(K, K) > 0$  for all sufficiently large  $r$ . The circular tangent direction is therefore eventually spacelike and cannot be normalized to a timelike dust four-velocity.  $\square$

The literal retained-order metric is even more restrictive. If the remainder terms in Eqs. (248) and (249) are set to zero, i.e.  $\mathcal{E}_{t\phi} = 0$ ,  $\mathcal{E}_\phi = 0$ , Proposition 13 already excludes every  $\ell_0 \neq 0$ . If instead  $\ell_0 = 0$  while a nonzero constant  $v_0$  is retained all the way to spatial infinity, then  $g_{t\phi}^{(1)} = v_0 r$ . This linear growth is likewise incompatible with the asymptotic-flatness condition  $g_{t\phi} = O(1)$ . Thus, an exactly flat nonzero velocity profile cannot extend to infinity within the literal retained-order metric even when  $\ell_0 = 0$ . This stronger observation is not needed for Proposition 13, whose purpose is to isolate the obstruction produced specifically by the logarithmic term associated with a nonzero constant  $\ell_0$ . One might ask whether this logarithmic growth can instead be removed by a change of angular coordinates. Changing the reference scale  $r_0$  only shifts  $\log(r/r_0)$  by an additive constant and therefore leaves its radial growth unchanged. Likewise, a rigid angular transformation  $\tilde{\phi} = \phi - \omega_0 t$  can change the angular shift only by a constant. It cannot remove a term proportional to  $\log r$ . Moreover, the corresponding stationary generator is  $k_{\omega_0} = k + \omega_0 m$ . Using the asymptotic estimates established above gives

$$g(k_{\omega_0}, k_{\omega_0}) = \omega_0^2 r^2 + O(r) - 1 \quad (257)$$

as  $r \rightarrow \infty$ . For every  $\omega_0 \neq 0$ , this generator is therefore spacelike at sufficiently large radius. A rigidly rotating coordinate system cannot serve as an asymptotically inertial frame on the isolated end. Nor can the logarithmic dependence be removed by allowing the angular rotation to depend on radius. A transformation of the form  $\tilde{\phi} = \phi - F(r)t$  implies

$$d\phi = d\tilde{\phi} + F(r)dt + tF'(r)dr. \quad (258)$$

Whenever  $F'(r) \neq 0$ , substitution into the metric introduces explicit dependence on  $t$ , so the transformed coordinates are no longer adapted to stationarity. The logarithmic term therefore cannot be eliminated within the class of stationary, asymptotically inertial coordinate systems relevant to an asymptotically flat isolated end.

## B. Nonuniformity of the retained-order expansion and the exact obstruction

The preceding proposition is necessarily conditional because it tests a retained-order metric. It assumes that the displayed logarithmic term continues to give the leading asymptotic behaviour, with corrections smaller than  $r^2 \log r$ . The approximation used to derive Eq. (4.7) of Ref. [6], however, does not provide a uniform remainder estimate of this kind. Indeed, Eq. (4.6) of Ref. [6] uses the linearization

$$H = -e^{\ell_0 \eta} = -e^{\ell_0 v_0 r} \simeq -(1 + \ell_0 v_0 r), \quad (259)$$

whose dimensionless expansion parameter is

$$\varepsilon(r) := \ell_0 \eta = \ell_0 v_0 r. \quad (260)$$

For fixed  $\ell_0 v_0 \neq 0$ ,  $|\varepsilon(r)| \rightarrow \infty$  as  $r \rightarrow \infty$ . The linearization in Eq. (259) is therefore not uniform on an unbounded exterior end. The same loss of control is visible directly in the corresponding asymptotically orthonormal azimuthal component

$$\frac{g_{t\phi}^{(1)}}{\sqrt{g_{\phi\phi}^{(1)}}} = \frac{\ell_0}{2} r \log\left(\frac{r}{r_0}\right) + v_0, \quad (261)$$

which cannot remain a small perturbation at arbitrarily large radius when  $\ell_0 \neq 0$ . This leads to two possibilities. If the retained-order logarithmic term continues to dominate on the asymptotic end, then Proposition 13 excludes asymptotic flatness. If higher-order terms instead become comparable and modify the leading behaviour, then Eq. (4.7) no longer determines the asymptotic exterior. In neither case does the retained-order logarithmic formula establish an asymptotically flat constant- $\ell$  exterior. The exact constant- $\ell$  relations allow this issue to be settled without relying on the large-radius validity of the retained-order expansion. In particular, the exact metric itself excludes an asymptotically flat end whenever the equatorial speed approaches a nonzero constant.

**Corollary 10** (Exact asymptotic obstruction for a nonzero limiting speed). *Consider an exact constant- $\ell$  rotating-dust solution on an unbounded equatorial end, with  $\ell(\eta) \equiv \ell_0 \neq 0$ . Suppose that, in terms of the orbit-area radius  $r$ ,  $v(r, 0) \rightarrow v_\infty \in (0, 1)$  as  $r \rightarrow \infty$ . Then the exact metric cannot be asymptotically flat on that end.*

*Proof.* From Eq. (5),  $g_{\phi\phi} = (r^2 - \eta^2)/(-H)$ . For a constant- $\ell$  branch, Eq. (8) gives  $H = -Ce^{\ell_0\eta}$  with  $C > 0$ . Using  $\eta = rv$ , we therefore obtain the exact identity

$$\frac{g_{\phi\phi}}{r^2} = \frac{1 - v^2}{Ce^{\ell_0 rv}}. \quad (262)$$

By hypothesis,  $v(r, 0) \rightarrow v_\infty \in (0, 1)$ , and hence,  $rv(r, 0) \rightarrow +\infty$ , and  $1 - v(r, 0)^2 \rightarrow 1 - v_\infty^2 > 0$ . It follows from Eq. (262) that

$$\frac{g_{\phi\phi}}{r^2} \rightarrow \begin{cases} 0, & \ell_0 > 0, \\ +\infty, & \ell_0 < 0. \end{cases} \quad (263)$$

If the end were asymptotically flat, Lemma 5 would give  $g_{\phi\phi}/\varrho^2 \rightarrow 1$ . Moreover, Eq. (247) gives  $r/\varrho \rightarrow 1$ . Consequently,

$$\frac{g_{\phi\phi}}{r^2} = \frac{g_{\phi\phi}}{\varrho^2} \frac{\varrho^2}{r^2} \rightarrow 1, \quad (264)$$

which contradicts both alternatives in Eq. (263). Therefore, the exact metric cannot be asymptotically flat on the assumed end.  $\square$

The preceding corollary excludes a nonzero limiting equatorial speed. The same exact metric identity yields a stronger restriction. If a nonzero constant- $\ell$  branch is to remain asymptotically flat while staying uniformly subluminal, then its equatorial speed must decay at least as fast as the inverse radius.

**Corollary 11** (Inverse-radius decay required by asymptotic flatness). *Let an exact constant- $\ell$  branch with  $\ell_0 \neq 0$  extend along an asymptotically flat equatorial end, with  $r$  denoting the orbit-area radius. Assume that the equatorial speed is uniformly subluminal for sufficiently large  $r$ . More precisely, suppose that there exist  $\delta \in (0, 1)$  and  $R_0 > 0$  such that*

$$0 < v(r, 0) \leq 1 - \delta \quad (265)$$

for all  $r \geq R_0$ . Then,

$$rv(r, 0) = O(1), \quad v(r, 0) = O(r^{-1}) \quad (266)$$

as  $r \rightarrow \infty$ . In particular, an asymptotically flat end of the same nonzero constant- $\ell$  branch can neither approach a positive limiting speed nor exhibit a positive decay slower than inverse radius.

*Proof.* Asymptotic flatness and Lemma 5 give  $g_{\phi\phi}/\varrho^2 \rightarrow 1$ . Together with Eq. (247), which gives  $r/\varrho \rightarrow 1$ , this implies  $g_{\phi\phi}/r^2 \rightarrow 1$ . Taking the logarithm of the exact identity Eq. (262) gives

$$\ell_0 rv = \log(1 - v^2) - \log C - \log\left(\frac{g_{\phi\phi}}{r^2}\right). \quad (267)$$

The uniform subluminality condition Eq. (265) implies  $\delta(2 - \delta) \leq 1 - v^2 < 1$ , so  $\log(1 - v^2)$  remains bounded. Moreover,  $\log(g_{\phi\phi}/r^2) \rightarrow 0$ , while  $\log C$  is constant. The right-hand side of Eq. (267) is therefore bounded, and hence,  $\ell_0 rv = O(1)$ . Since  $\ell_0 \neq 0$  is constant,  $rv = O(1)$ , which immediately gives  $v = O(r^{-1})$  and proves Eq. (266).  $\square$

The boundedness conclusion can be sharpened to an exact locking law. This also exposes the role of the integration constant  $C$ , which cannot be discarded before the asymptotic normalization is imposed.

**Theorem 18** (Exact asymptotic locking). *Under the hypotheses of Corollary 11, the limit*

$$\eta(r, 0) = rv(r, 0) \rightarrow q := -\frac{\log C}{\ell_0} \quad (268)$$

exists. If the positive-speed dust branch itself reaches the end, then  $q > 0$  and timelikeness forces  $\Omega(q) = 0$ . If, in addition,

$$\frac{g_{\phi\phi}}{r^2} = 1 + O(r^{-2}),$$

then

$$\eta = q + O(r^{-2}), \quad v = \frac{q}{r} + O(r^{-3}), \quad H = -1 + O(r^{-2}), \quad \Omega = O(r^{-2}). \quad (269)$$

In particular, the normalization  $C = 1$  is incompatible with a positive constant- $\ell_0 \neq 0$  dust end.

*Proof.* Corollary 11 makes  $\eta$  bounded and hence  $v = \eta/r \rightarrow 0$ . Taking the limit of Eq. (267) gives  $\ell_0 \eta \rightarrow -\log C$ , proving Eq. (268). Positivity gives  $q \geq 0$ . If  $q = 0$ , Eq. (9) has logarithmic divergence, and the norm of  $k + \Omega m$  is eventually positive by the same comparison used in Corollary 9. Hence a positive timelike dust end requires  $q > 0$ . At a positive limiting  $q$ ,  $\Omega$  is smooth. If  $\Omega(q) \neq 0$ , asymptotic flatness makes the positive term  $\Omega^2 g_{\phi\phi}$  in  $H = g(k + \Omega m, k + \Omega m)$  grow quadratically, while the stationary and mixed terms remain lower order. This contradicts both timelikeness and  $H \rightarrow -Ce^{\ell_0 q} = -1$ . Therefore  $\Omega(q) = 0$ . In particular, one has  $\Omega^2 g_{\phi\phi} \rightarrow 0$  and  $\Omega g_{t\phi} \rightarrow 0$ .

Under the stronger displayed falloff, the right-hand side of Eq. (267) differs from  $-\log C$  by  $O(v^2) + O(r^{-2}) = O(r^{-2})$ . This yields the first two estimates in Eq. (269). Expanding  $H = -Ce^{\ell_0 \eta}$  about  $q$ , using  $Ce^{\ell_0 q} = 1$ , gives the third. Finally,  $\Omega(q) = 0$  and smoothness of  $\Omega$  at  $q > 0$  give the fourth. For  $C = 1$ , Eq. (268) gives  $q = 0$ , which was just excluded.  $\square$

These asymptotic results do not exclude a dust region of finite extent or a smooth transition away from the constant- $\ell$  branch before the asymptotic region is reached. The finite-boundary alternative requires a separate junction analysis, which is developed next.

## XII. HELICAL PLATEAUX, TOROIDAL BODIES, AND EXTERIOR COMPATIBILITY

The outer problem has two independent global features. Positivity can force a constant- $\eta$  region before infinity is reached, while an axis-avoiding compact body naturally has toroidal rather than simply connected exterior topology. Neither feature is visible in a purely local matching calculation.

### A. Eventual vacuum and the helical flat quotient

**Theorem 19** (Eventual vacuum and helical plateau). *Let a constant- $\ell_0 \neq 0$  regular sheet reach an asymptotically flat end and obey the uniform subluminality hypothesis of Corollary 11. Suppose  $\rho \geq 0$  on an open neighbourhood of every sufficiently large point of the sheet. Then  $\eta$  is constant on an outer open set and  $\rho = 0$  there. If that set lies in the same connected invertible sheet as the interior, then  $\eta$  is constant on the entire sheet. After asymptotic normalization, the plateau metric is*

$$ds^2 = -(dt - q_0 d\phi)^2 + r^2 d\phi^2 + dr^2 + dz^2, \quad \phi \sim \phi + 2\pi, \quad (270)$$

where  $q_0$  is the locking constant in Eq. (268).

*Proof.* The asymptotic decay  $v = O(r^{-1})$  makes both endpoints of the density interval in Eq. (33) tend to zero, whereas  $r\ell_0$  is unbounded. The algebraic density factor is therefore negative at every sufficiently large point. Since the full density is assumed nonnegative in a neighbourhood, Corollary 2 forces  $\nabla\eta = 0$  and  $\rho = 0$  on a neighbourhood of each such point. Theorem 3 propagates the resulting plateau through the connected invertible sheet.

On a constant- $\eta$  region,  $H$  and  $\Omega$  are constant and the reconstruction quadratures make  $\mu$  constant. Asymptotic normalization sets  $H = -1$ ,  $\Omega = 0$ , and  $\mu = 0$ . Substitution into Eq. (4) gives  $g_{tt} = -1$ ,  $g_{t\phi} = q_0$ , and  $g_{\phi\phi} = r^2 - q_0^2$ , which is precisely Eq. (270).  $\square$

**Corollary 12** (Regular-axis exclusion of a nontrivial helical plateau). *Under the hypotheses of Theorem 19, suppose the plateau metric extends within the same connected flat stationary-axisymmetric region to a regular fixed point of the axial action. Then  $q_0 = 0$ . If the positive dust sheet itself reaches the end, this contradicts the required locking value  $q_0 = q > 0$ . Thus a nonzero helical plateau can occur only on an axis-excised region or behind a change of domain, equations, or regularity; it is not a standard connected regular-axis completion of the dust sheet.*

*Proof.* At a regular fixed point,  $g(m, m) = 0$  and the orbit-area radius is  $r = 0$ . The continued identity  $g(m, m) = r^2 - q_0^2$  therefore gives  $q_0 = 0$ . For a positive dust sheet that itself reaches the end, Theorem 18 gives  $q_0 = q > 0$ . The two conclusions are incompatible.  $\square$

On the universal cover,  $T = t - q_0\phi$  makes Eq. (270) Minkowskian. The deck transformation acts by

$$(T, \phi) \mapsto (T - 2\pi q_0, \phi + 2\pi).$$

Thus the Lorentz holonomy is trivial but the affine time-translation holonomy is nonzero when  $q_0 \neq 0$ . The cylinder  $r = |q_0|$  is a chronology cylinder, not a horizon. An actual outer plateau restricted to  $r > r_0 > |q_0|$  is stably causal because  $g^{tt} < 0$  there; closed azimuthal timelike curves arise only in an inward extension to  $r < |q_0|$ . This is a helical flat quotient, geometrically the zero-deficit member of the usual spinning-string family, but no string source has been constructed.

### B. Toroidal topology and period obstructions

**Theorem 20** (Topology of an axis-avoiding compact body). *Let a spacelike slice carry an effective regular axial  $U(1)$  action with no exceptional orbits in the region considered, and let its orbit surface be a half-plane. Suppose a compact connected invariant matter body projects to a smoothly embedded closed disk  $B \subseteq \{r > 0\}$ . Then*

$$\Omega_{\text{mat}} \simeq D^2 \times S^1, \quad \partial\Omega_{\text{mat}} \simeq T^2,$$

and the vacuum orbit surface  $Q^+ = Q \setminus \text{int } B$  satisfies

$$\pi_1(Q^+) \simeq \mathbb{Z}, \quad H_{\text{dR}}^1(Q^+) \simeq \mathbb{R}.$$

Let  $q$  denote the vacuum orbit metric and  $\varpi = \sqrt{-\det G}$ . A global Weyl conjugate  $\zeta$  and a global twist potential  $\chi$  exist exactly when

$$P_\zeta := \oint_\gamma *_q d\varpi = 0, \quad P_\chi := \oint_\gamma \omega_k = 0 \tag{271}$$

for a generator  $\gamma$  of  $H_1(Q^+; \mathbb{Z})$ . Even when both periods vanish, global Weyl coordinates additionally require  $d\varpi \neq 0$  and global injectivity and properness of  $(\varpi, \zeta)$ .

*Proof.* The circle action is free over  $B$ . Its principal circle bundle is trivial because  $H^2(D^2; \mathbb{Z}) = 0$ , proving the solid-torus and boundary statements. The half-plane with the interior of one embedded disk removed deformation retracts onto a circle, which proves the homotopy and de Rham claims. In vacuum,  $*_q d\varpi$  and the descended twist form  $\omega_k$  are closed. A closed one-form on  $Q^+$  is exact precisely when its period on the single homology generator vanishes. Exactness supplies single-valued potentials, while noncriticality and global one-to-one proper behaviour are the separate conditions needed for those potentials to be coordinates.  $\square$

The theorem repairs an important logical point: requiring a simply connected vacuum orbit surface would exclude the natural compact nonzero- $\ell_0$  body by definition. It would not prove physical nonexistence. With  $n$  disjoint meridional disks the fundamental group becomes the free group  $F_n$ ; the corresponding physical-base Lax holonomies form a character variety only after the gauge and reality conditions have been specified.

The meridional reconstruction quadrature has the same global structure. If  $\alpha_\mu = M_r dr + M_z dz$ , the exact compatibility calculation gives

$$d\alpha_\mu = \frac{\eta_z A}{4r} \mathcal{LF} dz \wedge dr.$$

Thus  $\mathcal{LF} = 0$  is sufficient for local reconstruction, but a single-valued global  $\mu$  also requires all periods of  $\alpha_\mu$  to vanish. The converse implication fails on  $\eta_z A = 0$  and is not claimed.

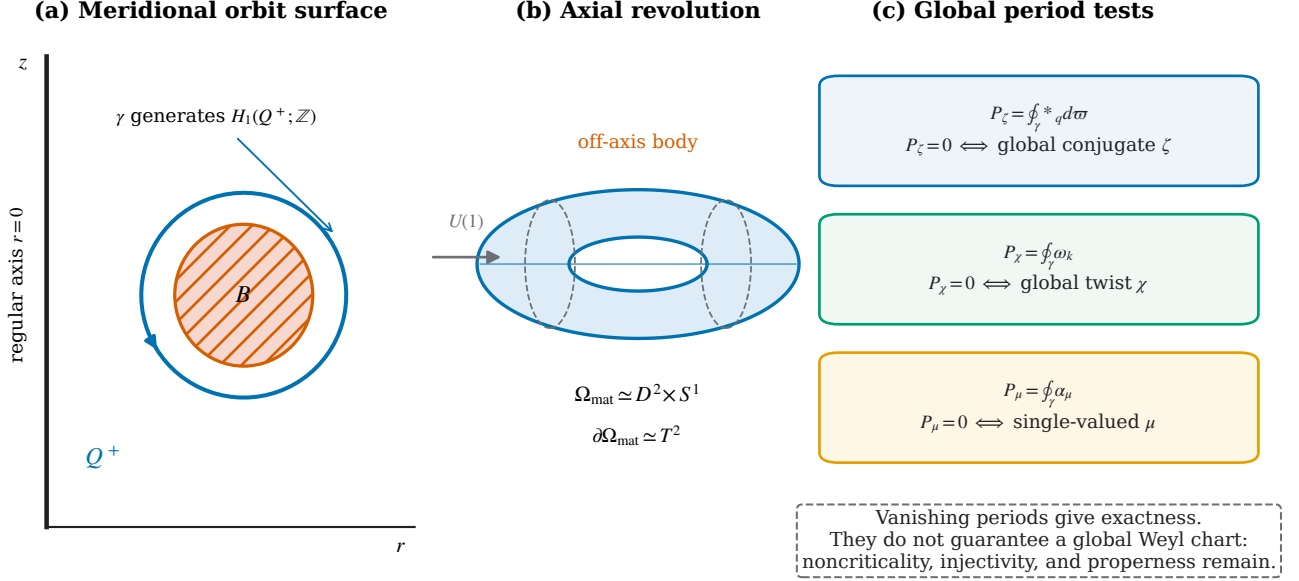


FIG. 3. Topology and periods for one compact axis-avoiding body. Its meridional disk  $B$  revolves to  $\Omega_{\text{mat}} \simeq D^2 \times S^1$ , while the exterior orbit surface  $Q^+$  has one homology generator  $\gamma$ . Under the local closedness equations,  $P_\zeta = 0$ ,  $P_\chi = 0$ , and  $P_\mu = 0$  are exactly the single-valuedness tests for  $\zeta$ ,  $\chi$ , and  $\mu$ , respectively. The first two period tests still do not make  $(\varpi, \zeta)$  a global Weyl chart: noncriticality, injectivity, and properness must be checked separately.

### C. A gauge-fixed linearized Ernst obstruction

The auxiliary potential  $\mathcal{F}$  is not the exterior Ernst potential. To state a rigorous first exterior test, fix a smooth compact vacuum orbit domain away from the axis, fix the boundary, asymptotic normalization, stationary and axial gauge, twist additive constant, and all topology sectors. On a region where a twist potential exists, use the Ernst variables  $X = (f, \chi)$  and action

$$I_E[f, \chi] = \frac{1}{2} \int_D \frac{r}{f^2} (|\nabla f|^2 + |\nabla \chi|^2) dr dz, \quad f > 0. \quad (272)$$

**Theorem 21** (Linearized Ernst Cauchy-data obstruction). *At the Minkowski point  $(f, \chi) = (1, 0)$ , the two Jacobi fields satisfy*

$$\partial_r(r\partial_r u) + \partial_z(r\partial_z u) = 0.$$

*On a fixed smooth boundary  $\Sigma$  with unique Dirichlet problem, their Cauchy data form the graph of a self-adjoint first-order Dirichlet-to-Neumann map with natural-conormal principal symbol  $r|\xi|$ . This graph is an exact Lagrangian subspace of*

$$H^{1/2}(\Sigma; \mathbb{R}^2) \oplus H^{-1/2}(\Sigma; \mathbb{R}^2).$$

*Consequently, the linearization of complete interior Darboux data can be matched only if its Ernst components lie on this graph and its independent orbit-area, conformal-factor, and meridional-curvature components satisfy their linearized constraints.*

*Proof.* The second variation of Eq. (272) at Minkowski is the direct sum of two weighted Dirichlet energies  $\frac{1}{2} \int r |\nabla u|^2$ . Its Euler equation is the displayed weighted Laplace equation. Green's identity makes the Dirichlet-to-Neumann map self-adjoint and its graph isotropic. Elliptic uniqueness and maximality of the Calderón space make it Lagrangian; the

on-shell quadratic action is a generating functional. Freezing the coefficient at a boundary point gives the principal conormal symbol  $r|\xi|$ . The last assertion is simply the linearized Darboux condition decomposed into its Ernst and independent geometric components.  $\square$

For a nonlinear candidate, let  $D$  and  $V$  denote the complete, gauge-fixed interior and vacuum boundary maps into one common boundary-data space. The matched moduli space is the fibre product

$$\mathcal{M}_{\text{dust}} \times_{\mathcal{B}_\Sigma} \mathcal{M}_{\text{vac}}.$$

If  $dD - dV$  is Fredholm and surjective modulo the explicitly fixed finite-dimensional gauge, the implicit-function theorem gives a local manifold with tangent space  $\ker(dD - dV)$ . This is a conditional theorem, not a claim that arbitrary Cauchy residuals form a Fredholm problem.

The vacuum system also has a Lax representation and twistor/Riemann–Hilbert descriptions [62–64]. The operational route is to convert complete Darboux data to a jump matrix, check its partial indices and reality conditions, and only then test finite-gap or finite-soliton sectors. Interior point linearisation is C-integrability; exterior Ernst inverse scattering is a distinct S-integrability. The positive-density source does not supply a Lax pair for the coupled dust–vacuum problem.

### XIII. FINITE BOUNDARIES AND VACUUM MATCHING

The asymptotic analysis leaves open the possibility that the constant- $\ell$  dust region terminates at finite radius and is joined to a different vacuum exterior. Reference [6] suggests that the radius at which the dust density vanishes may be interpreted as a characteristic galactic scale. Although  $\rho = 0$  provides a natural candidate location for a material boundary, it is not by itself a junction condition. A symmetry-preserving matching must instead satisfy the Darboux conditions, namely continuity of the first and second fundamental forms of the matching hypersurface [48–50]. If the induced metric is continuous while the second fundamental form has a jump, the matching carries an Israel surface stress tensor. We first identify the regular components of the zero-density set that can serve as candidate matching hypersurfaces. We then reduce the Darboux–Israel data to the stationary, axisymmetric variables used throughout this work and classify the resulting local matching alternatives. Finally, we return to the finite density cutoffs obtained earlier and distinguish the local junction conditions from the separate problem of constructing a globally regular, asymptotically flat vacuum exterior.

This local/global distinction has substantial precedent. Zingg, Aste, and Trautmann identified the overdetermination produced by prescribing both metric and normal data for a general non-rigid dust interior [10]. On the other hand, controlled Dirichlet existence and uniqueness results are available for restricted stationary axisymmetric vacuum boundary problems [65–67], and Marvan constructed a one-parameter Papapetrou-vacuum matching for every rigid van Stockum dust solution whose boundary exists [68]. These results are complementary: Dirichlet solvability does not realize arbitrary overdetermined Darboux data, while the rigid construction does not provide a constant- $\ell_0 \neq 0$  exterior theorem.

#### A. Zero-density sets and candidate matching surfaces

We begin by identifying which parts of the zero-density set can define a regular symmetry-preserving boundary. To this purpose, we recall the block form Eq. (1). The exact identity  $\det G = -r^2$  from Eq. (5) shows that, away from the rotation axis, the Killing block  $G$  has Lorentzian signature. At a regular point with  $r > 0$  and  $0 < v < 1$ , the density decomposition Eq. (32) together with the factorization Eq. (34) gives

$$\{\rho = 0\} = \mathcal{C} \cup \mathcal{Z}_- \cup \mathcal{Z}_+, \quad (273)$$

where

$$\mathcal{C} := \{(r, z) : \nabla_{rz}\eta = 0\}, \quad \mathcal{Z}_- := \left\{(r, z) : r\ell = -\frac{2v}{1-v^2}\right\}, \quad \mathcal{Z}_+ := \left\{(r, z) : r\ell = \frac{2v}{1+v^2}\right\}. \quad (274)$$

The critical set  $\mathcal{C}$  may intersect either algebraic branch, whereas  $\mathcal{Z}_- \cap \mathcal{Z}_+ = \emptyset$  for  $0 < v < 1$ . A point of the meridional zero set does not by itself define a matching surface. A regular symmetry-preserving candidate boundary is generated instead by a smooth embedded curve  $\gamma \subset \{\rho = 0\}$  in the meridional plane. Acting on  $\gamma$  with the stationary and axial symmetries produces the three-dimensional hypersurface

$$\Sigma = \mathbb{R}_t \times S_\phi^1 \times \gamma. \quad (275)$$

As long as  $\Sigma$  lies in a regular region with  $r > 0$ , it is timelike. Indeed, the Killing block has Lorentzian signature because  $\det G = -r^2 < 0$ , while the tangent direction along  $\gamma$  has positive norm in the meridional metric  $e^\mu(dr^2 + dz^2)$ . A nonsmooth component of the zero set, a component passing through a metric singularity, or the degenerate orbit  $r = 0$  does not define a standard timelike Darmois hypersurface and requires separate treatment. Equation (273) therefore provides an exhaustive classification of the zero-density set at regular points, but not yet a classification of admissible matchings. In particular, the scalar conditions defining  $\mathcal{Z}_-$  and  $\mathcal{Z}_+$  do not determine the full first and second fundamental forms of  $\Sigma$ . The required boundary data are derived in the next subsection.

### B. Symmetry-reduced Darmois–Israel data and local classification

Having identified the regular zero-density sets that can provide candidate boundaries, we now compute the geometric data required to test whether such a boundary can be matched to a vacuum exterior. Let  $\gamma$  be a smooth meridional curve generating the timelike hypersurface  $\Sigma$  introduced in Eq. (275). We parametrize  $\gamma$  by arclength  $\lambda$  with respect to the auxiliary flat meridional metric  $\delta = dr^2 + dz^2$ , so that  $\gamma(\lambda) = (r(\lambda), z(\lambda))$  and  $\dot{r}^2 + \dot{z}^2 = 1$ . The corresponding unit tangent and a chosen unit normal with respect to  $\delta$  are

$$T_E = \dot{r} \partial_r + \dot{z} \partial_z, \quad N_E = -\dot{z} \partial_r + \dot{r} \partial_z. \quad (276)$$

We define the signed Euclidean curvature  $\kappa_E$  by

$$\frac{dT_E}{d\lambda} = \kappa_E N_E. \quad (277)$$

The metric induced on the meridional  $(r, z)$ -plane is  $q = e^\mu \delta$  with  $\delta = dr^2 + dz^2$ . If  $\sigma$  denotes proper arclength along  $\gamma$ , then

$$d\sigma = e^{\mu/2} d\lambda, \quad e_s = e^{-\mu/2} T_E, \quad n = e^{-\mu/2} N_E, \quad (278)$$

where  $e_s$  is the unit meridional tangent to  $\Sigma$  and  $n$  is its spacelike unit normal. For the second fundamental form we use the convention

$$K_{ab} = h_a^\mu h_b^\nu \nabla_\mu n_\nu. \quad (279)$$

Reversing the orientation of  $n$  reverses the sign of every component of  $K_{ab}$ . The matching conditions are unaffected provided the normal orientation is chosen consistently on the two sides.

**Lemma 6** (Fundamental forms of a symmetry-preserving timelike boundary). *Let  $\Sigma$  be a smooth symmetry-preserving timelike hypersurface of the form (275), contained in a regular region with  $r > 0$ . In the intrinsic basis  $(\partial_t, \partial_\phi, e_s)$ , with  $A, B \in \{t, \phi\}$ , its first fundamental form is*

$$h_{AB} = G_{AB}, \quad h_{As} = 0, \quad h_{ss} = 1, \quad (280)$$

and its second fundamental form is

$$K_{AB} = \frac{1}{2} n(G_{AB}), \quad (281)$$

$$K_{As} = 0, \quad (282)$$

$$K_{ss} = e^{-\mu/2} \left[ \frac{1}{2} N_E(\mu) - \kappa_E \right]. \quad (283)$$

*Proof.* Restricting the block metric Eq. (1) to  $\Sigma$  gives  $g(\partial_A, \partial_B) = G_{AB}$  and  $g(\partial_A, e_s) = 0$ . Because  $e_s$  is normalized with respect to the physical meridional metric,  $g(e_s, e_s) = 1$ . This proves Eq. (280). The normal  $n$  has only meridional components, and all metric coefficients are independent of  $t$  and  $\phi$ . Hence,  $[n, \partial_A] = 0$ . For vectors tangent to  $\Sigma$ , we have  $(\mathcal{L}_n g)(X, Y) = g(\nabla_X n, Y) + g(X, \nabla_Y n)$ , so the symmetry of the second fundamental form gives  $2K_{AB} = (\mathcal{L}_n g)(\partial_A, \partial_B) = n(G_{AB})$ . This proves Eq. (281). Similarly,  $g(\partial_A, e_s) = 0$  and the commutator  $[n, e_s]$  is meridional. Block orthogonality therefore gives  $(\mathcal{L}_n g)(\partial_A, e_s) = 0$ , and hence,  $K_{As} = 0$ . It remains to determine  $K_{ss}$ . Let us set  $\omega := \mu/2$ , so that the physical meridional metric is  $q = e^{2\omega} \delta$ . Its Levi–Civita connection is related to the Euclidean connection by

$$\nabla_X^q Y = \nabla_X^E Y + X(\omega)Y + Y(\omega)X - \delta(X, Y)\nabla^E \omega. \quad (284)$$

Since  $e_s = e^{-\omega} T_E$  and  $n = e^{-\omega} N_E$ , we find

$$\nabla_{e_s}^q n = e^{-2\omega} [\nabla_{T_E}^q N_E - T_E(\omega) N_E] = e^{-2\omega} [\nabla_{T_E}^E N_E + N_E(\omega) T_E]. \quad (285)$$

From Eq. (277),  $\nabla_{T_E}^E N_E = -\kappa_E T_E$ . Therefore,

$$\nabla_{e_s}^q n = e^{-2\omega} [N_E(\omega) - \kappa_E] T_E. \quad (286)$$

Taking the scalar product with  $e_s = e^{-\omega} T_E$  gives

$$K_{ss} = q(\nabla_{e_s}^q n, e_s) = e^{-\omega} [N_E(\omega) - \kappa_E]. \quad (287)$$

Using  $\omega = \mu/2$  yields Eq. (283).  $\square$

We now apply these formulas to a dust–vacuum matching. Let a superscript  $-$  denote the rotating-dust interior and a superscript  $+$  a stationary, axisymmetric, circular vacuum exterior. We assume a fixed symmetry-preserving identification of the two boundary hypersurfaces, so that the stationary and axial group orbits are identified and the period of the axial orbits is preserved [69]. A common Killing basis is chosen under this identification. The induced tensors  $h_{ab}^\pm$  and  $K_{ab}^\pm$  may then be regarded as tensors on the same abstract boundary. We orient the unit normals on both sides consistently from the dust region toward the vacuum region and define the jump of any boundary quantity  $Q$  by

$$[Q] := Q^+ - Q^-. \quad (288)$$

Before normalising either boundary curve, let  $u$  be one common intrinsic parameter and write its two meridional embeddings as  $\gamma_\pm(u) = (r_\pm(u), z_\pm(u))$ . The complete first-form comparison is

$$G_{AB}^+|_\Sigma = G_{AB}^-|_\Sigma, \quad (289)$$

$$e^{\mu+} [(\dot{r}_+)^2 + (\dot{z}_+)^2] = e^{\mu-} [(\dot{r}_-)^2 + (\dot{z}_-)^2]. \quad (290)$$

The mixed components vanish by block orthogonality. Only after Eqs. (289)–(290) hold does their common induced metric provide one proper-arclength parameter  $\sigma$  and one abstract unit meridional tangent  $e_s$ . Normalising the two curves separately would make  $h_{ss} = 1$  tautological and would not prove that the boundary identification preserves physical arclength. With the common normalisation now established, the formulas of Lemma 6 can be compared component by component.

**Proposition 14** (Local matching alternatives at a zero-density boundary). *Let  $(\mathcal{M}^-, g^-)$  be a  $C^2$  stationary, axisymmetric, orthogonally transitive rotating-dust solution of the family considered above, and let  $\Sigma^- \subset \{\rho = 0\}$  be a smooth timelike symmetry-preserving hypersurface contained in a regular region with  $r > 0$ . Let  $(\mathcal{M}^+, g^+)$  be a  $C^2$  stationary, axisymmetric, circular vacuum spacetime with a timelike symmetry-preserving boundary  $\Sigma^+$ . We fix a symmetry-preserving identification of  $\Sigma^-$  with  $\Sigma^+$  and use the common orientation specified above. Then exactly one of the following local alternatives occurs.*

(I) **No standard distributional matching.** *If  $[h_{ab}] \neq 0$ , the induced metrics do not agree under the chosen identification. The resulting gluing is not a standard Darmois–Israel thin-shell matching. In particular, the discontinuity cannot be accounted for merely by assigning a finite surface stress tensor to  $\Sigma$ .*

(II) **Darmois matching without a thin shell.** *If*

$$[h_{ab}] = 0, \quad [K_{ab}] = 0, \quad (291)$$

*the two spacetimes admit a Darmois matching across  $\Sigma$ . In the symmetry-adapted variables of Lemma 6, these conditions are equivalent to*

$$[G_{AB}] = 0, \quad (292)$$

$$[n(G_{AB})] = 0, \quad (293)$$

$$\left[ e^{-\mu/2} \left( \frac{1}{2} N_E(\mu) - \kappa_E \right) \right] = 0. \quad (294)$$

*The mixed components impose no additional condition because  $K_{As}^\pm = 0$  on both sides.*

(III) **Matching with a nonzero Israel thin shell.** If  $[h_{ab}] = 0$  and  $[K_{ab}] \neq 0$ , the gluing carries a nonzero surface stress-energy tensor

$$S_{ab} = -\frac{1}{8\pi G_N} ([K_{ab}] - h_{ab}[K]), \quad [K] := h^{ab}[K_{ab}]. \quad (295)$$

Conversely, once the first fundamental form has been matched,  $S_{ab} = 0$  if and only if  $[K_{ab}] = 0$ .

The condition  $\rho|_\Sigma = 0$  does not distinguish alternatives (II) and (III). For a symmetry-preserving dust boundary, the dust four-velocity is tangent to  $\Sigma$ , so the normal matter projections entering the Gauss–Codazzi constraints vanish independently of the value of  $\rho$ . Imposing  $\rho = 0$  strengthens this statement to  $T_{\mu\nu}|_\Sigma = 0$ , but it still does not impose Eqs. (292)–(294) on the vacuum exterior. More specifically,

- (a) if  $\Sigma$  lies inside an open vacuum region of an already constructed  $C^2$  solution, so that the same vacuum metric is defined on both sides of the cut, restricting that metric to the two sides gives a local Darmois matching;
- (b) if  $\Sigma$  is an isolated regular component of  $\mathcal{Z}_+$  or  $\mathcal{Z}_-$ , an isolated smooth component of  $\mathcal{C}$ , or a smooth intersection of  $\mathcal{C}$  with one of the algebraic branches, then  $\Sigma$  is only a candidate matching surface. It admits a shell-free vacuum exterior precisely when a vacuum solution exists whose boundary data satisfy Eqs. (292)–(294). If the induced metric can be matched but these derivative conditions fail, the resulting continuous gluing carries the nonzero surface stress tensor in Eq. (295).

*Proof.* For a non-null matching hypersurface, equality of the first fundamental forms is the preliminary condition required for a continuous metric under the chosen identification. If  $[h_{ab}] \neq 0$ , the standard Darmois–Israel thin-shell construction does not apply. This gives alternative (I). Suppose next that  $[h_{ab}] = 0$ . The common first fundamental form then defines the intrinsic metric  $h_{ab}$  on  $\Sigma$ . A Darmois matching requires equality of the second fundamental forms,  $[K_{ab}] = 0$ . Applying Lemma 6 on the two sides gives  $[K_{AB}] = [n(G_{AB})]/2$ , and  $[K_{As}] = 0$ , together with

$$[K_{ss}] = \left[ e^{-\mu/2} \left( \frac{1}{2} N_E(\mu) - \kappa_E \right) \right]. \quad (296)$$

Together with  $[h_{AB}] = [G_{AB}] = 0$ , these relations show that Eq. (291) is equivalent to Eqs. (292)–(294). This proves alternative (II). If instead  $[h_{ab}] = 0$  but  $[K_{ab}] \neq 0$ , the singular part of the Einstein tensor is described by the Israel surface tensor Eq. (295), with the conventions for the normal and the jump fixed above. It remains only to verify that  $S_{ab} = 0$  cannot occur when  $[K_{ab}] \neq 0$ . Taking the trace of Eq. (295) with the three-dimensional inverse metric  $h^{ab}$  gives  $h^{ab}S_{ab} = [K]/(4\pi G_N)$ . Hence,  $S_{ab} = 0$  implies  $[K] = 0$ . Substitution back into Eq. (295) then gives  $[K_{ab}] = 0$ . The converse is immediate. Thus,  $S_{ab} = 0$  if and only if  $[K_{ab}] = 0$ , once  $[h_{ab}] = 0$ , proving alternative (III). It remains to determine what additional information follows from the condition  $\rho = 0$ . The dust four-velocity is

$$u = (-H)^{-1/2} (\partial_t + \Omega \partial_\phi). \quad (297)$$

Every symmetry-preserving hypersurface of the form Eq. (275) is tangent to both Killing directions. Therefore,  $u$  is tangent to  $\Sigma$  and

$$u \cdot n = 0, \quad T_{\mu\nu} n^\nu = \rho u_\mu (u \cdot n) = 0. \quad (298)$$

In particular,  $T_{\mu\nu} n^\mu n^\nu = 0$  and  $T_{\mu\nu} n^\mu e_a^\nu = 0$ , irrespective of whether  $\rho$  itself vanishes on  $\Sigma$ . For a Darmois matching, the Gauss and Codazzi equations require the corresponding Einstein-tensor projections to agree across the boundary, that is

$$[G_{\mu\nu} n^\mu n^\nu] = 0, \quad [G_{\mu\nu} n^\mu e_a^\nu] = 0. \quad (299)$$

The tangency of the dust flow therefore satisfies these necessary normal constraints against a vacuum exterior even before imposing  $\rho = 0$ . On a zero-density surface one has the stronger relation  $T_{\mu\nu}|_\Sigma = 0$ . Neither fact, however, fixes the exterior values of  $G_{AB}$ ,  $n(G_{AB})$ , or the meridional contribution in Eq. (294). Consequently, the scalar condition  $\rho = 0$  neither proves that a Darmois matching exists nor implies that a thin shell is unavoidable. Finally, if the cut is made inside an open vacuum region of an already constructed  $C^2$  solution, the restrictions of the same metric to the two sides have identical first and second fundamental forms. This proves statement (a). If instead the zero-density component is isolated, no such vacuum collar is available. The alternatives above then show that a shell-free matching exists precisely when compatible vacuum boundary data satisfying Eqs. (292)–(294) can be realized. This proves statement (b).  $\square$

The preceding classification determines when a thin shell is present. When the induced metric has been matched, the shell stress tensor can also be written explicitly in terms of the symmetry-reduced jumps in the second fundamental form.

**Corollary 13** (Symmetry-adapted form of the Israel surface tensor). *Assume that the first fundamental forms have been matched, so that  $h_{AB} = G_{AB}$  defines a common Killing block on  $\Sigma$ . Define*

$$\Delta_{AB} := [K_{AB}], \quad \Delta_s := [K_{ss}], \quad \Delta_G := G^{AB} \Delta_{AB}, \quad (300)$$

where  $G^{AB}$  is the inverse of the common matrix  $G_{AB}$ . Then, the Israel surface stress tensor has components

$$8\pi G_N S_{AB} = -\Delta_{AB} + G_{AB}(\Delta_G + \Delta_s), \quad (301)$$

$$S_{As} = 0, \quad (302)$$

$$8\pi G_N S_{ss} = \Delta_G. \quad (303)$$

Thus, the shell is completely determined by the four independent components  $(\Delta_{AB}, \Delta_s)$  and vanishes exactly when all of these components vanish.

*Proof.* Because the first fundamental forms have been matched,  $h_{AB} = G_{AB}$ ,  $h_{ss} = 1$ , and  $h_{As} = 0$ . Hence,  $[K] = h^{ab}[K_{ab}] = G^{AB}\Delta_{AB} + \Delta_s = \Delta_G + \Delta_s$ . Substitution into Eq. (295) gives Eqs. (301)–(303). The mixed component vanishes because  $K_{As}^+ = K_{As}^- = 0$  by Lemma 6. Finally, Proposition 14 shows that, once the first fundamental form has been matched,  $S_{ab} = 0$  if and only if  $[K_{ab}] = 0$ . This is equivalent to  $\Delta_{AB} = 0$  and  $\Delta_s = 0$ .  $\square$

Thus, a finite jump from positive dust density to vacuum does not by itself constitute a thin shell. The shell content is controlled by the jump in the second fundamental form.

The Darmois conditions also impose a useful invariant restriction on the orbit-area radius. This is particularly important when the dust interior is compared with a vacuum exterior written in canonical Weyl coordinates.

**Corollary 14** (Matching of the orbit-area radius). *In the common stationary-axial Killing basis fixed above, let us define  $\varpi := \sqrt{-\det G}$ . Every Darmois matching described in Proposition 14 satisfies*

$$[\varpi] = 0, \quad [n(\varpi)] = 0. \quad (304)$$

For the rotating-dust interior,  $\varpi^- = r$  by Eq. (5). Hence, an exterior written in canonical Weyl coordinates must satisfy

$$\varpi^+|_\Sigma = r|_\Sigma, \quad n(\varpi^+)|_\Sigma = n(r)|_\Sigma. \quad (305)$$

*Proof.* The condition  $[G_{AB}] = 0$  implies  $[\det G] = 0$ , and therefore,  $[\varpi] = 0$ . Since  $\det G < 0$  on the regular timelike boundary, differentiation of  $\varpi = \sqrt{-\det G}$  along the normal gives

$$2n(\log \varpi) = \text{tr}(G^{-1}nG). \quad (306)$$

For a Darmois matching, both  $G$  and  $nG$  agree on the two sides by Eqs. (292) and (293). Hence,  $[n(\log \varpi)] = 0$ . Together with  $[\varpi] = 0$ , this gives  $[n(\varpi)] = 0$ , which proves Eq. (304). Finally,  $\varpi^- = r$  for the rotating-dust interior, so Eq. (305) follows immediately.  $\square$

For later comparison with a finite radial cutoff, it is useful to record the corresponding specialization to a cylindrical boundary. Let us suppose the dust boundary is the straight cylinder  $r = R_\Sigma$ , and write the vacuum exterior in canonical Weyl coordinates, using  $\varpi$  as its canonical radial coordinate. By Corollary 14, the corresponding exterior boundary is  $\varpi = R_\Sigma$ . Let us choose the normal to point toward increasing canonical radius, namely  $r$  on the interior side and  $\varpi$  on the exterior side. Since the generating curves are straight lines in their respective meridional coordinate planes,  $\kappa_E^- = \kappa_E^+ = 0$ . The Darmois conditions therefore reduce to

$$G_{AB}^+ = G_{AB}^-, \quad (307)$$

$$e^{\mu_+} \left( \frac{dz_+}{du} \right)^2 = e^{\mu_-} \left( \frac{dz_-}{du} \right)^2, \quad (308)$$

$$e^{-\mu_+/2} \partial_\varpi G_{AB}^+ = e^{-\mu_-/2} \partial_r G_{AB}^-, \quad (309)$$

$$e^{-\mu_+/2} \partial_\varpi \mu_+ = e^{-\mu_-/2} \partial_r \mu_-, \quad (310)$$

with all quantities evaluated on the boundary and  $u$  one common intrinsic parameter. Orient that parameter so that the two canonical longitudinal Weyl coordinates are identified by  $z_+(u) = z_-(u) = -u$ . Then, Eq. (308) reduces to  $\mu_+ = \mu_-$  on  $\Sigma$ . Indeed, for this oriented straight generating curve the Euclidean curvature vanishes and Eq. (276) gives  $N_E^- = \partial_r$  and  $N_E^+ = \partial_\varpi$ , consistently with the chosen direction toward increasing canonical radius. Equations (281) and (283) then give

$$K_{AB}^- = \frac{1}{2}e^{-\mu_-/2}\partial_r G_{AB}^-, \quad K_{AB}^+ = \frac{1}{2}e^{-\mu_+/2}\partial_\varpi G_{AB}^+, \quad (311)$$

and

$$K_{ss}^- = \frac{1}{2}e^{-\mu_-/2}\partial_r \mu_-, \quad K_{ss}^+ = \frac{1}{2}e^{-\mu_+/2}\partial_\varpi \mu_+. \quad (312)$$

Equating the full induced metric and these components of the second fundamental form yields Eqs. (307)–(310). If the induced metric has been matched but any of the derivative conditions fails, the resulting continuous gluing carries the nonzero Israel surface tensor Eq. (295).

### C. Finite cutoffs and global vacuum compatibility

For the edge calibration,  $R_p^{\text{exact}} \in \mathcal{Z}_+$  is an exact density-positivity cutoff, not a constructed material boundary. Proposition 1 excludes the spatially constant ansatz on an open annulus, whereas Theorem 11 provides only local nonconstant realizations. A shell-free completion would additionally require a regular three-dimensional boundary carrying the full data

$$(G_{AB}, n(G_{AB}), K_{ss}) \Big|_\Sigma \quad (313)$$

and a compatible vacuum exterior. If a particular asymptotically flat stationary vacuum exterior is specified, its induced metric must first satisfy Eq. (292) under the chosen boundary identification. If the induced metric agrees but one or more of Eqs. (293) and (294) fail, the matching carries the nonzero surface stress tensor Eq. (295). If Eq. (292) itself fails, the proposed identification does not define a standard Israel thin-shell matching. For the global statements in this paper, we use the following standard isolated-body class.

**Definition 2** (Admissible isolated vacuum completion). An admissible isolated vacuum completion of the dust interior across  $\Sigma$  is a connected exterior spacetime  $(\mathcal{M}^+, g^+)$  satisfying all of the following conditions

1.  $g^+$  is  $C^2$  up to  $\Sigma$  and smooth in the exterior,  $R_{\mu\nu}[g^+] = 0$ , and the exterior is stationary, axisymmetric, and circular, with commuting Killing fields  $k^+$  and  $m^+$  whose axial orbits have period  $2\pi$ ;
2.  $\Sigma^+$  is a smooth connected timelike symmetry-preserving boundary, the identification with the dust boundary preserves the Killing basis and axial period, and the Darmois conditions (292)–(294) hold, so that no surface layer is present;
3. the exterior domain contains no additional boundary component, horizon, curvature singularity, or conical defect, and every axis segment in its closure is regular;
4. the orbit space  $Q^+$  of the  $\mathbb{R} \times U(1)$  action is allowed to be multiply connected. With  $\varpi = \sqrt{-\det G^+}$ , every period of  $*_q d\varpi$  and of the descended twist one-form vanishes. The resulting potentials are single valued,  $d\varpi \neq 0$  in the canonical-coordinate region, and  $(\varpi, \zeta)$  is injective and proper wherever global Weyl coordinates are invoked;
5. the exterior has one asymptotically flat end satisfying Eq. (242), with  $k^+$  normalized to unit time translation and the angular coordinate asymptotically nonrotating.

The local matching conditions must be distinguished from existence in this global class. Proposition 14 gives exact conditions at  $\Sigma$ , but satisfying them does not construct an exterior obeying Definition 2. In the Ernst formulation, the matched metric and its normal derivatives encode both Dirichlet- and Neumann-type boundary information. These data cannot be prescribed independently. Even for an exterior Dirichlet problem, arbitrary boundary values need not be admissible [70]. The global results cited here have different scopes. Mars and Senovilla prove uniqueness of the asymptotically flat stationary, axisymmetric vacuum exterior generated by a given finite rotating body within their standard isolated-body and harmonic-map setting, once the matching data and the remaining additive normalization

of the twist potential are fixed [71]. This is a uniqueness theorem conditional on existence, not a general nonlinear existence theorem. Hauser and Ernst give a constructive procedure for a given stationary, axisymmetric perfect-fluid interior, but the resulting vacuum exterior is not assumed to be asymptotically flat and the method does not by itself establish all of the regularity and global conditions in Definition 2 [72]. MacCallum, Mars, and Vera derive necessary and sufficient compatibility conditions for the overdetermined boundary data only in a perturbative problem through second order about a static isolated body [73]. Accordingly, none of these references supplies a general existence theorem for the zero-density branches considered here.

An isolated regular zero-density candidate surface must therefore pass two logically distinct tests

1. the local Darmois conditions (292)–(294);
2. existence of a vacuum solution satisfying every global condition in Definition 2 and realizing the matched boundary data.

Failure of the first test has the consequences classified in Proposition 14. Passing the local test does not guarantee the global one. If no admissible isolated vacuum completion realizes the matched data, then no exterior exists in the class of Definition 2. Introducing a thin shell changes the boundary data but does not ensure global existence.

#### XIV. OBSERVABLE SCOPE: ZAMO SPEED AND SPECTROSCOPIC FREQUENCY SHIFTS

Reference [6] explicitly notes that the ZAMO speed  $v = \eta/r$  need not coincide with the velocity inferred observationally for an external galaxy. The subsequent observational proposal [23] therefore considered photon frequency shifts. Direct stationary-axisymmetric redshift formulae and their inverse reconstruction limitations were developed in Ref. [22], while a recent locally inertial and radially locked construction provides a complementary observer prescription [24]. Recent work has also calculated gravitational-lensing observables and image asymmetries directly in stationary, axisymmetric rotating-dust geometries [26]. Here we examine this distinction using only the exact consequences of stationarity, axisymmetry, and geometric optics, without introducing an additional weak-field ray model. We first derive the invariant frequency ratio between a circular emitter and a circular observer in terms of the Killing constants of the connecting null geodesic. We then show that the scalar field  $v$  alone does not determine this observable, because the measured shift also depends on the emitter and observer frames, the metric functions, and the null-geodesic boundary-value problem. Finally, we explain why a spectroscopic prediction for an isolated system necessarily depends on the global completion of the dust region.

##### A. Invariant emitter–observer frequency shift

Let  $k$  and  $m$  denote the stationary and axial Killing fields, respectively, with their normalization and the period of the axial orbits fixed as above. Let  $\gamma$  be a future-directed, affinely parametrized null geodesic connecting an emission event  $p_e$  to an observation event  $p_o$ , and let  $\dot{\gamma}$  denote its tangent vector. For any future-directed unit timelike observer  $u$ , the measured photon frequency is the invariant scalar

$$\omega(u, \dot{\gamma}) := -g(u, \dot{\gamma}) > 0. \quad (314)$$

We consider circular emitter and observer worldlines whose four-velocities have the form

$$u_a = U_a(k + \Omega_a m), \quad a \in \{e, o\}, \quad (315)$$

where

$$U_a = [-g(k + \Omega_a m, k + \Omega_a m)]^{-1/2} = [- (g_{tt} + 2\Omega_a g_{t\phi} + \Omega_a^2 g_{\phi\phi})_a]^{-1/2}. \quad (316)$$

The normalization is well defined precisely where the circular direction  $k + \Omega_a m$  is timelike.

**Proposition 15** (Invariant emitter–observer frequency ratio). *Along  $\gamma$ , let us define the Killing constants*

$$E := -g(\dot{\gamma}, k), \quad L := g(\dot{\gamma}, m). \quad (317)$$

*Then,  $E$  and  $L$  are constant along the null geodesic. If  $E \neq 0$ , we define the conserved impact parameter  $b := L/E$ . For the circular emitter and observer of Eq. (315), let  $\omega_a := \omega(u_a, \dot{\gamma})$  with  $a \in \{e, o\}$ . Their exact frequency ratio is*

$$1 + z := \frac{\omega_e}{\omega_o} = \frac{U_e}{U_o} \frac{1 - \Omega_e b}{1 - \Omega_o b}. \quad (318)$$

For a dust emitter, Eq. (3) gives  $U_e = (-H_e)^{-1/2}$ , and hence

$$1 + z = \frac{1}{\sqrt{-H_e} U_o} \frac{1 - \Omega_e b}{1 - \Omega_o b}. \quad (319)$$

If an asymptotically flat completion exists and the observer approaches physical rest in the asymptotically nonrotating frame, assume explicitly that

$$g(k, k)_o \rightarrow -1, \quad \Omega_o^2 g_{\phi\phi}|_o \rightarrow 0, \quad \Omega_o g_{t\phi}|_o \rightarrow 0. \quad (320)$$

For a family of rays, also assume  $\Omega_o b \rightarrow 0$ . Then  $U_o \rightarrow 1$  and the exact ratio has the limiting form

$$1 + z \rightarrow \frac{1 - \Omega_e b}{\sqrt{-H_e}}. \quad (321)$$

*Proof.* Because  $\dot{\gamma}$  is tangent to an affinely parametrized null geodesic and  $k$  is a Killing field, we find

$$\dot{\gamma}^\mu \nabla_\mu g(\dot{\gamma}, k) = \dot{\gamma}^\mu \nabla_\mu (\dot{\gamma}^\nu k_\nu) = k_\nu \dot{\gamma}^\mu \nabla_\mu \dot{\gamma}^\nu + \dot{\gamma}^\mu \dot{\gamma}^\nu \nabla_\mu k_\nu = 0. \quad (322)$$

The first term vanishes by the geodesic equation. The second vanishes because  $\dot{\gamma}^\mu \dot{\gamma}^\nu$  is symmetric, whereas the Killing equation gives  $\nabla_{(\mu} k_{\nu)} = 0$ . Hence,  $E = -g(\dot{\gamma}, k)$  is constant along  $\gamma$ . The same argument applied to the axial Killing field  $m$  shows that  $L = g(\dot{\gamma}, m)$  is constant. At either endpoint, Eqs. (314) and (315) give

$$\omega_a = -g(\dot{\gamma}, u_a) = -U_a [g(\dot{\gamma}, k) + \Omega_a g(\dot{\gamma}, m)] = U_a (E - \Omega_a L). \quad (323)$$

For  $E \neq 0$  and  $b = L/E$ , this becomes  $\omega_a = E U_a (1 - \Omega_a b)$ . Taking the ratio of the emitted and observed frequencies yields Eq. (318). For a dust emitter, Eq. (3) gives  $U_e = (-H_e)^{-1/2}$ , which yields Eq. (319). Finally,

$$U_o^{-2} = -g_{tt} - 2\Omega_o g_{t\phi} - \Omega_o^2 g_{\phi\phi}.$$

The three conditions in Eq. (320) give  $U_o \rightarrow 1$ , and  $\Omega_o b \rightarrow 0$  controls the remaining ray factor. This proves the limiting relation Eq. (321).  $\square$

A spectroscopic rotation curve requires a family of such rays. Accordingly, the impact parameter  $b = L/E$  is fixed by the global source–observer boundary-value problem rather than by the local dust state.

## B. Non-identifiability and global completion

The insufficiency of  $v$  as observational input is already visible in a local orthonormal frame. Let  $u_Z$  denote the ZAMO four-velocity at the emission event, and let  $e_{\hat{\phi}}$  be the unit azimuthal spatial vector in the ZAMO frame. The dust four-velocity can then be written as

$$u_e = \Gamma(u_Z + v e_{\hat{\phi}}), \quad \Gamma = (1 - v^2)^{-1/2}. \quad (324)$$

At the same event, we can decompose the future-directed null tangent as

$$\dot{\gamma} = \omega_Z(u_Z + \hat{\nu}), \quad g(\hat{\nu}, \hat{\nu}) = 1, \quad g(u_Z, \hat{\nu}) = 0, \quad (325)$$

where  $\omega_Z := -g(u_Z, \dot{\gamma}) > 0$  is the photon frequency measured by the ZAMO. Let us define the azimuthal direction cosine by

$$\cos \alpha := g(\hat{\nu}, e_{\hat{\phi}}) \in [-1, 1]. \quad (326)$$

It follows that

$$\omega_e = -g(u_e, \dot{\gamma}) = \Gamma \omega_Z (1 - v \cos \alpha). \quad (327)$$

Thus, even the local Doppler factor depends on the photon direction, not only on the magnitude of the ZAMO speed  $v$ .

**Proposition 16** (Spectroscopic non-identifiability from the ZAMO speed). *For the class of stationary, axisymmetric observations described above, the scalar field  $v = \eta/r$  does not by itself determine the observed frequency shift. In particular, there is no functional  $\mathcal{Z}$  of  $v$  alone, independent of the observer and of the connecting null ray, such that  $z_{\text{obs}} = \mathcal{Z}[v]$  for every admissible emitter, observer, and connecting null geodesic. More specifically, knowledge of  $v$  alone does not in general determine the additional data  $H_e$ ,  $\Omega_e$ ,  $u_o$ , and  $b = L/E$ , or the map from a detector-screen position to the corresponding emission event. Consequently, exact or approximate flatness of  $v(r, 0)$  does not by itself imply exact or approximate flatness of a spectroscopically inferred rotation curve. The converse implication does not follow either.*

*Proof.* Equation (327) already shows that the local emission data are not determined by  $v$  alone. For a fixed rest-frame spectral-line frequency  $\omega_e$ , we have

$$\omega_Z = \frac{\omega_e}{\Gamma(1 - v \cos \alpha)}. \quad (328)$$

Thus, even with the emitter and its ZAMO speed fixed, the photon energy measured in the ZAMO frame depends on the emission direction  $\alpha$ . Observer dependence provides an independent obstruction. Let us fix the spacetime, the emission event, the connecting null geodesic, and the observation event. At the observation event, future-directed unit timelike circular observers form one branch of the unit hyperbola in the Lorentzian Killing plane spanned by  $k$  and  $m$ . Choose an orthonormal basis  $(e_{\hat{0}}, e_{\hat{\phi}})$  of that plane. Such observers may be parametrized locally as  $u(\beta) = \cosh \beta e_{\hat{0}} + \sinh \beta e_{\hat{\phi}}$ . Their measured photon frequency is  $-g(\dot{\gamma}, u(\beta)) = A \cosh \beta + B \sinh \beta$  for constants  $A$  and  $B$  determined by  $\dot{\gamma}$ . This expression could be constant on an interval of  $\beta$  only if  $A = B = 0$ . In that case,  $\dot{\gamma}$  would be orthogonal to the entire Killing plane. Because the metric is block diagonal between the Killing and meridional planes,  $\dot{\gamma}$  would then lie entirely in the positive-definite meridional plane. A nonzero vector in that plane cannot be null, giving a contradiction. Hence, two admissible circular observers at the same observation event can measure different photon frequencies while the spacetime, emitter, ray, and scalar field  $v$  are unchanged. No observer-independent functional of  $v$  alone can therefore reproduce every admissible frequency ratio. Even after the observer and viewing configuration have been fixed, Eq. (318) still contains the ray constant  $b = L/E$ . Its value is selected by the null-geodesic boundary-value problem and depends on the metric and on the endpoint conditions. In the rotating-dust parametrization, specifying  $v$  determines  $\eta = rv$ , but it does not by itself determine the additional function  $H$  or the corresponding angular velocity  $\Omega$ . The complete metric is also required to determine the connecting null ray and the screen-to-source map. Thus, the additional data  $H_e$ ,  $\Omega_e$ ,  $u_o$ , and  $b = L/E$  are not contained in the scalar speed profile alone, which proves the proposition.  $\square$

The proposition is a statement about insufficiency of observational input, not a solution-space degeneracy theorem. It does not assert the existence of two globally regular solutions of the full Einstein–dust system having the same field  $v$  and the same complete observational setup but producing different redshifts. Establishing such a stronger statement would require a separate existence and nonuniqueness analysis for the full boundary-value problem. What has been proved is the logically prior fact that  $v$  alone does not specify the observer four-velocity, the photon four-momentum, or the global ray connecting the emitter to the detector. It therefore cannot determine the spectroscopic frequency shift by itself.

This non-identifiability becomes especially important when the dust region is embedded in an isolated spacetime. The asymptotic and matching results of Sects. XI and XIII show that the missing propagation data cannot be supplied by the constant- $\ell$  interior alone.

**Corollary 15** (Global-completion alternatives for the observable). *Consider a nonzero constant- $\ell$  dust branch and a spectroscopic observation intended to be made by an observer at rest in an asymptotically nonrotating isolated end. Then, the following alternatives have distinct observational consequences.*

1. *If the retained-order logarithmic branch of Eqs. (239)–(240) is promoted to the actual behaviour on the unbounded end, then Proposition 13 excludes asymptotic flatness. The asymptotic specialization  $U_o \rightarrow 1$  and  $\Omega_o \rightarrow 0$  in Eq. (321) is therefore not available within that branch.*
2. *If the dust region terminates at a finite symmetry-preserving hypersurface and is Darmois matched to a stationary, axisymmetric vacuum exterior, then Eq. (318) must be evaluated in the globally matched spacetime. The exterior geometry, together with the endpoint conditions, enters the null-geodesic boundary-value problem and therefore affects both the conserved ratio  $b = L/E$  selected for the observed ray and the screen-to-source map. The interior data  $(v, \ell)$  alone do not determine the observable.*
3. *If the solution instead leaves the constant- $\ell$  branch smoothly before reaching the asymptotic end, then the transition region and the eventual exterior likewise form indispensable parts of the forward model. The constant- $\ell$  interior alone again does not determine the observed frequency shift.*

*If a thin shell is present at a matching surface, its surface data add to the information required by the global propagation problem rather than eliminating that dependence.*

*Proof.* The first alternative follows directly from Proposition 13. If the retained-order logarithmic branch is extended as the actual asymptotic metric for  $\ell_0 \neq 0$ , the required asymptotically flat end does not exist. The asymptotic observer limit used in Eq. (321) therefore cannot be taken within that branch. For the second alternative, a symmetry-preserving Darmois matching identifies the stationary and axial Killing fields across the boundary. The quantities  $E$  and  $L$  of Eq. (317) are consequently the conserved Killing quantities of the global null ray. Their ratio  $b = L/E$  is selected by the complete boundary-value problem connecting the emitter to the observer. Proposition 14 shows that the required vacuum exterior is not determined merely by the condition  $\rho = 0$ . Hence, neither the exterior geometry nor the resulting spectroscopic observable is fixed by the interior data  $(v, \ell)$  alone. In the third alternative, the constant- $\ell$  description ceases to apply before the asymptotic region is reached. The transition geometry and the subsequent exterior then enter the same global null-geodesic problem and must therefore be included in the forward model. This proves all three alternatives.  $\square$

### C. The spectroscopic forward problem

A complete observational prediction requires a forward model that maps the spacetime geometry, the emitter and observer worldlines, the source geometry, and the observational prescription to a frequency-shift map on the detector screen. This procedure can be represented schematically as

$$(g, u_e, u_o, \mathcal{S}, \iota, \mathcal{P}) \mapsto \{\gamma_\Theta\} \mapsto z_{\text{obs}}(\Theta), \quad (329)$$

where  $\Theta$  labels a detector-screen position,  $\gamma_\Theta$  is the null geodesic selected by the corresponding source–observer boundary-value problem,  $\mathcal{S}$  denotes the source and emission geometry,  $\iota$  is the disk inclination, and  $\mathcal{P}$  specifies the treatment of systemic, peculiar, and cosmological frequency shifts together with the line-profile inference procedure. If the dust region has a finite boundary, the geometry entering this forward problem must include the matched exterior and any transition region or surface layer that is present. Likewise, replacing the exact null propagation by an approximate ray model requires quantitative control of the resulting error in the screen-to-source map over the radial domain of interest. To transfer the approximate-flatness bounds of Sect. X to a spectroscopically inferred velocity profile, one would need an additional comparison between the observable profile and the local ZAMO speed. For example, a sufficient route would be to establish, on the relevant interval,

$$\left| \frac{d \log v_{\text{spec}}}{d \log R_{\text{obs}}} - \frac{d \log v(r, 0)}{d \log r} \right| \leq \delta \quad (330)$$

for some prescribed tolerance  $\delta \geq 0$  measuring the allowed difference between the two logarithmic slopes. Here,  $R_{\text{obs}}$  denotes the projected observational radius and  $r$  the emission radius. Such an estimate would have to be supplemented by controlled monotone bounds relating  $R_{\text{obs}}$  to  $r$ . Both velocity profiles must also remain positive on the interval so that the logarithmic slopes are defined. No comparison of this kind follows from the velocity-field equation alone. It requires the completed spacetime, the emitter and observer frames, and the null-geodesic propagation problem developed above. Constructing and testing this full spectroscopic forward model is therefore a natural subject for subsequent numerical work.

## XV. CONCLUSIONS AND OUTLOOK

The outcome is a branch-specific phase diagram rather than a blanket no-go statement for rotating dust. The broad finite-radius and matching tensions were already visible in earlier work; the advance here is to make the constant- $\ell$  mechanisms sharp, to identify their singular strata, and to connect local realizability to the genuinely global tests that a galaxy model must pass.

Locally, the proposed nonzero constant field fails both the exact and retained-order VFEs on every open radial region. Nonconstant fields can nevertheless realize an exactly flat equatorial trace away from the axis. The auxiliary map explains both facts: its regular sheet linearizes the VFE, whereas its lower density boundary is a Whitney fold at which the Legendre map and PDE symbol degenerate. Classical regular sheets are automatically analytic, but their elliptic Cauchy continuation is Hadamard unstable. Runge approximation enlarges a local germ only on a protected compact core; it does not manufacture a global physical solution.

The sharp local freedom is therefore compatible with strong global rigidity. A regular metric axis forces  $H'(0) = 0$ , excluding every nonzero constant- $\ell$  sheet that reaches it. Under complete-half-plane and decay hypotheses, the

remaining  $\ell_0 = 0$  branch is Minkowski vacuum. On axis-avoiding annuli, density and the invariant vertical-maximum condition give finite span bounds. The nonpositive- $\ell_0$  bound and the positive- $\ell_0$  exact-flat limit are sharp through genuine analytic PDE solutions; for positive tolerance and positive  $\ell_0$ , the new strict gap proves that the earlier comparison estimate was not optimal and isolates a precise state-constrained control problem that remains open.

At infinity, the retained-order logarithmic branch is not a standard asymptotically flat end. An exact uniformly subluminal nonzero constant- $\ell_0$  tail instead locks  $rv$  to one constant. Nonnegative density then forces an eventual vacuum plateau whose local geometry is a flat helical quotient. Nonzero affine time holonomy is possible only on an axis-excised domain; it cannot extend within the same flat region to a regular axial fixed point. A compact axis-avoiding body is naturally a solid torus, so its vacuum orbit surface carries period obstructions that a simply connected exterior definition would erase by assumption.

At a finite boundary,  $\rho = 0$  supplies neither an exterior nor a shell-free matching. The complete first and second fundamental forms, equivalently the GHY boundary momentum together with the induced metric, must agree. Even then the nonlinear vacuum exterior must be constructed; the Calderón graph derived here is only its gauge-fixed linearized compatibility test. The weighted scalar action, its Noether currents, Poisson cocycle, and multisymplectic form organize the auxiliary equation, but they are not the Einstein–dust phase space. The latter requires the vortical Brown action, gravitational constraints, physical boundary terms, and covariant charges. The neutral EMDA rigidity and duality analysis is conditional on a different field theory and supplies no hidden completion of the dust model.

Finally,  $v$  is a local ZAMO speed, not a spectroscopic rotation curve. A prediction requires the completed metric, emitter and observer frames, and the connecting null-geodesic boundary-value problem. Thus the constant-field edge construction does not by itself define a globally viable isolated galaxy. The results do not exclude nonconstant- $\ell$  interiors, separately constructed finite exteriors, pressure-supported or kinetic matter, nonstandard asymptotics, or an observer-complete model that survives all of the tests above.

#### A. Natural continuation: from admissibility to physical prediction

The present analysis supplies admissibility conditions rather than a completed galaxy model. A natural continuation is therefore not to begin by fitting another prescribed ZAMO-speed profile, but to construct a complete spacetime, establish its dynamical viability, and only then compare its predicted observables with data. Such a program has several distinct components.

1. *Close the remaining sharp control problem.* Determine the exact exit-time supremum for the positive- $\ell_0$ , positive- $\varepsilon$  system in Eq. (220), including boundary arcs and density tangencies, and prove that any extremal trace lifts to an admissible local PDE solution. This is the one unresolved sharpness question in the present annular theory.
2. *Construct globally admissible backgrounds.* A finite parametrization of  $H$  or  $\ell$ , together with an appropriate matter closure, should be chosen so that a finite-thickness galaxy can be constructed subject to the field equations rather than imposed kinematically. Pressureless dust remains an important control case, but anisotropic-fluid or Einstein–Vlasov descriptions provide natural extensions when velocity dispersion, multistreaming, or pressure support cannot be neglected. Regular-axis and equatorial conditions, density positivity, causality, and either a finite free boundary or a controlled transition to the outer region must be imposed as part of the same construction.
3. *Solve the interior and exterior as a coupled boundary-value problem.* For a finite matter configuration, the interior fields, the location of the matter boundary, and the stationary vacuum exterior should be determined consistently rather than obtained by attaching an exterior after fitting the interior. The symmetry-reduced Darboux data derived in this work provide the local interface conditions. A surface layer should be introduced only when it is part of the physical model, not as an implicit replacement for incompatible boundary data. Numerical continuation from analytically controlled limits, using spectral or high-order finite-element methods, is one possible approach to the resulting nonlinear free-boundary problem.
4. *Establish numerical reliability and global regularity.* Candidate solutions should be tested by convergence under domain and resolution refinement, residuals of all independent Einstein and matter equations, and explicit checks of the axis and junction conditions. Sensitivity to boundary placement and parametrization should also be quantified. Curvature invariants, energy conditions, causal pathologies, and the appropriate global charges provide further independent diagnostics.
5. *Determine dynamical stability.* The existence of circular motion in a stationary geometry is not sufficient for an astrophysical galaxy. Radial and vertical epicyclic stability provide useful first diagnostics where those notions are applicable, but the matter and geometry ultimately require a coupled perturbation analysis. For dust, caustic

formation and multistreaming deserve particular attention. Fluid and kinetic extensions introduce their own pressure, dispersion, and phase-space modes. Selected stationary configurations should eventually be tested by nonlinear evolution.

6. *Compute observables from the complete spacetime.* Implement the forward map formulated in Sect. XIV, including ray tracing through the interior, transition region, and exterior. In particular, quantify the relation between the inferred spectroscopic speed and the local ZAMO speed before applying the approximate-flatness bounds derived here.
7. *Compare viable relativistic models with conventional mass models.* Only after the preceding consistency tests have been passed is a statistical comparison with galaxy data meaningful. Rotation, vertical-dynamics, photometric, gas, and lensing measurements should be treated with common baryonic nuisance parameters and controlled systematics. Approximate flatness should be defined quantitatively, for example through a logarithmic-slope interval, rather than by visual inspection. Parameter identifiability and model complexity should be included when comparing the relativistic construction with standard baryons-plus-halo baselines.

This program is deliberately falsifiable. It may show that no regular, positive-density, stable, globally completed solution produces an observable effect of the required magnitude. Such a result would extend the present mathematical obstructions into a stronger physical exclusion. Conversely, if viable configurations exist, they must make joint predictions for spectroscopic shifts, vertical dynamics, lensing, and other observables that distinguish them from conventional mass models. The present work supplies the consistency conditions that such future constructions must satisfy before either outcome can be assessed.

## Appendix A: Symbolic reproducibility checks

The accompanying package separates exact algebraic certificates from analytical proofs. From the revised manuscript directory, the complete symbolic suite is run with

```
make verify
```

and the vector figures are regenerated with

```
make figures
```

The suite contains the following independent layers.

1. `constant_ell_checks.mpl` verifies the original eleven exact reductions: the exact-minus-retained identity, both constant-field residuals, the Killing-block determinant, density factorization and roots, the point-linearization identity, the logarithmic trace identity, the fold factorization, and the optimizer equation.
2. `verify_variational.mpl` and `verify_variational_sympy.py` independently verify the pulled-back Euler–Lagrange equation, fibre Hessian, De Donder–Weyl Hamiltonian, Poincaré–Cartan pullback, multisymplectic and Noether identities, Pohozaev formula, exact modes, distributional reflected-layer coefficient, and the reduced Ernst equations.
3. `rifsimp_branches.mpl` records every denominator and treats  $r = 0$ ,  $\eta = 0$ ,  $v = 0$ ,  $A = 0$ , and  $\ell_0 = 0$  separately before applying `rifsimp`. It recovers the nonlinear VFE on the regular sheet, proves that no two-dimensional physical solution lies wholly in the fold, and verifies the exact Frobenius identity for the  $\mu$  quadrature.
4. `thomas_certificates.mpl` uses `DifferentialThomas` to obtain a disjoint four-component differential decomposition. It retains the two polynomial closures at  $\eta = 0$  instead of silently dividing them away, and independently proves the first-prolongation constant-field incompatibility and the symbol-rank jump.
5. `verify_approx_flat_sharpness.py` checks the exact control drift, all comparison differences used in Theorem 17, and a seeded set of 5000 admissible state comparisons. The randomized comparisons test the encoded inequalities; the proof of the theorem remains analytical.
6. `audit_core_symbolic.py` and `audit_physical_string_sympy.py` independently rederive selected density, trace, asymptotic, matching, covariant-action, frame, duality, and axidilaton identities without importing the primary verification routines.

7. `generate_figures.py` regenerates the three vector figures and the machine-readable density-boundary CSV. It checks the plotted roots and fold identity before writing any artifact; the curves labelled as representative radial paths remain explicitly schematic.
8. `check_manuscript.py` audits citation keys, labels, metadata, house-style inequalities and delimiters, and release-blocking build-log diagnostics. The atomic `revision_tracker.py` and the generated `REVISION_STATUS.md` provide a restart-safe claim–evidence record.

Every scientific certificate terminates with an error when an asserted identity or branch count fails. Maple 2022 uses a local kernel socket on some systems, but the scripts make no network request. These calculations certify the encoded local algebra and differential ideals. They do not certify axis regularity, global existence, positivity away from the tested hypotheses, nonlinear vacuum matching, stability, or observational viability.

### Code availability

The symbolic certificates, validation scripts, source data, and figure generators supporting this work are supplied with the manuscript. A public research package supporting Parts I and II is planned for release at

<https://github.com/dutykh/Constant-Ell-Rotating-Dust>

The repository is not required for the logical validity of the analytical proofs; it is intended to make the reported symbolic identities and numerical validation artifacts transparent and reusable.

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